

Hours-Biased Technological Change^{*}

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Abstract

In the United States, average hours worked have declined over the long run, but unevenly across the wage distribution: hours have fallen at the bottom and risen at the top, reversing the hours-wage correlation. Motivated by these facts, this paper proposes a matching model of the labor market in which workers choose hours and sort into jobs that differ in how they reward working time. Hours-Biased Technological Change (HBTC) increases the complementarity between hours and the worker-job bundle, so the return to hours increasingly operates through the allocation of workers to jobs. HBTC can overturn income effects when high-skill workers are allocated to jobs where their hours are especially productive. An estimated version of the model using U.S. data over the post-1980 period reveals that HBTC accounts for about a third of the model-implied rise in income inequality and explains the rise in the hours-wage correlation.

Keywords: Hours; Sorting; Technological Change; Income Inequality.

JEL codes: J22, J31, C78, O33.

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1 Introduction

Over the last century, hours worked have declined substantially. For example, the average American worker now spends 26% less time at work than she did a century ago. Given that sustained economic growth has increased productivity and wages, this suggests that workers devote more time to leisure as they grow richer (Boppart and Krusell, 2020).¹

However, recent cross-sectional evidence suggests that *who* spends more time at work changes as countries develop. In the U.S. of the 1950s, for example, those with the lowest incomes worked the longest hours. Nowadays, this correlation has reversed: it is CEOs, top lawyers, and other high-wage workers who work the most. This observation is hard to explain in light of the long-run evidence: If higher wages induce to work less, why is it the case that higher-wage workers work more, and increasingly so?

This paper proposes a new mechanism to reconcile these facts, and explores its implications. The theory builds on the observation that the systematic increases in the hours-wage correlation have been accompanied by unprecedented technological progress. It has long been recognized that technological change has been behind changes in the wage structure over the past decades, with computers and ICTs as prominent examples. But new technologies have also changed what an hour of work can do: they made work less tied to a particular workplace or a fixed schedule, and more easily continued across workers, ultimately making it easier to adjust the work schedule with extra hours if required. For example, a consultant in New York can now use shared cloud files or instant communication to coordinate early in the morning with Europe and stay online late to resolve issues with Asia, turning what used to be separate business days into one continuous production window.² These changes are pervasive in modern labor markets, but are unlikely to affect all workers equally: a late-night online meeting between the top managers of a ride-sharing company via smartphone brings more value to the firm than an additional ride from one of the drivers.³

This project provides a framework to study how technological change affects the returns to working hours across workers and jobs – and the associated implications for inequality

¹The fact that the rich used to consume more leisure has attracted the attention of sociologists since at least the end of the 19th century: Veblen (1899) discusses how the wealthiest were spending large parts of their time in leisure activities, while low-class people worked long hours.

²Organizational research has studied the “autonomy paradox”, a phenomenon by which the introduction of mobile devices has offered flexibility on one hand, and at the same time, increased availability expectations, causing high-skilled professional workers to be effectively less flexible in their working time (Mazmanian *et al.*, 2013). Presser (2003) provides an early account of the shift in economic activity increasingly moved into evenings, nights, weekends, and across time zones as service demand expanded and communication costs fell.

³While most evident across workers at opposite ends of organizations, the same logic can apply even within skilled work. Shared patient records and clinical workflows can make some skilled tasks easier to transfer while extending the reach of senior decision-makers: residents and pharmacists can pass routine steps across shifts, while the same system lets an attending physician see the full history remotely and intervene when a complex case requires a decision.

and the macroeconomy. The paper makes three contributions.

The first contribution is theoretical. The theory is a matching model of the labor market in which workers sort into jobs based on skill as well as how their time is valued in that job.⁴ In the model, an income effect induces high-skilled workers to work fewer hours; at the same time, complementarities in production raise the returns to hours worked in more productive jobs. When high-skilled workers are allocated to those jobs, this implies that their returns to hours are higher, and the more so the stronger the complementarities between hours and job/skill attributes. When such complementarities are strong enough, high-skilled workers supply more hours despite the large income effects. The same trade-off between income effects and hours complementarities determines the equilibrium sorting condition: when better jobs reward hours more, the hours choice of high-skilled workers reinforces the direct type-complementarities, and can overcome income effects that push in the opposite direction. As part of the theoretical contribution, I characterize the equilibrium link between the skill-job allocation and hours' choice for a wide class of preferences and production functions. Furthermore, I show that the framework nests several models typically used in the literature on labor supply, and highlight that these models typically fail to account for a positive and rising hours-wage correlation.

Technological change affecting working hours (a phenomenon we label Hours-Biased Technological Change, HBTC) is what magnifies the link between hours and sorting. In the model, a change that increases the complementarity between hours and the skill-job bundle raises the value of an additional hour especially in better jobs. With positive assortative matching, stronger hours complementarities make high-skilled workers choose longer hours and increase their earnings by more, because they are allocated to the jobs where hours are most valuable. In this sense, the model's key insight is that the true returns to hours reflect not only a within-job return, but also include the effect that working more has on the allocation of workers to jobs. This channel is especially strong when new technologies make jobs increasingly differ in how they reward working time.

As a second contribution, the paper uses U.S. data over the post-1979 period to estimate the model. Estimated parameter changes reveal that well-known forces, such as increases in marginal returns to skill (i.e. Skill-Biased Technological Change, [Katz and Murphy \(1992\)](#), [Krusell, Ohanian, Ríos-Rull and Violante \(2000\)](#), [Autor and Acemoglu \(2011\)](#)), have been accompanied by a change in the relative value of hours worked across skills and jobs, which are interpreted as HBTC.⁵ Thus, the post-1980 technology changes were not only

⁴For a framework featuring endogenous hours with sorting, see [Calvo, Lindenlaub and Reynoso \(2024\)](#) and [Michelacci and Pijoan-Mas \(2016\)](#).

⁵In particular, estimates reveal that in 1980, hours and skills/jobs were roughly Cobb-Douglas in production. Starting from the 1980's, and as a consequence of HBTC, these complementarities have become stronger and estimates suggest that hours and skills/jobs are complements.

skill-biased, but also hours-biased, in the sense that the value of an additional hour increasingly depended on the worker-job match in which it was supplied.

The estimated model is used to decompose the rise in income inequality. SBTC and HBTC both increase inequality, but through different channels. SBTC is the main force behind the rise in wage dispersion. HBTC is a new source of inequality because it raises the covariance between wages and hours: high wages and long hours increasingly occur together. This channel is essential because the large income effects embedded in preferences push in the opposite direction. Without HBTC, SBTC would raise wages at the top but induce high-wage workers to reduce hours, causing the wage-hours covariance to fall. Conversely, without SBTC, the covariance effect of HBTC would be larger, but the rise in wage dispersion would be much smaller.⁶ The estimates also show that the quantitative force of HBTC comes through sorting: When the dispersion in job types and type-complementarities are held fixed at their 1979 levels, changing the return to hours has a much smaller effect. Once the estimated changes in type-complementarities and dispersion in job productivities are allowed to take place, high-skilled workers' additional hours are most productive, and the effect of HBTC on income inequality becomes much larger.

As a third contribution, I provide suggestive evidence on the main mechanism. Building on [Denning, Jacob, Lefgren and vom Lehn \(2022\)](#), I replicate in my CPS-ORG sample their distinction between within-occupation returns to actual hours and across-occupation prices of expected hours. The results mirror theirs: within occupations, the wage-hours elasticity remains close to zero, while the across occupation elasticity is large and rises over time. This pattern points to sorting as an increasingly important source of the return to hours.

To conclude, two further exercises illustrate broader implications of the mechanism. First, I extend the model with preference shocks, extending [Galichon, Kominers and Weber \(2019\)](#) and [Bandiera, Kotia, Lindenlaub, Moser and Prat \(2024\)](#), to study constraints on long hours when job assignment is stochastic. When HBTC makes hours especially productive in top matches, these constraints generate larger output losses and can move high-skill workers away from the jobs where their hours are most valuable. Second, I compare fixed-assignment labor-supply elasticities with full-assignment elasticities that allow hours to affect job allocation, in the spirit of [Scheuer and Werning \(2017\)](#). The main insight is that HBTC changes the earnings elasticity relevant for progressivity calculations: when hours affect access to better jobs, the response to incentives reflects both hours choices and movement along the assignment schedule, with the assignment margin becoming especially important in the middle and upper parts of the earnings distribution.

⁶These effects are quantitatively significant: In counterfactual exercises, HBTC is shown to account for about 34% of the model-implied increase in income inequality, thus confirming the importance of correctly modeling hours in production to fully understand the drivers of income inequality in U.S.

Literature A vast literature studies the relationship between hours and wages across countries and time periods. [Boppart and Krusell \(2020\)](#) introduce a new general preference class in which income effects dominate substitution effects. Relatedly, [Bick, Fuchs-Schündeln and Lagakos \(2018\)](#) and [Bick, Fuchs-Schündeln, Lagakos and Tsujiyama \(2022b\)](#) highlight the role of structural change in accounting for patterns of hours worked along the development spectrum. [Rachel \(2021\)](#) and [Kopytov, Roussanov and Taschereau-Dumouchel \(2023\)](#) propose theories to explain the decline in hours worked based on improvements in leisure technologies.⁷ Relative to all these papers, this study argues that technological change can systematically change the returns to hours across jobs as countries develop, and reconciles the aggregate and cross-sectional evidence by complementing preference-based theories with a theory in which technological complementarities play a central role.

This paper also closely relates to the literature that has attempted to explain diverging trends in labor market outcomes between high- and low-skilled. [Wolcott \(2021\)](#) proposes a model to explain the diverging trends in employment rates across skills. [Boppart and Ngai \(2021\)](#) reconcile diverging trends in leisure and increasing inequality with a mechanism based on intertemporal substitution. [Heathcote, Storesletten and Violante \(2010\)](#) analyze the macroeconomic and welfare implications of rising wage inequality and increasing hours-wage correlation. [Auray, Fuller and Guillaume \(2021\)](#) study multiple jobs and comparative advantage in a model that embeds a similar tension between cross-sectional and time series facts on hours worked, as in this paper. [Alon, Coskun and Doepke \(2019\)](#) propose a family channel to explain the reversal in the hours-wage correlation. [Ferraro and Valaitis \(2025\)](#) reconcile the flat cross-sectional employment rates with a model of quality choice. Relative to these papers, I contribute a new mechanism and show its quantitative importance in explaining rising inequality.

With respect to models of sorting ([Lindenlaub, 2014](#); [Chade and Lindenlaub, 2022](#); [Eeckhout and Kircher, 2018](#); [Vereshchagina, 2021](#)), this paper explicitly models the hours decision. In this respect, the following contributions are closest to this framework: [Calvo, Lindenlaub and Reynoso \(2024\)](#), [Michelacci and Pijoan-Mas \(2016\)](#) and [Shao, Sohail and Yurdagul \(2026\)](#). The first studies how the interplay between marriage and labor market decisions, shaped by the endogenous hours decision, affects inequality and hours in equilibrium. The second provides a competitive growth model where the pace of technological progress affects hours and the sorting of workers to jobs. The third studies the role of hours complementarities in production on the size distribution of firms, and their impact on earnings inequality. Relative to these papers, this paper characterizes sorting for a general class of production and utility functions and proposes a new mechanism to reconcile the data.⁸

⁷[Hochmuth \(2024\)](#) focuses on the movements in hours worked among entrepreneurs.

⁸Moreover, I differ in the quantitative application of the paper, in that I estimate how the production tech-

2 Motivating Evidence

In this section, I present two sets of facts (one aggregate and one cross-sectional) that motivate the theory.

Data The main data source I use for hours worked in U.S. is the CPS, which contains detailed information on hours worked and wages for the period of interest. I complement hours data from CPS with hours data from [Kendrick \(1961\)](#) and [Kendrick \(1973\)](#) to be able to trace total hours worked back in time⁹. In Appendix A, I show that the main messages delivered by the analysis below are very similar across other datasets of hours worked (e.g. ATUS).¹⁰ Data on hours worked across countries are from Penn World Tables.

Average Hours Worked The first motivating fact highlights the long-run behavior of hours per worker.¹¹ Figure 1 plots, for the U.S. and for selected advanced economies, the long-run behavior of hours worked per worker, together with an index of real average wages for the U.S. Over the long-run, average hours worked exhibit a clear downward trend across countries. For the U.S., the downward trend in hours is accompanied by a steady rise in average wages (left panel). The magnitudes are also significant, with some countries observing declines in average hours of roughly ten hours (see e.g. Denmark, Germany, U.S.).

These simple patterns, interpreted through the lens of neoclassical models of labor supply, point towards preferences where - along the balanced growth path - income effects are stronger than substitution effects, so that an increase in average wages due to productivity implies lower hours worked. This is the main intuition behind models that aim to characterize the forces behind long-run decline in hours worked, as in [Boppart and Krusell \(2020\)](#) and [Bick et al. \(2022b\)](#).¹² Recent evidence on hours trends along the development spectrum is also consistent with the mechanism: [Bick et al. \(2018\)](#) show that hours per worker decline with GDP per capita, and that these patterns cannot be explained by systematic differences

nology has changed over time, with a particular focus on how hours enter in production. In this sense, another related paper is [Erosa, Fuster, Kambourov and Rogerson \(2022\)](#), which develops a Roy-style model of occupational choice to study the implications of gender differences in home production possibilities for the gender wage gap.

⁹The latter datasets have been used extensively in the labor supply literature to capture long-run trends in hours worked (see e.g. [Francis and Ramey, 2009](#)) and [Cociuba et al., 2018](#))).

¹⁰To construct a real wage index, I use labor productivity divided by hours worked and complement it with series from [Kendrick \(1973\)](#). The same approach is used by [Kopytov et al. \(2023\)](#).

¹¹Since the focus of the paper is on the intensive margin, the relevant measure to be considered is hours per worker. Moreover, as noted in [Boppart and Krusell \(2020\)](#), long run movements in hours worked per worker (the intensive margin) are more important than participation rates (the extensive margin). Nevertheless, a similar picture emerges if we consider hours per capita over the very long term (see also [Francis and Ramey \(2009\)](#))).

¹²The idea that, over long run periods of productivity and wages growth, people would dedicate more time to leisure activities and less to work traces back at least to [Keynes \(1930\)](#).

in age, educational attainment or sectoral composition across countries.¹³

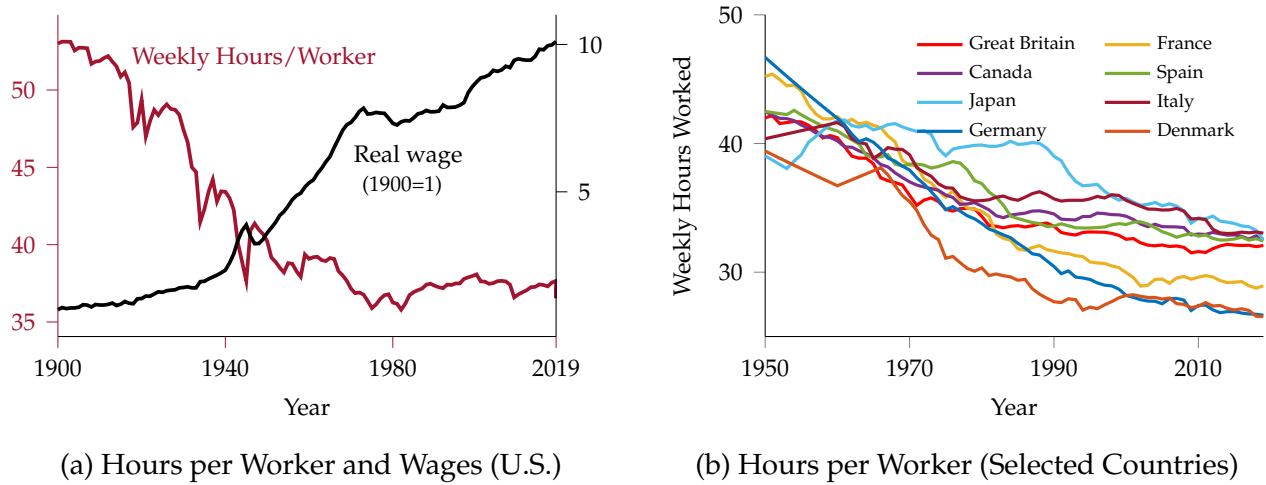


Figure 1: Working time trends in the United States and other advanced economies.

Notes: Panel (a) plots average weekly hours worked per worker in the United States together with a real wage index (normalized to 1 in 1900). For 1900-1958, total annual hours and total employment are from [Kendrick \(1961\)](#) and [Kendrick \(1973\)](#), taken from the [Francis and Ramey \(2009\)](#) replication files, with average weekly hours computed as total annual hours divided by total employment, divided by 52; for 1959-2019, total annual hours are from the CPS-based series in [Cociuba et al. \(2018\)](#), divided by the annual mean of monthly civilian employment from the BLS, divided by 52. The wage index is constructed following [Kopytov et al. \(2023\)](#), computed as real GDP (1900-1928) and real compensation of employees (1929-2019), divided by employment and by the hours-per-worker series. Panel (b) plots average weekly hours worked per employed worker in selected economies (1950-2019), constructed from average annual hours worked per person engaged in the Penn World Table ([Feenstra, Inklaar and Timmer, 2015](#)), divided by 52 (retrieved from FRED).

Hours and Wages in the Cross-Section The second set of facts concerns the cross-sectional relationship between hours worked and wages. Using CPS-ORG data, I divide the wage distribution into ten deciles and plot the mean wages and hours for three representative deciles. Figure 2 clearly shows diverging patterns for hours and wages across the distribution: while wages for high-wage workers rose, wages at the bottom stagnated, consistent with rising wage inequality ([Autor and Acemoglu, 2011](#)) driven by wages disproportionately increasing at the top. Hours worked also diverged across the distribution: high-wage workers started working longer hours, with an increase concentrated in the 1980's and the 1990's; viceversa, low-wage workers saw their hours worked steadily decrease.¹⁴

¹³Moreover, these trends are consistent with papers using time use data, which show that hours worked have declined (and leisure time increased) in U.S., as in [Aguiar and Hurst \(2007\)](#). Similar evidence is also provided in [Kopytov, Roussanov and Taschereau-Dumouchel \(2023\)](#), which show that the overall decrease of hours as wages grow is a trend that characterizes all OECD countries in the post-war period.

¹⁴These patterns are consistent with [Costa \(2000\)](#), who showed that the U.S. experience is characterized by a sign reversal in the hours-wage elasticity, from negative to positive: low-wage workers used to work the longest hours in 1960's and 1970's; starting from the 1980's, however, high-wage workers work significantly more hours than low-wage workers.

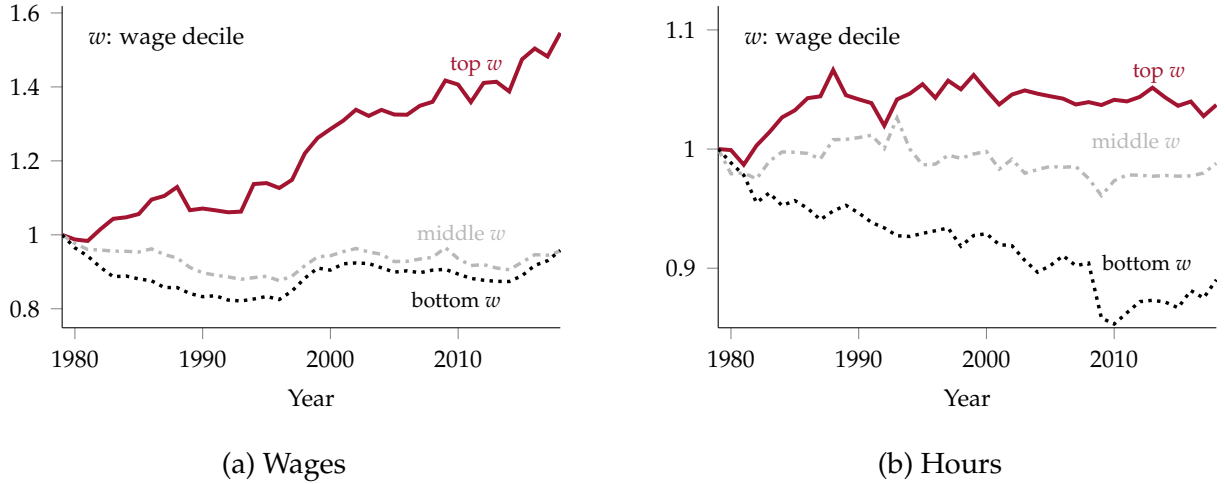


Figure 2: Wages and Hours by Wage Decile (US).

Notes: The left (right) panel displays mean wages (hours) by wage deciles. The red continuous line represents the top decile; the grey dashed line represents the 50th percentile; the dotted black line represents the bottom decile. Series are normalized by dividing by their value at the beginning of the sample. Raw series are plotted in Figure A.5. Source: CPS-ORG and own calculations.

To formalize the qualitative results displayed in Figure 2, I analyze the cross-sectional hours-wage elasticity by running the following regression year-by-year in the baseline sample:¹⁵

$$\log(h_i) = \alpha + \beta \log(w_i) + X_i + \epsilon, \quad (1)$$

where h_i is individual hours worked, w_i is individual wages, and X_i are demographic controls¹⁶. The estimated coefficient $\hat{\beta}_t$ summarizes the relationship between hours and wages in the cross-section. The estimated coefficient for the U.S. over time is depicted in Figure 3 (left panel). The right panel shows, using data from Bick *et al.* (2018), the coefficient obtained from the regression (1) performed across countries, and it is plotted against the level of per capita GDP, denoted by $\hat{\beta}_{cc}$.¹⁷

For the U.S., the coefficient is negative at the beginning of the sample, and - starting from the 1980's - turns significantly positive and increasing, with a mild decrease only in the most recent years in the sample (left panel).¹⁸ This trend (rising, from negative to positive) is

¹⁵This is the same specification used in Costa (2000) but updated to more recent years, and the same specification adopted in Bick *et al.* (2018).

¹⁶In the baseline specification, I consider age and age squared as controls to account for the systematic variation in hours worked across the life-cycle.

¹⁷For cross-sectional analysis, we restrict the sample to males-only, but as shown in Bick *et al.* (2018), these patterns hold across genders.

¹⁸Similar evidence is provided by Heathcote *et al.* (2010) and Boppart *et al.* (2024) using U.S. data. Moreover, data from time use sources (see in particular Aguiar and Hurst (2007)) are consistent with the evidence presented thus far: starting from the mid-1980's, leisure time has increased for the low-skilled workers more than for the high-skilled workers (while importantly, it increased uniformly across skill categories between 1965

robust to hours definitions and sample selection, as shown in Appendix A.2.

As first shown in Bick *et al.* (2018), the U.S. experience is replicated across countries along the development spectrum: as the right panel shows, the hours-wage elasticity is negative for low-income countries, while it gets progressively smaller and even positive for the richest countries. In other words, as technology evolves, it is high-wage workers that progressively work the longest hours, while low-wage workers work fewer hours (and increasingly so).

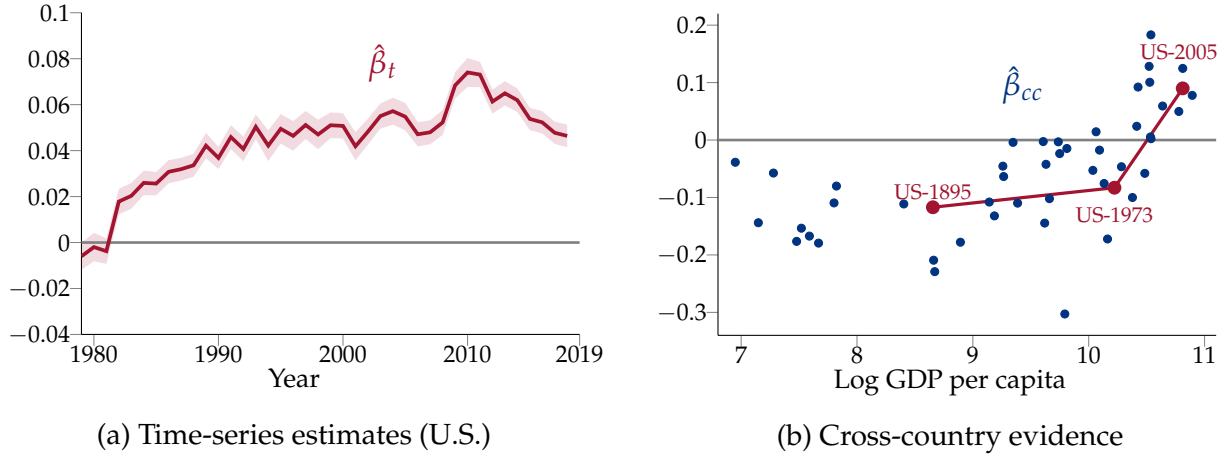


Figure 3: Elasticity of hours with respect to wages.

Notes: Panel (a) plots the estimated coefficient $\hat{\beta}_t$ from the regression of log hours on log wages over time run with US data (CPS-ORG), as in (1); the shaded area denotes the 95% confidence interval. Panel (b) plots the cross-country elasticity estimates from Bick *et al.* (2018) against log GDP per capita. The red line connects the historical U.S. estimates from Costa (2000) for 1895 and 1973 with the U.S. 2005 estimate from the Bick *et al.* (2018) dataset.

Summary Figure 2 and Figure 3 have shown that the negative relationship emerging from aggregate data (and suggestive of large income effects in workers' preferences) seems to be counteracted by other forces in the cross-section, and this is especially true for the U.S. in recent years and for countries at the technological frontier. This observation suggests a link between technological progress and the cross-sectional relationship between hours and wages. Importantly, the concurrent increase in wage inequality (especially in the U.S.) only makes these patterns harder to explain, since the forces that typically explain declining hours in the aggregate would predict large decreases in hours for high-wage workers.

Taken together, the evidence motivates a theory that builds on preference specifications as in Boppart and Krusell (2020) and at the same time attempts to reconcile aggregate and cross-sectional evidence on hours and wages over time. Theoretical and quantitative models

and 1985). Figure A.6 in Appendix A.2 shows this explicitly by replicating the results in Boppart and Ngai (2021).

explaining wage inequality have shown how technological change can affect the marginal product of workers' skills (Krusell *et al.*, 2000) and the allocation of skills to jobs (Autor and Acemoglu, 2011). In light of the evidence it is natural to build on these views and ask whether technological changes concurrent with the rise in the hours-wage elasticity can affect the marginal product of one more hour worked across workers and jobs. The next section develops a theory to investigate this relationship.

3 Theory

This section develops a new assignment framework, in which workers sort into jobs based on their skills and hours.¹⁹ After characterizing the theory, I show how it nests several models of the labor supply as special cases. Finally, I do comparative statics to illustrate the mechanism.

3.1 Framework

Environment I consider a static competitive labor market, composed of heterogeneous workers $x \in \mathcal{X} = [0, \bar{x}]$ and jobs $y \in \mathcal{Y} = [0, \bar{y}]$; we denote by $Q(x)$ and $G(y)$ the distribution of workers and jobs, respectively, and $q(x)$ and $g(y)$ as their densities. Agents on both sides of the market have mass equal to one. The attribute of individuals x can be thought of as time-invariant skill or ability; the productivity of the job y can be thought of as a skill requirement of a particular job. Workers and jobs match in a one-to-one fashion to produce output, denoted by $f(x, y, h)$. Output f depends, therefore, on both the quality of jobs y and workers x assigned to them, and on the hours supplied by workers, h .

Individuals are endowed with one unit of time to be allocated between market work and leisure (there is no home production). Individuals get utility from consumption and leisure, and we denote utility by $u(c, h)$, which is allowed to be non-separable in its inputs.

Jobs' problem Jobs maximize profits by choosing the type x and hours h ; profits are defined as output net of income w to be paid to the worker²⁰:

$$\max_{x, h} f(x, y, h) - w \tag{2}$$

¹⁹For the purposes of the theoretical section, one can view y as firms or jobs, interchangeably, as the theoretical model allows for both interpretations.

²⁰To save on notation, I use w as income but it should be understood that income is a function of the matched pair x, y and the hours choice, i.e. $w = w(x, y, h)$. In equilibrium, with endogenous assignment $\mu(x)$ and hours $h(x)$, income will be a function of workers' skills $w(x)$.

The choice of worker type x by the job will determine, in equilibrium, the assignment function $\mu(x)$ that maps workers to jobs as well as income and profit functions.

Workers' problem Workers choose time allocation taking income w as given:

$$\max_h u(c, h) \quad s.t. \quad c = w \quad (3)$$

Together with the jobs' problem, this determines optimal choice of hours h as a function of skill, $h^*(x)$. Hours choice then becomes the key link between the worker's problem and the jobs' problem. If output f only depends on skills x and job attributes y (that is, $f = f(x, y)$), this is a standard assignment problem between workers and jobs in the tradition of [Becker \(1973\)](#). Precisely because hours enter in production, the labor supply choice by the worker affects the equilibrium sorting pattern and with it, the equilibrium levels of inequality and output. Therefore, hours choice and equilibrium assignment are intertwined and jointly determined by the jobs' and workers' problems.

Market clearing The model is closed by specifying a market clearing condition, essentially requiring that the workers and jobs match in a measure-preserving way. Under Positive Assortative Matching (PAM), better workers are assigned to better jobs and the market clearing condition can be written as²¹:

$$\int_{\mu(x)}^{\bar{y}} g(s) ds = \int_x^{\bar{x}} q(s) ds \quad (4)$$

where $\mu(x)$ is the equilibrium matching function, assigning workers to jobs.

Equilibrium We are now ready to define a competitive equilibrium of this economy.

Definition 1. *A competitive equilibrium of this economy is a tuple of functions $(w(x), \mu(x), h(x))$ such that:*

- $w(x)$ and $h(x)$ solve problems (2) and (3) (optimality)
- equation (4) holds (market clearing)

3.2 Conditions for Assortative Matching

Towards a complete characterization of the equilibrium of this economy, we want to seek conditions under which assortative matching arises. This is most directly achieved by noticing that this is effectively a matching problem with imperfectly transferable utility (ITU)

²¹Here and throughout the text, I refer to PAM as Positive Assortative Matching, indicating that better workers are matched in equilibrium with better jobs (analogously for Negative Assortative Matching, NAM).

between matched agents. This class of problems has been analyzed in the literature (see in particular, [Legros and Newman \(2007\)](#) and [Chade *et al.* \(2017\)](#)).

To apply the results from the literature on matching with ITU, it is useful to rewrite the problem by substituting the wage in the worker problem (3), assuming that firm y gets at least a value $V \in [0, V^{max}]$ when matched with worker x . The pair's problem then becomes:

$$\Phi(x, y, V) = \max_h u(f(x, y, h) - V, h) = u(f(x, y, h^*) - V, h^*) \quad (5)$$

where $\Phi(x, y, \cdot)$ traces out the Pareto utility frontier of the match, for every possible value of V . Of course, in equilibrium, V will be a function of the firm's attribute ($V = V(y)$) and is determined endogenously with the equilibrium assignment.

The usefulness of writing the problem as in (5) is that in order to know whether PAM arises in equilibrium, it is enough to check that the Generalized Indifference Condition (GID) provided in [Legros and Newman \(2007\)](#) is satisfied. Because in this setting agents' types are continuous, we will use the differential version provided in [Chade *et al.* \(2017\)](#). In particular, PAM arises in equilibrium if (and only if):

$$\Phi_{xy} - \Phi_{xV} \frac{\Phi_y}{\Phi_V} > 0 \quad (6)$$

Equation (6) makes clear that two types of complementarities drive equilibrium sorting.²² There is a complementarity between types, Φ_{xy} : if better workers do disproportionately better with better jobs ($\Phi_{xy} > 0$), this is a force pushing towards PAM. In the case of transferable utility (TU), $\Phi_{xy} = f_{xy}$ and $\Phi_{xV} = 0$, therefore (6) boils down to $f_{xy} > 0$ (as in [Becker \(1973\)](#)). In this ITU setting, however, what matters for sorting is also how the surplus of the match varies across worker-firm pairs, which in turn depends on the complementarity between worker type and partner's (jobs') utility, captured by Φ_{xV} . Intuitively, this term measures how easy it is to transfer utility between higher x workers and higher y jobs. The easier it is to transfer utility (the larger Φ_{xV}), the more likely it is that (6) is satisfied (recall that $\Phi_V < 0$). In this setting, this force will depend on both characteristics of preferences (income effects) and on technological complementarities between hours, jobs and worker types. The next step is to highlight these characteristics by writing (6) in terms of primitives of the model (preferences and technology).²³ This leads us to the first proposition:

²²Equation (6) can be derived from the FOC of (5) with respect to job type y , $\Phi_y + \Phi_V V_y = 0$, and then differentiating this along the equilibrium matching $\mu(x)$ combined with the requirement that the SOC must be negative for this to be an optimum.

²³Alternatively, the problem can be characterized as a joint, simultaneous choice of jobs and hours by the workers, that gives rise to a multidimensional second order condition; then, PAM/NAM arises depending on conditions derived from the Hessian of the problem, as in [Eeckhout and Kircher \(2018\)](#). I show this in Appendix B.1.

Proposition 1. *A necessary and sufficient condition to have Positive Assortative Matching (PAM) for any distribution of types is*

$$f_{xy} + f_{hy}h_x^* > 0 \quad \forall (x, y, V) \quad (7)$$

where $h_x^* = \frac{\partial h^*}{\partial x}$ and $h^* = h^*(x, y, V)$ is the equilibrium hours function.²⁴

Proof: see Appendix B.1. ■

Condition (7) has an intuitive interpretation. The term f_{xy} is standard in the literature and captures complementarities between types x and y in production. If f_{xy} is positive and large, high types x produce relatively more with high types y , pushing towards a PAM allocation. The second term, $f_{hy}h_x^*$, is new and captures the effect of the hours decision on the sorting patterns. When $f_{hy} > 0$, and $h_x^* > 0$ or $h_x^* < 0$ (but not too negative), the equilibrium allocation is pushed towards PAM. The condition says that if high type jobs produce relatively more with longer hours ($f_{yh} > 0$), and high-skilled workers work more than their low-skilled counterparts ($h_x^* > 0$), PAM is more likely to arise in equilibrium (for a given f_{xy}). In this case, hours are a way for high-skilled workers to secure top jobs. If, on the other hand, low- x workers work sufficiently more than high- x workers (h_x^* is sufficiently negative), they can compensate for their lower ability and be matched with the high types y , as long as their hours input is particularly valuable in top jobs ($f_{yh} > 0$). The opposite intuition is at work when $f_{yh} < 0$. Importantly, if $f_{yh} = 0$, hours are equally valuable across jobs, so working more or less hours has no impact on sorting.

The key takeaway from Proposition 1 is that the endogenous labor supply decision can create forces towards or against PAM, beyond standard type-complementarities embedded in f_{xy} ; these forces crucially depend on whether longer hours are more productive in high type jobs (which determines f_{yh}). Condition (7) is expressed in terms of an endogenous object, h_x^* . The advantage of this condition is that it's simple and intuitive, but leaves the question of how to express positive sorting in this model purely in terms of primitives. For this purpose, we derive the following proposition:

Proposition 2. *A necessary and sufficient condition on primitives to have Positive Assortative Matching (PAM) for any distribution of types is*

$$f_{xy} [-(u_{cc}f_hf_h + 2u_{ch}f_h + u_cf_{hh} + u_{hh})] + f_{hy} [(u_{cc}f_xf_h + u_cf_{xh} + u_{hc}f_x)] > 0 \quad \forall (x, y, h, c) \quad (8)$$

²⁴Throughout the text, we denote partial derivatives with lowercase letters, e.g. $f_x = \frac{\partial f}{\partial x}$, and similarly for cross-partial derivatives.

The opposite inequality provides a condition for Negative Assortative Matching (NAM). If utility is separable ($u_{ch} = 0$), then (8) simplifies to

$$f_{xy} [-(u_{cc}f_hf_h + u_cf_{hh} + u_{hh})] + f_{hy} [(u_{cc}f_xf_h + u_cf_{xh})] > 0 \quad \forall(x, y, h, c) \quad (9)$$

Proof: see Appendix B.1. ■

Inspecting closely this condition reveals further insights into the model. The first term on the left hand side of (8) is composed of the type-complementarity term f_{yx} , and the sum of terms $-(u_{cc}f_hf_h + 2u_{ch}f_h + u_cf_{hh} + u_{hh})$ which from optimality we know is positive.²⁵ Therefore, the sign of the second term is crucial to determine the allocation: If the second term $f_{hy}(u_{cc}f_xf_h + u_cf_{hx} + u_{hc}f_x)$ is negative and large enough, the allocation will be NAM (even if $f_{xy} > 0$). For example, if $f_{hy} > 0$, the key term becomes $(u_{cc}f_xf_h + u_cf_{hx})$, which captures the key trade-off between income and substitution effects that drive the hours choice and with it, the equilibrium sorting. The first term $(u_{cc}f_xf_h)$ is a by-product of the skill and hours premium f_xf_h combined with income effects coming from preferences, u_{cc} , and is therefore negative. The larger the curvature in utility (the more negative u_{cc}), the more negative this term and the less likely that high-skilled individuals will work longer hours. The term f_{xh} captures the incentive to work longer hours for high-skilled workers and is akin to a type-dependent substitution effect. The last term, $u_{hc}f_x$, captures the complementarity between hours worked and consumption. It follows that the sign of $(u_{cc}f_xf_h + u_cf_{hx} + u_{hc}f_x)$ is determined by the relative strength of income effects embedded in preferences on one hand, and hours complementarities (captured by $f_{xh} > 0$) on the other.²⁶

Hours Choice As the previous paragraph made clear, the key endogenous outcome that shapes equilibrium sorting is the hours choice by workers' skill type. The following proposition characterizes this choice explicitly.

Proposition 3. *In equilibrium, high-skilled workers choose higher hours ($h_x^* > 0$) if*

$$-f_hf_x\left(-\frac{u_{cc}}{u_c} + \frac{u_{ch}}{u_h}\right) + f_{xh} > 0 \quad \forall(x, y, h, c) \quad (10)$$

Proof: see Appendix B.1. ■

Condition (10) highlights once more that at the heart of the model lies a key trade-off be-

²⁵This is because $-(u_{cc}f_hf_h + 2u_{ch}f_h + u_cf_{hh} + u_{hh})$ corresponds to the second order optimality condition of the hours problem.

²⁶Figure B.1 in Appendix B.2 plots the equilibrium allocation of the model for several combinations of technology and preference parameters, to help visualize how the allocation changes when the different forces in (8) change.

tween income effects in preferences, acting through $u_{cc} < 0$, and technology, acting through f_{xh} , which push h_x in opposite directions. As we have seen in Proposition 2, the same forces directly affect equilibrium sorting - and can reinforce or push against PAM. These opposing forces are similar to those arising in a static labor supply model.²⁷ The following corollary shows when the Proposition 3 holds in a wide class of commonly used preference specifications.

Corollary 1. *Assume that utility is separable in consumption and hours ($u_{hc} = 0$), then condition (10) simplifies to*

$$(u_{cc}f_hf_x + u_cf_{hx}) > 0 \quad \forall (x, y, h, c) \quad (11)$$

or, equivalently

$$(f(x, y, h) - V) \left(-\frac{u_{cc}}{u_c} \right) < \frac{f_{xh}(f(x, y, h) - V)}{f_xf_h} \quad \forall (x, y, h, c) \quad (12)$$

It follows that if $f - V$ is log-supermodular in (x, h) , then condition (12) is satisfied if the relative risk aversion coefficient $-c \frac{u_{cc}}{u_c} < 1$.

Proof: see Appendix B.1. ■

From the previous corollary, it is clear that $-c \frac{u_{cc}}{u_c} < 1$ is a sufficient condition for hours worked to be increasing in workers' attributes, x . A direct implication is that when $-c \frac{u_{cc}}{u_c} > 1$, sufficiently strong complementarities in production between skills and hours are required for the condition to be satisfied. This is of particular interest when considering preferences of the CRRA form, where $-c \frac{u_{cc}}{u_c}$ is simply given by a parameter σ . It follows that in this widely used preference class, assuming $\sigma > 1$ ($\sigma < 1$) is enough to determine whether income effects are stronger (weaker) than substitution effects, and hence whether a higher wage pushes towards working more (less). In other words, hours-skill complementarities in production can reinforce incentives to work long hours for high type workers even with large income effects.²⁸

²⁷The key distinguishing feature here is that substitution effects can be heterogeneous across skills, i.e. the marginal product of one more hour worked is higher across skills (this happens when $f_{xh} > 0$).

²⁸When utility is not separable in consumption and labor ($u_{ch} \neq 0$), condition (10) is more likely to hold; the reason is that complementarities between hours and consumption can drive up the incentives to work for high wage workers.

3.3 Equilibrium Properties and Comparative Statics

We now illustrate the model's properties assuming a specific functional form for production and utility functions. We will assume a nested CES for technology

$$f(x, y, h) = A \left(\beta (\alpha x^\rho + (1 - \alpha) y^\rho)^{\frac{\gamma}{\rho}} + (1 - \beta) h^\gamma \right)^{\frac{1}{\gamma}} \quad (13)$$

and a [MaCurdy \(1981\)](#) utility function

$$u(c, h) = \frac{c^{1-\sigma}}{1-\sigma} - \psi \frac{h^{1+\frac{1}{\theta}}}{1+\frac{1}{\theta}}. \quad (14)$$

The advantage of this production technology is that it allows for flexible complementarity patterns, and allows for changes in the skill premium f_x and changes in skill-job complementarities f_{xy} through changes in α and ρ , as well as changes in hours-complementarities f_{xh}, f_{yh} through changes in γ .²⁹

3.3.1 Technological Change

We want to characterize how different types of technological change allowed in the model affect equilibrium incomes $w(x)$ and hours $h(x)$.³⁰ Figure 4 plots these outcomes for several values of γ . When γ gets more negative, hours turn from decreasing to increasing in skill type x (left panel), despite the large income effects operating in the background. In fact, income $w(x)$ becomes steeper in skills: better types are not only matched to better jobs, but they also work more, therefore increasing their marginal valuation in the market. In sum, a decrease in γ leads to higher types working relatively more hours, and because of that, getting rewarded more for their skills.

How do these effects compare with changes in the return to skill? It is useful to start from an increase in the weight of skill in production (α), but we show results for both ρ and α (in Figure 5 for α and Figure B.2 for ρ). In both cases, hours become decreasing in skills for large values of α and for more negative values of ρ , and this effect is driven by high-skilled working less.³¹ The reason is precisely the income effects operating in the background.³² In sum, both changes in returns to skill (through α and ρ) and hours-complementarities

²⁹Changes in hours-complementarities will be interpreted as Hours-Biased Technical Change in the quantitative application.

³⁰For these comparative statics exercises, and consistently with the quantitative application, we will consider points in the parameter space where PAM is satisfied for any type distribution, so the matching function $\mu(x)$ is unchanged along these specifications.

³¹In fact, the effect for low-skilled is much more muted.

³²The difference between ρ and α is that they change the equilibrium income function $w(x)$ in different ways: ρ mostly increases convexity, while α makes the income function steeper in x throughout the domain.

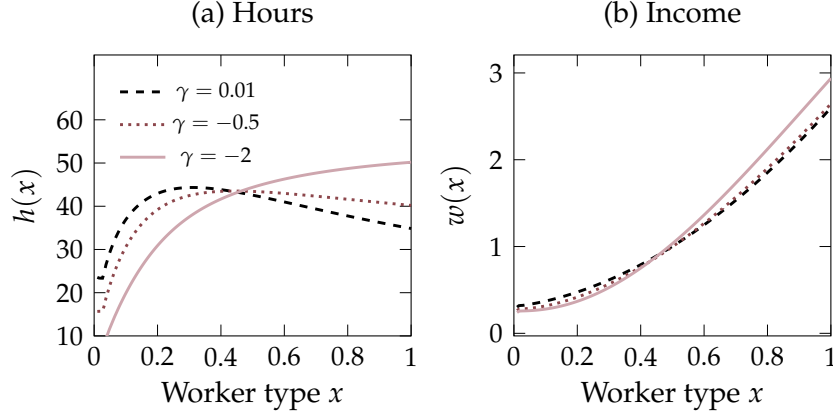


Figure 4: Equilibrium policy functions for alternative values of γ .

Notes: Income $w(x)$ is in thousands. Hours are weekly. The remaining parameter values are those corresponding to year 2010 in the estimated model (see Figure 9).

(through γ) increase income disparities across workers, but they do so with opposite effects on their labor supply choices.

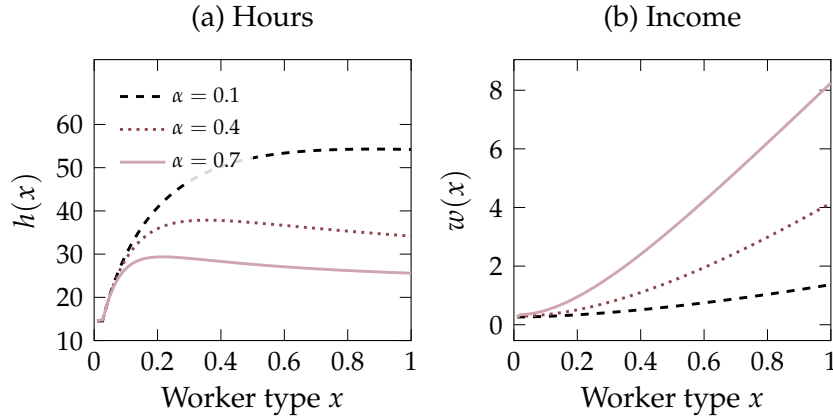


Figure 5: Equilibrium policy functions for alternative values of α .

Notes: Income $w(x)$ is in thousands. Hours are weekly. The remaining parameter values are those corresponding to year 2010 in the estimated model (see Figure 9).

Decomposition As a last step of the analysis, we unpack these comparative statics responses and illustrate the mechanism behind them. To do so, note that the terms in Condition 9: an income effect term $f_{yh}u_{cc}f_x f_h$, a substitution effect term $f_{yh}u_c f_{xh}$, and a sorting term $-f_{xy}(u_{cc}f_h^2 + u_c f_{hh} + u_{hh})$. We plot each of these terms (normalized by a baseline value of γ) separately to understand how changes in technology drive the equilibrium responses shown in Figure 4 and Figure 5. As Figure 6 makes clear, a lower γ makes the income effect term more negative at the top of the skill distribution (left panel), while it increases the substitution effect term for high-skilled (especially relative to low-skilled). The third term

shows that the sorting term is also pushed up by a lower γ , even though its effect is smaller in magnitude than the substitution effect.³³ In sum, the mechanism through which a lower γ leads high-skilled workers to work longer hours is through a combination of higher returns to hours in better jobs (the substitution effect term), and stronger gains from matching with those jobs (the sorting term); these terms combined outweigh the negative effect coming from preferences, pushing in the opposite direction.

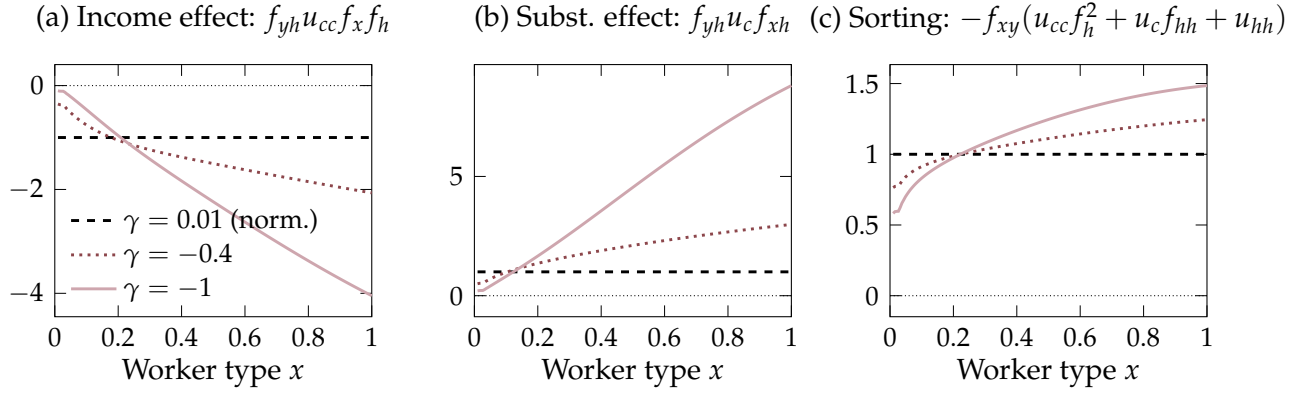


Figure 6: Decomposition of (9) for alternative values of γ .

Notes: Each panel plots, along the equilibrium matching path $y = \mu(x)$, one of the three terms that govern the sign of positive assortative matching from (9): the income effect $f_{yh} u_{cc} f_x f_h$ (Panel a), the substitution effect $f_{yh} u_c f_x h$ (Panel b), and the sorting term $-f_{xy}(u_{cc} f_h^2 + u_c f_h h + u_{hh})$ (Panel c). Each curve is normalized pointwise by the absolute value of the corresponding term evaluated at the baseline value of $\gamma = 0.01$, so the baseline (dashed) is flat at ± 1 and other curves show the force relative to baseline. The remaining parameter values are those corresponding to year 2010 in the estimated model (see Figure 9).

3.3.2 The Role of Preferences

We conclude our comparative statics exercises section by showing how preferences can interact with technological changes to drive the equilibrium relationship between hours and skills in the cross-section, summarized here by the slope of (log) hours over (log) skills. Figure 7 plots this slope over a range of values of σ , for two values of the technology parameter γ . For either value of γ , the slope is decreasing, initially positive when $\sigma = 0$ and then turning negative for large enough values of σ . Interestingly, the value of σ required to make the slope negative is much higher when γ is lower. In other words, when σ is large, whether the slope is negative or positive depends on whether hours are sufficiently complementary to skills and jobs in production. When $\gamma = 0.01$, so that hours and the skill-job bundle are essentially Cobb-Douglas, a $\sigma > 1$ generates a negative slope; in this case, the model behaves similarly to a textbook model where $\sigma > 1$ implies income effects larger than sub-

³³To save on space, the decomposition Figures for α and ρ are presented in Appendix B.2.

stitution effects; when γ is such that hours and skills-jobs are Hicks complements (in the Figure, $\gamma = -1$), the slope shifts upward for any given level of σ .³⁴

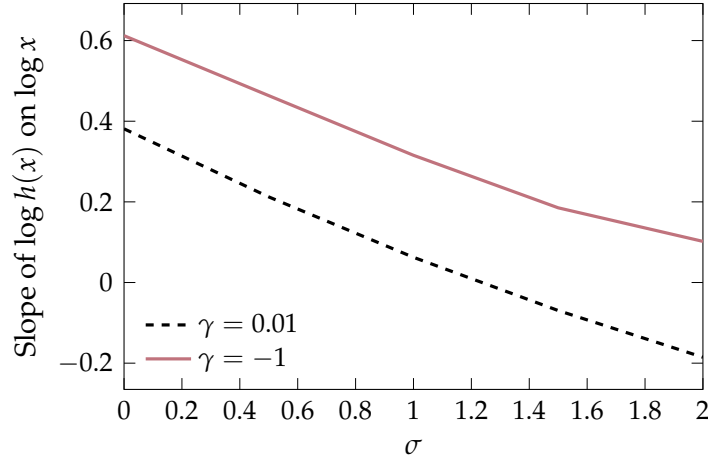


Figure 7: Slope of $\log h(x)$ on $\log x$ as a function of σ .

Notes: The figure plots, as a function of the consumption curvature parameter σ , the slope coefficient from a population-weighted OLS regression of $\log h(x)$ on $\log x$ across worker types, with one line per value of γ . The remaining parameter values are those corresponding to year 2010 in the estimated model (see Figure 9).

3.4 Special Cases and Relationship to the Literature

The purpose of this section is to discuss how the framework relates to existing models, deriving the latter as special cases whenever possible.

Neoclassical Macro Frameworks The first link is the one with neoclassical growth frameworks with endogenous labor supply. Recently, [Boppart and Krusell \(2020\)](#) propose a general class of preferences where income effects exceed substitution effects. In their framework, the central object to be characterized is the Marginal Rate of Substitution (MRS) between consumption and hours, $MRS = -\frac{u_h}{u_c} = w$.³⁵ In this paper, the MRS condition is pair-specific and takes the form

$$MRS(c(x), h(x)) = -\frac{u_h(x)}{u_c(x)} = f_h(x, \mu(x), h(x)) \quad (15)$$

Equation (15) already highlights that in this framework, hours are chosen to balance a

³⁴This result is reminiscent of [Berger, Herkenhoff, Jeong and Mongey \(2025\)](#), which show that the equilibrium sorting in a simplified version of their model depends on the relative values of σ and the worker-job complementarity terms.

³⁵In [Boppart and Krusell \(2020\)](#), the MRS takes the form $MRS = -\frac{u_h}{u_c} = c^{\frac{1}{1-\nu}} q(hc^{\frac{\nu}{1-\nu}})$ for some function $q(\cdot)$. The implication is that on a balanced growth path, if consumption c grows at a rate $g^{1-\nu}$ and hours at a rate $g^{-\nu}$, the argument of $q(\cdot)$ stays constant and the MRS grows one for one with the wage, implying income effect is stronger than the substitution effect.

preference term MRS and a technology term f_h . The comparison with [Boppart and Krusell \(2020\)](#) can be made directly from (15). In the homogeneous-job benchmark, the marginal product of hours is common across workers; By contrast, in this assignment environment the return to an extra hour is match specific. As shown in Appendix B.3.1, under standard regularity conditions and whenever better jobs raise the marginal product of hours ($f_{hy} > 0$), they induce higher hours.³⁶ Sorting therefore amplifies the substitution margin higher up the job ladder and provides a countervailing force to income effects.

Efficiency Units Efficiency units of labor are often used to model differences in labor inputs in macro-labor models. The common feature of these models is that the job cares about the labor h (hours) input scaled by the efficiency (or quality) of the input x , i.e. the jobs care about xh . In the current setting, production would take the form $f(x, y, h) = \tilde{f}(xh, y) = \tilde{f}(z, y)$ where z is efficiency units of labor. It can be shown that in this setting, for PAM it is sufficient to have $\tilde{f}_{zy} > 0$. The intuition behind this is that in equilibrium, high-skilled workers always choose enough hours so that efficiency units are higher than low-skilled workers (even if the latter work more). It follows that if those units are more valued in better jobs ($\tilde{f}_{zy} > 0$), high-skilled workers will always be allocated to those jobs. In other words, in this special case the hours' role in driving sorting is very limited due to the exact technological substitutability of hours and skills.

In the context of technological change and inequality, a prominent framework featuring efficiency units is [Krusell *et al.* \(2000\)](#). In that framework, the labor input of skilled workers s in production is given by $s = \psi H$, where H is the total hours input and ψ captures the efficiency of the worker (e.g. human capital), and similarly for low-skilled workers l . In Appendix B.3.2, I show that even extending this framework to include hours in production in a possibly non-linear way (so that $s = \psi g(h)$) and a labor supply decision still implies that who works more hours depends only on the preference parameter σ , unless $g(h)$ is exogenously imposed to be type-specific ($g_s(h) \neq g_l(h)$).³⁷

It is also worth discussing the way the current framework differs from [Krusell *et al.* \(2000\)](#) and other works that model technological change as an exogenous shift in the price of equipment capital. In a production function as in (13), such changes can be represented by shifts in A, α, β . The additional generality of this framework is that it also allows for technical changes represented by changes in ρ, γ , which govern the returns to hours relative to skill across jobs; these changes then determine which jobs have the steepest returns to hours and,

³⁶If, in addition, f_h is log-supermodular in (y, h) , then the elasticity $\partial_y \log f_h$ is increasing in hours, so that a marginal improvement in job quality has a larger proportional effect on the return to an extra hour in the higher-hours matches associated with better jobs. Under the production structure in (13), this is obtained when $\gamma < 0$.

³⁷In Appendix B.3.2, I provide a formal argument and characterization of how this framework relates to [Krusell *et al.* \(2000\)](#) and formally show the statements in this section.

because of sorting, which workers are assigned to such jobs in equilibrium.

Hours and Skills as a Bundle This framework nests the assignment structure of a few models in the literature where skills and hours are ‘bundled’ together; in these models, only the bundle $g(x, h)$ enters the production function, so that production becomes $f = \tilde{f}(g(x, h), y)$. This class of models have been used frequently in the optimal taxation literature (see [Scheuer and Werning \(2017\)](#) or [Ales and Sleet \(2016\)](#)) but also recently in the literature on marriage and labor market sorting ([Calvo et al., 2024](#)). With separable utility, it is easy to show³⁸ that Condition (9) reduces to $\tilde{f}_{gy}[-g_x u_{hh} + u_c \tilde{f}_g(g_h g_{xh} - g_x g_{hh})] > 0$; hence, if $\tilde{f}_{gy} > 0$, so that better jobs value effective labor more, a sufficient condition for positive assortative matching is $g_{hh} < \frac{g_h}{g_x} g_{xh}$.³⁹

It is also instructive to specialize the condition for hours to be increasing in skills, Proposition (3), to this case. Under separable utility, hours are increasing in skills if and only if $\frac{g_{xh}}{g_x g_h} > \tilde{f}_g \left(-\frac{u_{cc}}{u_c} \right) - \frac{\tilde{f}_{gg}}{\tilde{f}_g}$.⁴⁰ Hence, in these models, an upward-sloping hours schedule requires sufficiently strong complementarity between skills and hours inside the bundle.⁴¹ In sum, because jobs value only the scalar $g(x, h)$, the bundle restriction therefore mechanically ties job-skill and job-hours complementarities. The technology used in the quantitative section, instead, can separately govern worker-job sorting and the return to long hours in a match, allowing the model to separate SBTC, which encompasses worker-job complementarity, from HBTC, which changes hours-match complementarity.⁴²

Multiplicatively Separable Hours Frameworks such as [Erosa, Fuster, Kambourov and Rogerson \(2022\)](#) feature heterogeneity in the effects of working hours by occupation and comparative advantage of workers across occupations, extending [Roy \(1951\)](#): earnings e in occupation j take the form $e = a_j \cdot g_j(h)$, where a_j is the occupation-specific productivity of a worker and g_j the occupation-specific function that governs the elasticity of earnings to hours in that occupation. In the present framework, this corresponds to a multiplicatively separable technology of the form $f(x, y, h) = \tilde{f}(x, y)z(y, h)$, where $\tilde{f}(x, y)$ captures comparative advantage across worker-job pairs and $z(y, h)$ the occupation-specific returns to hours.

³⁸See Appendix B.3.3.

³⁹A similar result arises in a model with sorting and taxation but with risk-neutral workers, as proven by [Vereshchagina \(2021\)](#).

⁴⁰If, in addition, $\tilde{f}$ is concave in the bundle, $\tilde{f}_{gg} \leq 0$, then $g_{xh} > 0$ is necessary for hours to increase with skills.

⁴¹More generally, the bundle restriction implies that both sorting and hours depend on how $g(x, h)$ combines skills and hours, while ruling out any direct job-specific trade-off between them, since $\frac{f_h}{f_x} = \frac{g_h}{g_x}$ is independent of y .

⁴²This technology also allows changes over time in hours premia holding sorting forces fixed, and changes in sorting forces holding the hours technology fixed, which will be an important property in the counterfactuals.

Under this production structure and separable CRRA preferences,⁴³

$$\frac{f_{xh}(f - V)}{f_x f_h} = \frac{f - V}{f} \leq 1,$$

so that by Proposition (3) and Corollary (1), $h_x > 0$ is impossible for any $\sigma \geq 1$.⁴⁴ This result affects the properties of the allocation. In the multiplicatively separable case, PAM is obtained when

$$\bar{f}_{xy}z(y, h) + \bar{f}_x z_y(y, h) + (\bar{f}_y z_h(y, h) + \bar{f} z_{hy}(y, h))h_x > 0.$$

Therefore, whenever better jobs also have steeper returns to hours, so that $f_{hy} = \bar{f}_y z_h + \bar{f} z_{hy} > 0$, the hours component pushes away from PAM for the empirically relevant case $\sigma > 1$, since in that case $h_x < 0$. A PAM allocation then requires a sufficiently strong direct sorting force, captured by $\bar{f}_{xy}z + \bar{f}_x z_y$, relative to the negative hours component.⁴⁵ In that sense, the multiplicative separability can generate PAM only through the direct sorting force, effectively shutting down the mechanism at the core of the general model, namely that sufficiently strong hours complementarities can overturn the income effect and make high-skill workers work more.

Supervision Time Models Supervision time models such as [Garicano and Rossi-Hansberg \(2006\)](#)⁴⁶ have been used extensively to study inequality and trade ([Caliendo, Mion, Opro-molla and Rossi-Hansberg, 2020](#)). At the core of their setting is a production structure where workers of skill x solve a fraction $F(x)$ of problems at work, and ask help to the manager (which in this case, can be interpreted as the managerial side of the match y) for the remaining fraction $1 - F(x)$. Each interaction involves a fixed amount τ of supervisory time. Consider a slight departure from their framework (fully outlined in Appendix B.3.5) and assume that worker and supervisory structure are active for an h amount of hours; this gives a supervision time constraint that takes the form $h \geq \tau n h [1 - F(x)]$, where n is the amount of workers to be supervised. This setting essentially implies a production structure where $f(x, y, h) = \hat{f}(A(x, y), h) = \frac{1}{\tau} \frac{yh}{1 - F(x)}$, which already produces positive type- and hours-complementarities f_{xy}, f_{xh}, f_{yh} , and can therefore be considered a special case of the

⁴³See Appendix B.3.4 for details.

⁴⁴More precisely, $\sigma > 1$ implies $h_x < 0$, while $\sigma = 1$ implies $h_x \leq 0$, with equality only if $V = 0$.

⁴⁵This mirrors the mechanism analyzed by [Vereshchagina \(2021\)](#) in the context of progressive taxation, where sufficiently strong direct complementarity preserves PAM under progressive taxation. More precisely, Proposition 4 in her paper requires $z(y, h)z_{hh}(y, h) \leq z_h(y, h)^2$ and $f_{xy} \geq (f_x/f_h)f_{hy} \geq 0$. When $f_y > 0$, the latter is equivalent to requiring that the Hicks elasticity of complementarity between x and y , $ff_{xy}/(f_x f_y)$, weakly exceeds that between h and y , $ff_{hy}/(f_h f_y)$.

⁴⁶For a summary of these models, see [Garicano and Rossi-Hansberg \(2015\)](#).

general environment of this paper.⁴⁷ The logic leverages on the core insight of these models, and implies that hours are more valuable in better matches: a longer workday is especially productive when the worker rarely needs help and the managerial side of the job is high value. In Appendix B.3.5, I show that a richer extension to this basic knowledge-based production structure setting can provide a useful microfoundation for how technological change might have increased the hours complementarities f_{yh}, f_{xh} .

4 Estimation

The goal of this section is to estimate the production function, as well as the skill and job types distributions, and assess the quantitative relevance of the mechanism in the paper. The model is estimated year-by-year for the years 1979-2018, a period of rapid technological advancements in the workplace and the period where the hours-wage correlation reversed sign and showed the largest increases.

4.1 Functional Forms

Production Function I assume the same production function used in the comparative statics section, given by (13).⁴⁸ This gives us five technological parameters to be estimated year-by-year: $A_t, \alpha_t, \beta_t, \rho_t, \gamma_t$. Section 4.2 provides an identification argument for these parameters.

Preferences I make use of the same utility function (MaCurdy, 1981) as in the comparative statics section, given by (14). To calibrate this utility function, the values for σ and θ are taken from the literature, and ψ is normalized to 1. In particular, σ is set to 1.4, which is in the mid-range of values from the literature that aims at matching the overall decline in hours worked, both across time (Boppart and Krusell, 2020) and across the development path (Bick *et al.*, 2022b). The curvature in hours disutility cost θ is set to 0.5, a value well within the range of commonly used values to calibrate intensive-margin elasticities (see Keane (2011) for a survey).⁴⁹ Perhaps the most important feature of this calibration is that the curvature

⁴⁷Of course, the full knowledge-based hierarchy model involves a much richer characterization of how layers are formed, which is absent in my setting. Likewise, the nested-CES specification used in the quantitative section involves more flexible complementarities among inputs than those implied in this simple extension, which implies that production is multiplicatively separable.

⁴⁸Note that this is a slightly more general version of the multiplicatively separable production function, of the form $f(x, y, h) = A(\alpha x^\rho + (1 - \alpha)y^\rho)^{\frac{\beta}{\rho}} h^{1-\beta}$. As noted for example in Chade and Lindenlaub (2022), the latter is perhaps one of the most commonly used production function for empirical applications, and it is a special case of the production function I employ (in particular, it can be obtained by letting $\gamma \rightarrow 0$).

⁴⁹Notice that the calibration I adopt for the utility function is very close to available estimates of the same functional form from cross-sectional studies; in particular, Heathcote *et al.* (2014) employ a similar specification

in consumption, σ , being above 1 implies that the substitution effect is being dominated by the income effect.⁵⁰

Distributions Both the skills and the jobs distributions, $Q(x)$ and $G(y)$ are assumed to be unobserved, and hence they need to be estimated.⁵¹ In particular, skills x and jobs y are assumed to follow log-normal distributions, with time-dependent parameters, given by $\log\mathcal{N}(a_t^x, b_t^x)$ and $\log\mathcal{N}(a_t^y, b_t^y)$, respectively. A similar approach has been used extensively in the literature, see e.g. [Lise et al. \(2016\)](#). A key advantage of this approach is that it allows for a year-by-year estimation of the model parameters without the need for data on (observable) distributions, which are rarely available at yearly frequency. The flexibility of this strategy therefore allows for an analysis of the underlying trends behind the parameter estimates, rather than only changes between two points in time.⁵²

4.2 Moments and Identification

Data CPS-ORG is used as the main sample, since it is available for the whole period of interest and it contains information on hours worked, earnings, and hourly wages. When computing the estimation targets, I will restrict the focus to the population of full-time males.⁵³ I focus on the male population only because the analysis abstracts from the participation margin.

Moments I estimate the model by Simulated Method of Moments ([Pakes and Pollard \(1989\)](#), [McFadden \(1989\)](#)). In practice, I pick a set of moments m to identify the set of model parameters θ . For every year, the estimation procedure uses a combination of a global and a local search algorithm to seek for the parameter vector θ_t that minimizes the weighted square distance between model moments $m_t(\theta_t)$ and data moments $\bar{m}_t$:

$$\min_{\theta_t} (\bar{m}_t - m_t(\theta_t))' \Omega_t (\bar{m}_t - m_t(\theta_t)),$$

as in (14) and find values of σ and θ of 1.71 and 0.46, respectively.

⁵⁰This specification has the additional feature that leisure and consumption are separable, i.e. $u_{ch} = 0$. This formulation is convenient in this paper since - as the ultimate objective is to assess the role of technological complementarities in explaining patterns of hours inequality - it effectively shuts down complementarities between hours and consumption (or income) coming from the utility function, and hence isolates the role of technology in explaining the data.

⁵¹I interpret y in my model as jobs rather than firms. This is motivated by the application of the model, and specifically the technological changes at the center of the mechanism. This is a common interpretation in sorting models trying to understand rising inequality, see e.g. [Lindenlaub \(2014\)](#).

⁵²Analyzing these trends (rather than just points in time) is useful because SBTC and HBTC need not have accelerated (or decelerated) during the same years, and in general, because these trends contain valuable information on the identification of the production parameters, as will become clear in the identification argument presented later on.

⁵³For a full description of the data used in the estimation and all the sample restrictions used, see Appendix A.1 For details on the estimation algorithm, see Appendix C.1.

where Ω_t is the weighting matrix.⁵⁴ Table 1 contains a summary of the targeted moments. For the details on the estimation algorithm, see Appendix C.1.

Table 1: Targeted Moments

Notation	Description
$\mathbb{E}(w)$	Avg. Hourly Wages
$w90/w50$	90/50 Perc. Wage Ratio
$w90/w10$	90/10 Perc. Wage Ratio
$h(wdec9)$	Avg. Hours worked at 9th wage decile
$h(wdec5)$	Avg. Hours worked at 5th wage decile
$h(wdec2)$	Avg. Hours worked at 2nd wage decile
$std(w)$	Std of (log) wages

Identification I now discuss how each moment is related to the parameter to be estimated, with the understanding that the parameters are jointly estimated. The TFP level A informs the average income level in the model. The parameter β informs average hours worked in the 5th wage decile (median), as it determines the importance of hours in production (compared to job/skill) through $1 - \beta$. The mechanism is that β directly shifts up (or down) the marginal product of an hour worked, f_h . The parameters α , ρ and γ crucially determine the income and hours inequality in the model; however, key to identification is that they do so differently, as the comparative statics in the previous section have shown; in fact, while increases in α (and similarly, decreases in ρ) tend to reduce average hours in top wage deciles (and increase them in bottom wage deciles), the opposite is true for decreases in γ .⁵⁵ Moreover, α and ρ inform inequality at different points in the income distribution. The former increases overall inequality, while the latter governs the convexity in the wage function. Finally, the means of the worker and job type distributions a_t^x and a_t^y are normalized at 0 for all t . The job type dispersion parameter b_t^x is normalized to 1, and we estimate the job type dispersion parameter b_t^y ; this argument is motivated by the fact that in this setting, only the ratio of these two parameters is identified.⁵⁶ The parameter b_t^y is estimated yearly together with the production function parameters, and directly informs dispersion in wages, captured by the standard deviation of log wages.

⁵⁴Following the literature, I use a diagonal weighting matrix given by the inverse of the (bootstrapped) variances of the empirical moments, i.e. $\Omega = \text{diag}(\widehat{\text{Var}}(\bar{m}))^{-1}$.

⁵⁵The assumed preference specification, and in particular the fact that income effects are stronger than substitution effects in preferences ($\sigma > 1$) is crucial to this result.

⁵⁶In the positive assortative assignment equilibrium, the matching function is defined by $\mu(x) = G^{-1}(Q(x))$. Under lognormality, $Q_x(x) = \Phi\left(\frac{\log x}{b_t^x}\right)$, $G_y^{-1}(u) = \exp(b_t^y \Phi^{-1}(u))$, where $\Phi(\cdot)$ is the CDF of a normal distribution; therefore $\mu_t(x) = \exp\left(b_t^y \Phi^{-1}\left(\Phi\left(\frac{\log x}{b_t^x}\right)\right)\right) = \exp\left(\frac{b_t^y}{b_t^x} \log x\right) = x^{b_t^y/b_t^x}$. Hence the matching function depends on (b_t^x, b_t^y) only through the ratio b_t^y/b_t^x .

With this identification strategy, there are in total six parameters to be estimated: five from the production function ($A, \alpha, \beta, \gamma, \rho$) and one from the distributions of worker and job types (b^y). Table 2 contains a summary of the parameters to be estimated.

Table 2: Estimated parameters (time-varying)

Parameter	Meaning
A_t	Total Factor Productivity (TFP)
β_t	Weight of hours in production
α_t	Weight of skill in production (relative to job type)
γ_t	Elasticity of substitution between hours and skill/job
ρ_t	Elasticity of substitution between skill and job types
b_t^y	Variance of distribution of job types rel. to skill types

Measurement Error In order to align the model and data moments' construction, and in particular alleviate concerns about division bias (Borjas, 1980), we augment the model with measurement error, and assume that hours are reported with noise. In particular, we assume that - denoting the true value of hours in the model in year t , H_t - we observe two noisy hours measures with multiplicative (log-normal) reporting errors,

$$\begin{aligned}\log H_t^{\text{obs}} &= \log H_t + \varepsilon_{h,t}, & \varepsilon_{h,t} &\sim \mathcal{N}\left(-\frac{1}{2}\sigma_h^2, \sigma_h^2\right), \\ \log H_t^{\text{obs},w} &= \log H_t + \varepsilon_{w,t}, & \varepsilon_{w,t} &\sim \mathcal{N}\left(-\frac{1}{2}\sigma_h^2, \sigma_h^2\right),\end{aligned}$$

with $\text{corr}(\varepsilon_{h,t}, \varepsilon_{w,t}) = \rho_{hw}$.⁵⁷ Hours-based moments are computed in the model from H_t^{obs} . Observed hourly wages are instead constructed by division $W_t^{\text{obs}} = Y_t / H_t^{\text{obs},w}$ where $H_t^{\text{obs},w}$ denotes the hours measure used in the wage denominator and Y_t is earnings.

The advantage of this simple procedure is that the model-implied OLS relationship between hours and wages becomes a close counterpart of its empirical analogue, because it mimics the use of two distinct hours measures to mitigate the division bias problem (Bick et al., 2018; Heathcote et al., 2010), while leaving the equilibrium solution unchanged.⁵⁸ The value of σ_h^2 is set to 0.03.⁵⁹ We also set $\rho_{hw} = 0$, implying independent errors in reporting for the two hours measures.⁶⁰

⁵⁷The mean shifts $-\sigma_h^2/2$ ensure $E[H_t^{\text{obs}} | H_t] = H_t$, so measurement error affects dispersion but not average hours.

⁵⁸For consistency, we adopt an analogous procedure in cleaning the data used for estimation, and use two different hours measures available in CPS, as explained in the data paragraph.

⁵⁹ σ_h^2 induces an equivalent variance of measurement error in (log) wages, since $\log W_t^{\text{obs}} = \log Y_t - \log H_t - \varepsilon_{w,t}$, and hence $\text{var}(\log W_t^{\text{obs}} - \log W_t) = \sigma_h^2$.

⁶⁰A value of $\rho_{hw} = 0$ eliminates mechanical division bias from the model-implied cross-sectional relationship between hours and wages $\hat{\beta}$, while leaving active classical attenuation from measurement error in the denominator.

4.3 Results

Model Fit The model and data moments targeted in the estimation are plotted in Figure 8. The model, despite being parsimonious, fits the data well. The model matches well the hours moments despite the hours choice being deterministic and the only source of randomness is measurement error. The ability of the model to match wage inequality measures is also reassuring in light of the counterfactual exercises of the next section.

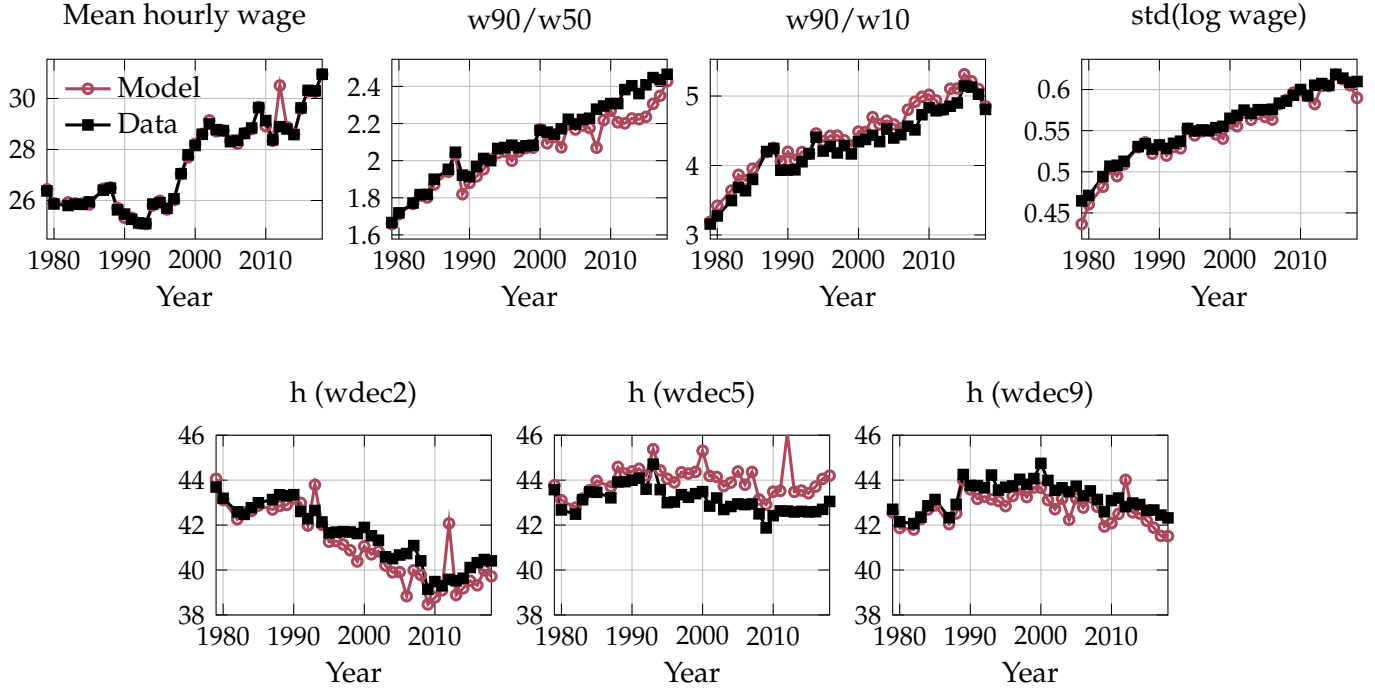


Figure 8: Model vs. data: targeted moments.

Notes: Black squared line plots the evolution of data moments, while red circled line plots model moments.

Estimates The estimated parameters are presented in Figure 9. The estimates reveal several interesting patterns, some new to this model, and some common given previous work focused on understanding the determinants of inequality in U.S. First, there has been an increase in the marginal product of skill in production, captured by α : from 0.17 (s.e. 0.01) to 0.23 (s.e. 0.002). Furthermore, complementarity between skills x and jobs y rose, i.e. ρ has decreased from -0.83 (s.e. 0.26) to -7.74 (s.e. 0.17), with most of the decrease concentrated in the first decades.⁶¹ These patterns are at least qualitatively similar to what the literature on wage inequality in U.S. has found.⁶² Furthermore, the estimated model reveals

⁶¹The targeted moments are imputed as raw data in the estimation algorithm, and therefore the parameter estimates inherit the unavoidable randomness in the year-to-year movements of data targets.

⁶²For example, estimating a version of her model with one-dimensional skills, [Lindenlaub \(2014\)](#) finds that returns to skill have increased (see Table 18 in her paper).

that the complementarity between skills/jobs and hours has also increased significantly: γ decreased, from a value of 0.04 (s.e. 0.02) to -0.51 (s.e. 0.04). These estimates imply that the economy has moved from a world that is close to Cobb–Douglas in the skill-job bundle and hours to one in which hours are much less substitutable with that bundle. As a result, long hours are rewarded disproportionately at the top of the skill and occupation distribution: extra hours matter more when supplied by the workers in the highest-value matches.⁶³ Finally, the estimates reveal an increase in the dispersion of job types relative to worker types which, given our normalization, is represented by an increase in b^y .⁶⁴ This means that the matching function $\mu(x)$ has become steeper, so that the same differences in worker types translate into larger differences in assigned job quality.⁶⁵

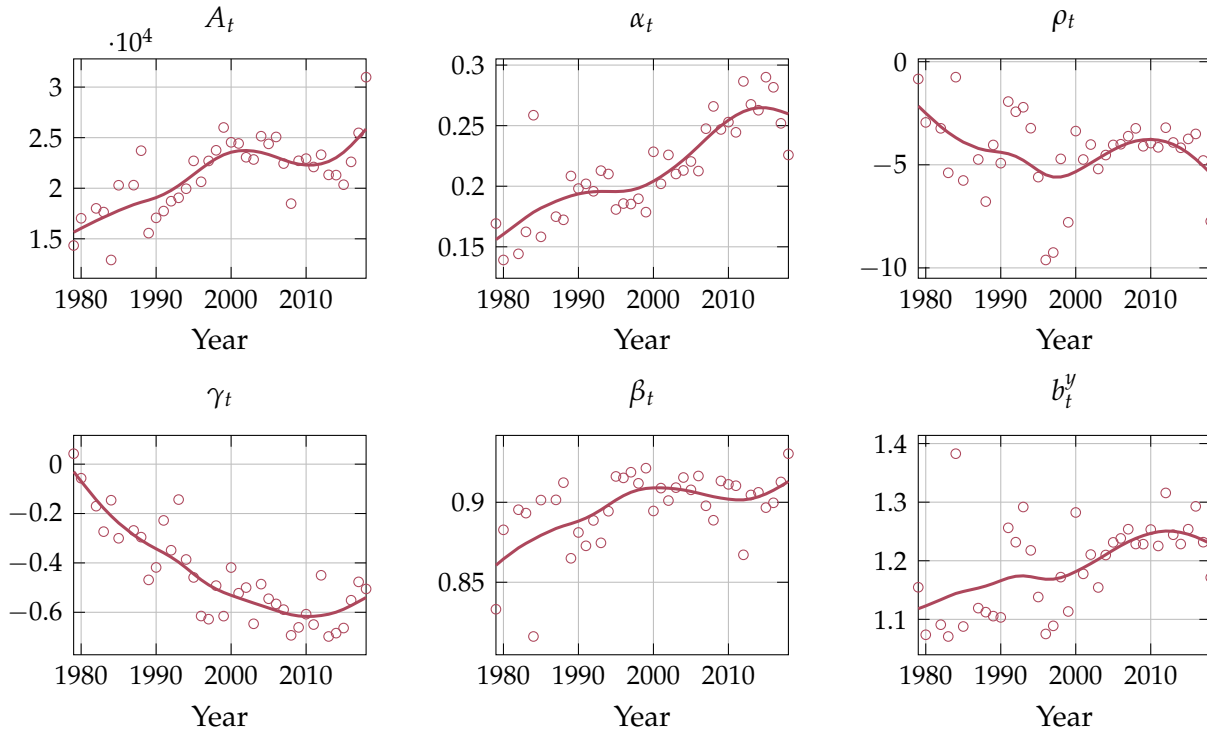


Figure 9: Estimated parameters by year.

Notes: Red circles plot the raw series and the red line plots the HP-filtered series, obtained with $\lambda = 100$.

Validation Figure 10 plots a few untargeted moments in the model and in the data as a simple and transparent validation exercise. The model captures well both the level and the trend in the cross-sectional elasticity between hours and wages (left panel). Moreover, it cap-

⁶³In turn, the value of the parameter β , the weight of hours vs skills/jobs in production only showed a relatively mild increase over the time period considered.

⁶⁴Under the interpretation that y are occupations or bundles of tasks, this can be read as an increase in occupational heterogeneity, or a widening in the distribution of jobs available in the economy and larger differences across jobs in their productivity.

⁶⁵The estimation diagnostics and details on standard error computations are reported in Appendix C.1.

tures well both the increase and then decrease in the fraction of workers working more than 50 hours per week (middle panel). Finally, the right panel shows that the model captures a significant fraction of the dispersion in hours worked. On average, the model predicts between 55% and 60% of the overall dispersion (depending on the year considered). This is expected, since the model does not feature preference heterogeneity and all variation comes from either different work incentives or measurement error.⁶⁶

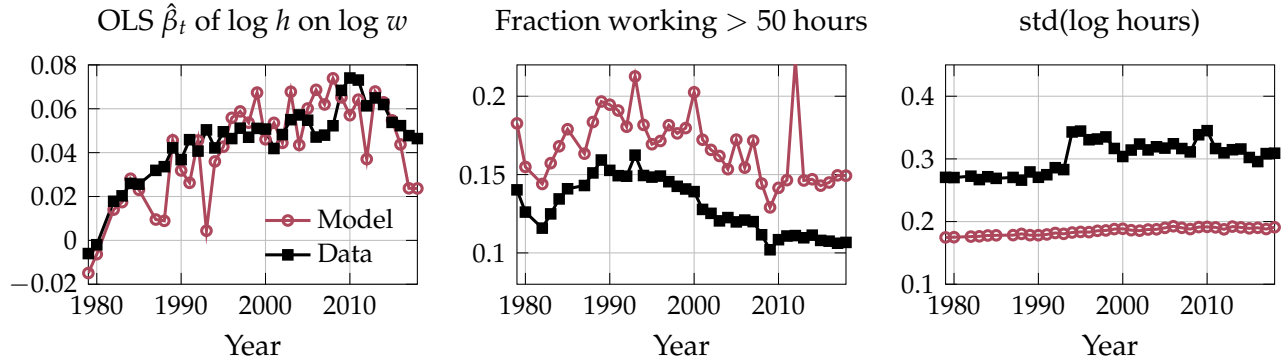


Figure 10: Model vs. data: untargeted moments.

Notes: Black squared line plots the evolution of data moments, while red circled line plots model moments.

5 Application: Income Inequality and the Role of Hours

The estimated model lends itself to a natural counterfactual exercise, which aims at looking at the evolution of income inequality in U.S. by explicitly taking into account the role of hours and sorting, and how technological change shapes this role.

5.1 Counterfactuals

Rise in Income Inequality In the following counterfactuals, we fix one set of estimated parameters to their initial level, and analyze the evolution of inequality in this counterfactual economy by feeding the model with all other parameters as estimated over time, effectively shutting down the channels of interest to gauge their importance. Skill-Biased Technological Change (SBTC) is referred to as changes in (α, b_y, ρ) , capturing increases in the return to skill x , broadly defined.⁶⁷ An advantage of using a broad definition of SBTC is that we

⁶⁶Figure C.1 in Appendix C.2, moreover, shows the model's ability to closely track average hours worked by wage decile, despite only three of those deciles are being targeted in estimation.

⁶⁷Increases in α raise the marginal product of skill x ; similarly, a lower ρ captures increases in complementarities between workers and jobs, while increases in b^y capture a stretching of the job-productivity distribution, which magnifies the gap in matched job quality between high- and low-skilled workers. We thus define SBTC in a broad sense to encompass both shifts in the production technology and shifts in the relative importance of

can immediately compare it with the effects of HBTC, which are by definition defined as changes in γ . The effects of shutting down technological changes on the overall rise in income inequality - defined as the variance of (log) income - is depicted in Figure 11.

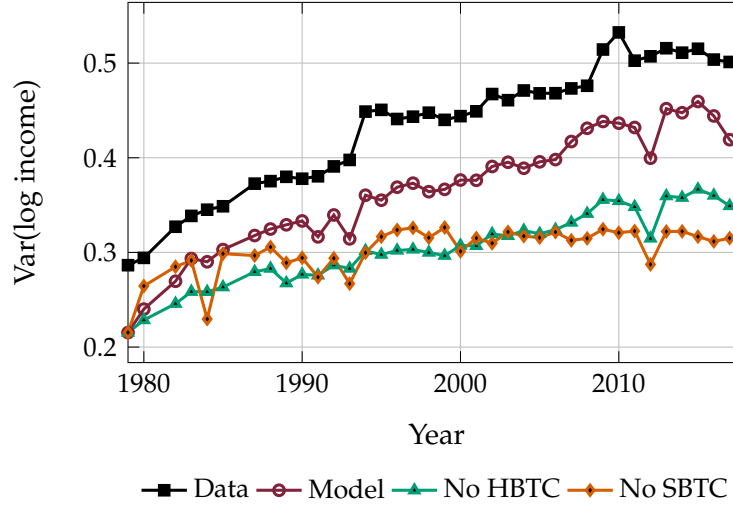


Figure 11: Variance of log income over time.

Notes: The figure displays the evolution of income inequality in the data ("Data"), in the estimated model ("Model"), and the counterfactual model evolution keeping γ at their initial estimated level ("No HBTC"), and keeping (α, b_y, ρ) at their initial estimated levels ("No SBTC").

The estimated model underpredicts the level of income inequality due to the fact that it only accounts for a fraction of the dispersion in hours, but it captures the evolution very well. Shutting down either SBTC or HBTC decreases the rise in income inequality, thus showing the importance of both types of technological changes for its evolution, with HBTC explaining about 34% of the model-implied rise in inequality. This exercise demonstrates the quantitative importance of HBTC in explaining the rise in income inequality, but leaves the question of what causes this increase, which is what the next set of counterfactuals show.⁶⁸

Income Inequality: Decomposition Taking as our measure of inequality the variance of (log) income, we have the decomposition:

$$\text{var}(\log(\text{income})) = \text{var}(\log(\text{wage})) + \text{var}(\log(\text{hours})) + 2 \text{cov}(\log(\text{wages}), \log(\text{hours})) \quad (16)$$

high- y jobs. Both notions have been used in the literature to capture changes in the return to skill over time. Changes in production technologies are closer to frameworks like [Krusell *et al.* \(2000\)](#), while changes in b^y are akin to occupation-demand notions of SBTC as in [Autor and Acemoglu \(2011\)](#) or [Costinot and Vogel \(2010\)](#).

⁶⁸SBTC explains roughly 40% of the model-implied rise in inequality. The share of the rise explained by HBTC is computed as the fraction of the model-implied increase in inequality between 1979 and 2018 that disappears when γ is kept at its 1979 level. Formally, $\text{Share}_{\text{HBTC}} = \frac{(I_{2018}^{\text{model}} - I_{1979}^{\text{model}}) - (I_{2018}^{\text{no HBTC}} - I_{1979}^{\text{no HBTC}})}{I_{2018}^{\text{model}} - I_{1979}^{\text{model}}}$, where I is defined as $\text{Var}(\log(\text{income}))$ and analogously for SBTC.

A key advantage of this model is that rising income inequality can be studied in all its three components, since in this setting all three components are endogenous outcomes of the model. Exploiting the decomposition in (16), the next Figure shows the channels through which SBTC and HBTC affect total inequality.

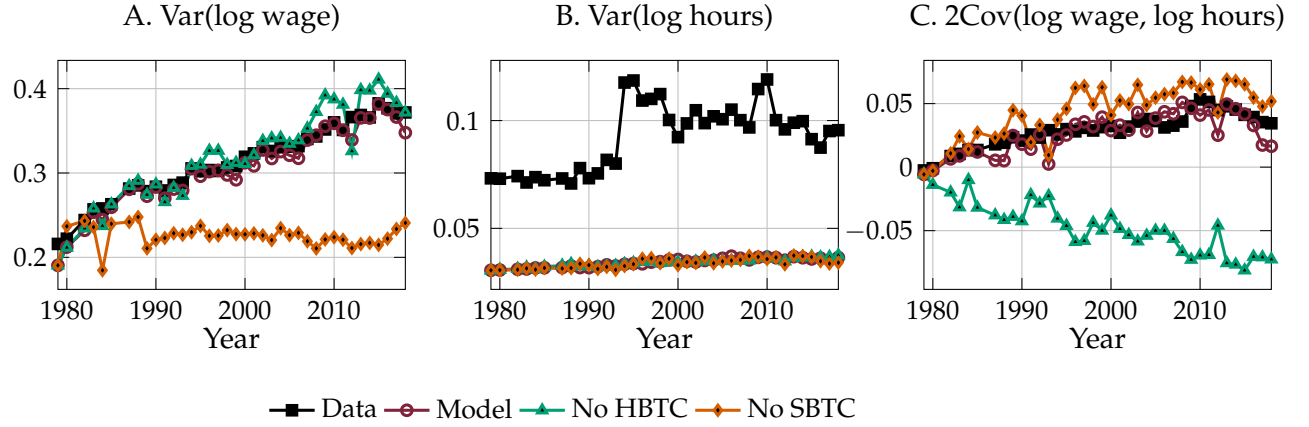


Figure 12: Rise in Income inequality: decomposition.

Notes: The figure displays the three components of income inequality in (16) (Panels A, B and C, respectively). For each panel, the figure plots the actual evolution of each component in the data (“Data”), the evolution in the estimated model (“Model”), and the evolution in the counterfactual model keeping γ at its initial estimated level (“No HBTC”), and keeping (α, b_y, ρ) at their initial estimated levels (“No SBTC”).

The results are plotted in Figure 12. Several results emerge. Focusing on Panel A., we notice that SBTC is the main driver of the increase in the variance of wages. Quantitatively, the effect of HBTC on the variance of wages is small since increases the returns to working long hours are accompanied by the rising hours worked of this group, thus offsetting their effect on wages. Panel B. shows that neither SBTC nor HBTC had a material effect on hours dispersion. The reason this happens is that most of the dispersion is left unexplained by technological forces, and therefore these changes have minimal effects on the overall dispersion relative to the data. Finally, Panel C. (in line with the predictions of the theory) shows that the two estimated parameter changes have had opposite effects on the counterfactual relationship between hours and wages: without SBTC, the increase in the covariance between hours and wages would have been even higher than observed. Conversely, without HBTC, increases in the return to work long hours would be absent and hence the covariance would have decreased significantly due to the combination of SBTC and large income effects in preferences. Preferences, therefore, play a key role not only in explaining aggregate movements in hours but also in identifying the different effects that technological changes considered have on inequality.

The implication is that considering changes in the marginal product of skills x on the

variance of wages hides important negative effects on inequality coming through the covariance term; likewise, ignoring changes in the marginal product of hours for certain types of workers is a new channel through which inequality increases, and becomes apparent only when the cross-sectional relationship between hours and wages is properly accounted for. This is a new result that has profound implications for our understanding of the deep causes of inequality, and complements the literature on Skill Biased Technical Change in that it shows that - in addition to the increased importance of skills in production - how these skills perform with more hours of work has first-order implications for inequality trends, both qualitatively and quantitatively.

5.2 Mechanism

We conclude the section with a summary of the mechanism, how it relates to existing work, and provide some suggestive evidence.

The Role of Sorting: Theory The model mechanism sheds light on the link between the nature of hours in production, summarized by the complementarity parameter γ , and the convexity of earnings in hours. This link operates through sorting. To see this, define $C(x, y) = (\alpha x^\rho + (1 - \alpha)y^\rho)^{1/\rho}$ and assume for simplicity that the assignment is positive assortative and equilibrium hours $h(x)$ are increasing in skill, so that the inverse $x = \eta(h) \equiv h^{-1}(x)$ is well defined. By optimality of the hours choice, $w_h = f_h(\eta(h), \mu(\eta(h)), h) = f_h(C(h), h)$. Differentiating this expression with respect to hours, we get

$$w_{hh} = f_{hh}(C(h), h) + f_{hC}(C(h), h) C'(h). \quad (17)$$

Equation (17) characterizes the global returns to hours in this economy. The first term is the local curvature in the returns to hours and is negative whenever f_{hh} ; the second term can be positive or negative depending on the shape of the production function f_{hC} and the equilibrium assignment $C'(h)$. The term f_{hC} is positive whenever $\gamma < 1$; the equilibrium term $C'(h)$ means that high-skilled workers work more, and are assigned better jobs, so hours are higher in the skill-job bundle and hence $C'(h) > 0$.⁶⁹ Therefore, the combination of complementarities in hours together with sorting drives the term $f_{hC}(C(h), h) C'(h)$ and hence constitutes a force that raises w_{hh} .

In this light, sorting constitutes a micro-foundation for why earnings might have become more convex in hours worked (Goldin, 2014): because working more allows them to get better jobs, workers' incentives for long hours increase. This is especially true for top workers, because the return to their skills (and their hours) is magnified in top jobs. Moreover, this

⁶⁹See Appendix B.4 for formal derivations.

model also relates to the recent literature assessing the underlying heterogeneity in returns to hours across hours and occupations. In particular, sorting makes it possible for weak within-occupation returns to long hours and steep global returns to coexist, and thus identifies a new mechanism for which the observed earnings-hours profile need not coincide with the return to hours relevant for labor supply (Bick, Blandin and Rogerson, 2022a).

The Role of Sorting: Quantitative The best way to gauge the quantitative relevance of the mechanism is to ask how changes in γ interact in equilibrium with changes in gains to matching to drive up income inequality. To do so, we compute two counterfactual effects. The first is the effect of HBTC, computed on its own by starting from the 1979 economy and changing only γ to its estimated value in year t , while keeping all other parameters fixed at their 1979 values. Denoting the resulting level of income inequality (variance of log income) by I_t^{HBTC} , the effect of HBTC alone is

$$I_t^{HBTC} - I_0, \quad (18)$$

where I_0 is the model-implied value of inequality in the estimated 1979 economy. The interaction between changes in gains to matching and HBTC is gauged by computing a second counterfactual, given by the effect of HBTC in an economy in which the matching function and the complementarity between types undergo the estimated changes. Changing b_y and ρ to their estimated values in year t , we obtain an economy with outcome $I_t^{\Delta\mu}$; additionally changing γ , we obtain an economy with outcome $I_t^{HBTC+\Delta\mu}$. The marginal effect of HBTC conditional on changes in gains to matching is therefore

$$I_t^{HBTC+\Delta\mu} - I_t^{\Delta\mu} \quad (19)$$

Comparing (18) and (19) reveals the nature of the interaction between changes in the matching function and HBTC across time.⁷⁰ Figure 13 plots these two objects. A positive interaction between HBTC and gains to matching emerges from the mid-1990's onwards, since the effect of the former in driving up inequality is much larger when the latter is also present. The intuition comes directly from the model's mechanism: because HBTC raises the returns to hours for high x , high y matches, when relative productivity in those matches increases, so does the effect of working one more hour relative to workers (and jobs) at the bottom of the distribution.⁷¹

⁷⁰In fact, it follows that if $I_t^{HBTC+\Delta\mu} - I_t^{\Delta\mu} > I_t^{HBTC} - I_0$, then $\Delta\mu$ amplifies the effect of HBTC on I . If instead $I_t^{HBTC+\Delta\mu} - I_t^{\Delta\mu} < I_t^{HBTC} - I_0$, then $\Delta\mu$ dampens it.

⁷¹As Figure C.2 in Appendix C.3 shows, this is due to the fact that HBTC now reflects in an increased wage dispersion and an increased covariance between hours and wages when both changes are present.

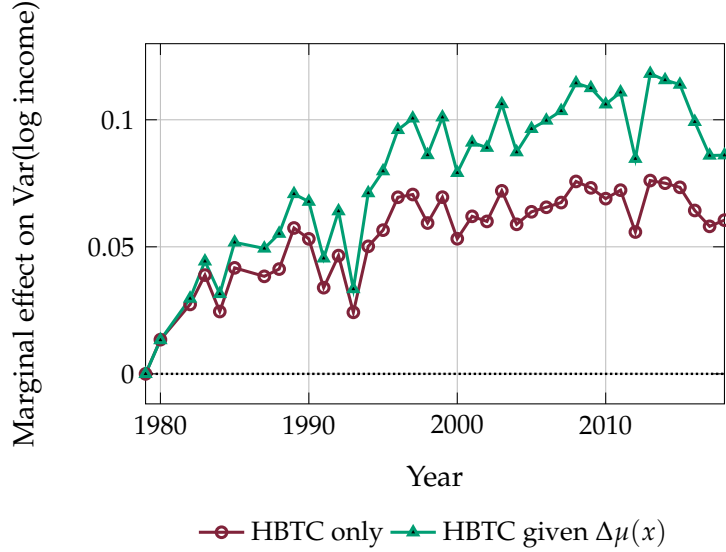


Figure 13: Interaction of HBTC with changes in $\mu(x)$.

Notes: The figure displays a) the increase in the variance of log income in the model economy when estimated changes in γ are fed into the 1979 estimated economy, leaving all other parameters unchanged, relative to the level of the variance of log income in 1979 ("HBTC only"); b) the increase in the variance of log income when both estimated changes in γ and in ρ, b^y are fed into the 1979 estimated economy relative to when only ρ, b^y are changing ("HBTC given $\Delta\mu$ ").

Evidence This section provides evidence that supports the mechanism. In particular, and following [Denning, Jacob, Lefgren and vom Lehn \(2022\)](#) analysis, I decompose the returns of hours within jobs from the price of expected hours across occupations, which captures the across-occupations hours-wage elasticity. First, I run an individual-level regression of log hourly wages on log hours with detailed occupation fixed effects, which identifies the within-occupation elasticity. Second, I collapse the data to occupation-year means of log wages and log hours and estimate the corresponding across-occupation elasticity. The results are plotted in Figure A.7. The resulting series closely mirrors [Denning et al. \(2022\)](#) qualitative pattern obtained with U.S. Census data: the within-occupation elasticity remains close to zero throughout the sample, while the across-occupation elasticity is large and rises sharply over time. In other words, the rise in the return to hours is primarily coming from the increasing allocation of high-hour workers into occupations in which long hours command a high premium. This evidence provides support to the mechanism, in that it suggests that the allocation that long hours can grant to workers is an increasingly important component of the true returns to hours.⁷²

⁷²Moreover, this new evidence is consistent with the literature that estimates returns to long hours across occupations and skills: [Goldin \(2014\)](#), [Cortés and Pan \(2019\)](#), and [Jang and Yum \(2022\)](#) emphasize that some occupations disproportionately reward long and continuous hours, which is consistent with the estimated decline in γ .

6 Implications

I now discuss two settings in which the mechanism proposed in this paper might have additional implications.⁷³

Hours Constraints The first setting is a stochastic extension of the model to study the misallocation effects of hours distortions. The extension replaces the deterministic assignment with an allocation of worker-job pairs and hours that not only depends on production complementarities and preferences, but also on EV-I type idiosyncratic shocks.⁷⁴ The extension is used to study a simple reduced-form distortion that raises the nonpecuniary cost of supplying hours, $\psi_\tau = (1 + \tau)\psi$, where $\tau \geq 0$ indexes the strength of the wedge. τ can be interpreted as capturing time-varying forces that affect social norms, child-care related costs, commuting costs, or other nonpecuniary burdens associated with long schedules. The full exposition of the model and the exercise can be found in Appendix D.1.

The exercise of interest is to compare the effect of changes in the wedge τ under two technology regimes, indexed by the strength of the hours-complementarity parameter γ .⁷⁵

The first result is that output losses from the hours wedge are systematically larger under HBTC. Intuitively, when long hours are more complementary to worker skill and job quality, distortions to hours destroy more surplus. The second, more surprising result is that the incidence of the wedge is highly unequal across workers. Income losses are largest for high-skilled workers, even though the direct reduction in long-hours caused by higher τ is more broadly distributed across the skill distribution. The wedge also reallocates top jobs away from top workers and toward lower-skill workers, because the effects of changing τ operate not only through the intensive margin of hours but also through the assignment of workers to jobs. These results seem particularly relevant when studying gender gaps and the misallocation of skilled women across jobs. If interpreted as unequal division of labor at home, the effects of a gender-specific wedge causing high-ability women to be allocated away from top jobs could be magnified by HBTC, resulting in higher losses in incomes and output - especially for workers whose long hours are most valuable in the market.⁷⁶ More generally, the extension suggests a potentially meaningful role for HBTC in amplifying exogenous differences in hours constraints, with heterogeneous effects across the skill distribution.

⁷³The full discussion and presentation of these extensions is developed in Appendix D.

⁷⁴The reference framework is Galichon, Kominers and Weber (2019), extended to have hours entering in production. For a recent application of a similar framework, see Bandiera, Kotia, Lindenlaub, Moser and Prat (2024).

⁷⁵In the first, which we label the HBTC regime, the hours-complementarity parameter γ is set to its value in the terminal year of the estimation. In the second, which we label no HBTC, all parameters are held fixed at their terminal-year values except for γ , which is reset to its 1980 estimate.

⁷⁶We fully pursue this extension in separate work, see Mantovani and Qiu (2025).

Labor Supply Elasticities The second setting relates to labor supply elasticities and optimal tax progressivity. Following the increase in income inequality in the U.S., the question is whether tax progressivity should respond to this increase, and if so, by how much (Heathcote *et al.*, 2021).⁷⁷ Therefore, it is interesting to look at how labor supply elasticities respond to HBTC in this setting, and how sorting affects these elasticities, both qualitatively and quantitatively.

To do so, in Appendix D.2, I show a comparison between compensated elasticities in this model and elasticities computed with fixed assignment (i.e., shutting down the effect that working more hours has on the allocation of workers to jobs), in the spirit of Scheuer and Werning (2017).⁷⁸ The full elasticity then adds the assignment margin. Importantly, this gap itself depends on the technology governing the return to hours, which we have argued has changed over time due to HBTC.

The analysis reveals that HBTC reshapes the cross-sectional profile of elasticities: in the baseline economy, compensated earnings elasticities rise with earnings, while in the no-HBTC counterfactual they decline at the top. Intuitively, with HBTC top earnings remain much more closely tied to the marginal return to hours. Appendix D.2 further shows that assignment amplifies the within-job responses, so that the elasticity relevant once workers are allowed to move along the assignment schedule exceeds the fixed-assignment elasticity. This amplification is proportionally larger under HBTC for the middle and upper parts of the distribution, and in the top tail the average adjustment is systematically larger in the estimated economy than in the no-HBTC counterfactual. In other words, HBTC not only changes the elasticity profile within jobs; it also increases the extent to which sorting affects those elasticities. Under the estimated changes in the production function, these adjustments are quantitatively significant.

These findings suggest a possible implication for discussions of tax progressivity. In standard sufficient-statistics approaches, the key behavioral input is the earnings elasticity. In this setting, however, that object is not invariant to technology: it depends on how hours enter production and on the extent to which higher-hour workers are sorted into better jobs. Fully characterizing how progressivity should change over time is beyond the scope of the paper⁷⁹, but these derivations suggest that the type of technological changes emphasized in

⁷⁷The literature has studied optimal taxation in the presence of superstar effects similar to those analyzed in this setting (Scheuer and Werning, 2017; Ales and Sleet, 2016), albeit with production functions different from the one used in the quantitative application of this paper, and none of these papers studies how technological change affects these margins.

⁷⁸This distinction is useful because, in the present model, the same force emphasized in the mechanism section - namely that hours are more valuable in better matches - affects not only inequality but also the elasticity of earnings with respect to incentives.

⁷⁹Establishing that tax progressivity should change in this setting would require solving the full normative problem, which is beyond the scope of the paper. The message here is that the technological changes em-

this paper may make unadjusted within-job elasticities increasingly misleading in environments where hours and sorting are strongly intertwined.⁸⁰

7 Concluding Remarks

This paper has developed a new theory of sorting in which workers are assigned to jobs based on their skills and the hours they provide. The main mechanism emerging from the theory is that the allocation of workers to jobs through hours is a key component of the true earnings returns to hours, and shows that this margin has become stronger over time due to pervasive technological advancements in the labor market. Empirical evidence based on [Denning *et al.* \(2022\)](#) supports the mechanism, and in particular the role of sorting as a key determinant of the returns to hours faced by workers, above and beyond the within-job returns. This echoes [Bick *et al.* \(2022a\)](#), who show that the market earnings-hours schedule cannot be inferred directly from the raw cross-sectional relationship between wages and hours; in this setting, the relevant return to hours also reflects how hours change the worker's assignment to jobs.

The application of an estimated version of the model suggests that the rise in income inequality in the U.S. can be explained by known forces (such as SBTC) in addition to new forces (HBTC) capturing a rise in the returns to hours in top jobs. These forces operate to raise the cross-sectional relationship between hours and wages, thus reconciling the negative aggregate relationship between hours and wages with a positive and rising cross-sectional relationship.

In the final part of the paper, I consider a stochastic version of the model to show that wedges to working long or flexible hours can misallocate talent across jobs, in the spirit of [Hsieh, Hurst, Jones and Klenow \(2019\)](#). The broader message is that, once technology makes the returns to working time more heterogeneous across jobs, inequality and aggregate productivity increasingly depend not only on who possesses more skill, but also on who can supply the hours needed to be allocated to the jobs where those skills are most valuable. Fully pursuing this extension represents a fruitful avenue for future research.

phasized in the paper alter the earnings-elasticity object entering sufficient-statistics tax calculations. While [Scheuer and Werning \(2017\)](#) show that superstar effects can be neutral conditional on the correct sufficient statistics, the stronger comparative-static cancellation result in that paper need not apply automatically here, because output depends separately on worker type, job type, and hours, and sorting and hours are jointly determined in equilibrium, as in [Vereshchagina \(2021\)](#).

⁸⁰[Vereshchagina \(2021\)](#) studies the effects of changes in tax progressivity on sorting when hours are relevant in production, finding that the reallocation of managers to jobs following tax reforms can have significant effects on output and inequality; these results further emphasize the importance of considering the sorting of workers to jobs when analyzing optimal tax rates.

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A Data Appendix

A.1 Sample Construction

Overview The main dataset used in the analysis is the Current Population Survey (CPS). In particular, I use the Ongoing Rotation Group(ORG) due to the detailed information on earnings and hours worked.⁸¹ I adopt the following sample restrictions: I only consider males, aged 25-64. Multiple-job holders, self-employed, and workers with usual weekly hours below 10 are also excluded from the sample. The dataset is then trimmed of the bottom 1% of the usual hours distribution. All statistics are computed using the provided sample weights.

Hours Unless otherwise noted and stated explicitly, all hours statistics are computed using *hourslw*, which is hours worked last week across all jobs.⁸² Hours moments (means, dispersion, percentiles, and hours-by-wage-decile profiles) are computed using *hourslw* and the CPS-ORG weights. In addition, the hours-wage elasticity is computed year-by-year from a weighted regression of $\log(\text{hourslw})$ on $\log(rw)$ controlling for age and age squared.

Wages Real hourly wages are measured by the CEPR variable *rw* (real hourly wage in 2019 dollars) and are used as provided in the extracts. Wage dispersion is measured using the within-year standard deviation of residual log wages: for each year I regress $\log(rw)$ on a quartic in age and compute the weighted standard deviation of the residuals. Wage percentiles and wage-decile group assignments are computed using weighted inverse-CDF (step) quantiles based on the CPS-ORG weights.

A.2 Robustness and Additional Evidence

In this section I provide additional evidence as robustness checks for the facts in Section 2.

A.2.1 Hours-Wage Elasticity Trends

In this subsection we provide robustness checks for Figure 3 (left panel) and show that the message holds across different specification and data sources.

⁸¹The main advantage of using CPS-ORG is that questions on work and hours refer to current pay or usual hours, unlike CPS-ASEC, which uses as a reference period of the last week or last year. For details on how variables are extracted and cleaned from CPS, see https://ceprdata.org/wp-content/cps/CEPR_ORG_Wages.pdf.

⁸²Since multiple-job holders are excluded from the sample, *hourslw* coincides with hours worked at the main job for the retained observations

ASEC Figure A.1 computes the same elasticity using the CPS-ASEC sample (March supplement), following the specification in Boppart *et al.* (2024). Consistently with the fact that the hours-wage correlation *level* can be subject to the type of hours measure used (see also the discussion in (Heathcote *et al.*, 2010)), the level of the estimated coefficient $\hat{\beta}$ is overall lower, ranging from -0.05 in 1976 to 0.02 in 2015. The trend is however clearly increasing, and it turns from negative to positive.

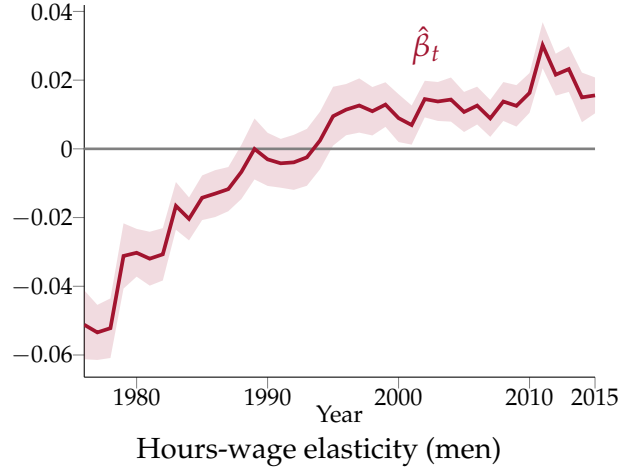


Figure A.1: Robustness I: hours-wage elasticity in the ASEC sample.

Notes: The figure reports the estimated coefficient from a year-by-year regression of log annual hours on log hourly wages using the ASEC sample, following the specification in Boppart *et al.* (2024). Sample is restricted to men aged 25-54, excludes individuals in the armed forces, individuals not in the labor force, individuals with unemployment during the year, and observations in the bottom 1 percent of the annual-hours distribution within year. Regressions control for age, age squared, education categories, and a race indicator, and are weighted using provided sample weights. The shaded area denotes the 95% confidence interval.

CPS-ORG We check that the rising hours-wage correlation in our main CPS-ORG sample is not driven by the sample selection procedures described earlier. In Figure A.2, Figure A.3, and Figure A.4, we plot the hours-wage elasticity computed using CPS-ORG as in the Figure 3, but with a higher restriction in terms of minimum working hours and a different definition of hours worked (usual hours), respectively. The results are essentially unchanged relative to the main figure.

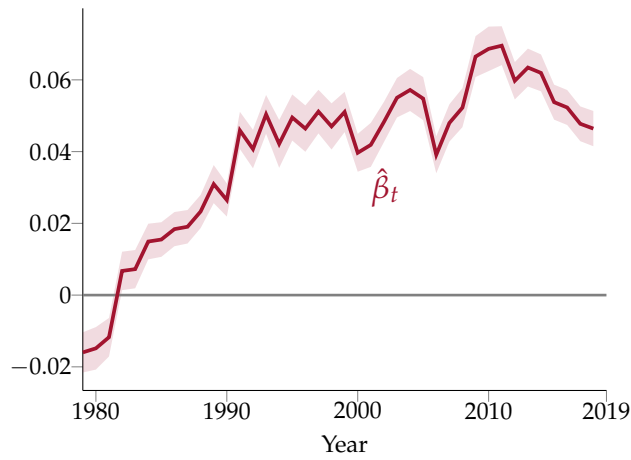


Figure A.2: Robustness II: hours-wage elasticity: minimum hours.

Notes: The figure reports the estimated coefficient from a year-by-year regression of log hours on log hourly wages. Relative to the baseline, the figure restricts workers with usual weekly hours below 20.

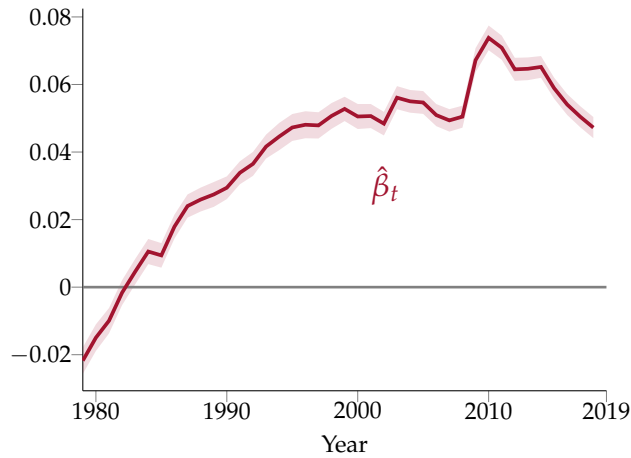


Figure A.3: Robustness III: hours-wage elasticity: usual hours.

Notes: The figure reports the estimated coefficient from a year-by-year regression of log hours on log hourly wages. Relative to the baseline, the figure uses, as hours concept, usual weekly hours on the main job (*uhourse*).

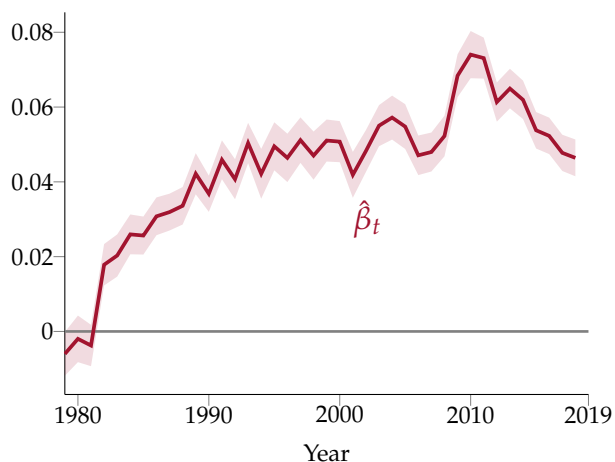


Figure A.4: Robustness IV: hours-wage elasticity: 5% trimming.

Notes: The figure reports the estimated coefficient from a year-by-year regression of log hours on log hourly wages. Relative to the baseline, the figure uses a higher trimming rule for hours at the bottom of the distribution (5%).

Raw Hours and Wages By Wage Decile In figure A.5, I divide the sample of workers in CPS by wage decile (i.e., I compute the wage decile year by year) and plot the average hours worked by each decile. To avoid cluttering, I only plot average hours worked for the top, the bottom, and the middle decile. The figure clearly shows that workers with the lowest wages (black line) decreased their hours worked; workers with highest wages (red line) increased their hours, albeit the increase is concentrated in the 1980's and 1990's.

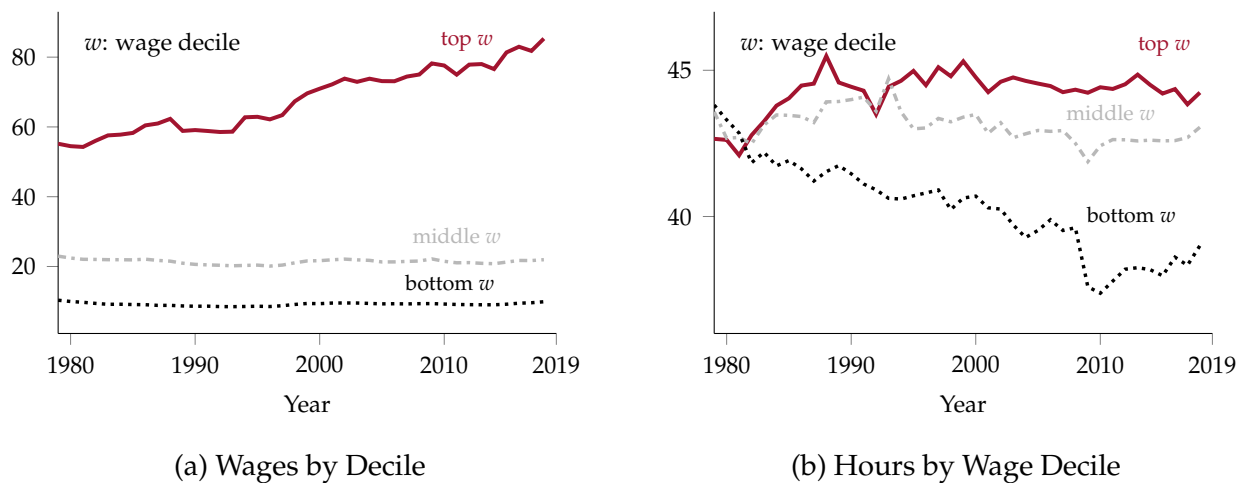


Figure A.5: Wages and Hours by Wage Decile.

Notes: Panel (a) plots raw hourly wages for the bottom, middle, and top wage deciles. Panel (b) plots raw hours for workers in the bottom, middle, and top wage deciles. In both panels, deciles are based on the wage distribution in each year. Source: CPS-ORG.

ATUS Next, I turn to another dataset to inspect whether trends in hours worked and leisure by skill category are unique to CPS. I use the American Time Use Survey (ATUS),

already used for studies regarding leisure trends, e.g. [Aguiar and Hurst \(2007\)](#)⁸³.

Taking data from [Boppart and Ngai \(2021\)](#) (see their paper for details on data construction), I plot average hours worked on the market and leisure hours by skill category. I approximate skill by years of education, and I consider four skill categories, from high-school or less (less than 12 years of schooling) to more than college (more than 16 years of schooling). I plot the evolution of market hours and leisure hours by educational category in Figure A.6 (left and right panel, respectively).

The figures show very interesting patterns. From the beginning of the sample until roughly 1985, workers of each education category consume more leisure hours and less market hours. Importantly, the movements are almost perfectly parallel along educational categories. Starting from 1985, however, highly educated workers diminish their leisure hours and increase their market hours; this is not true for low-educated workers, who continue to consume more leisure and provide lower market hours as years go by. Hence, evidence from ATUS data support the patterns highlighted using CPS: hours worked by skill display divergent trajectories, but only starting from the 1980's. I conclude that ATUS data support the mechanism and provide additional evidence to motivate the analysis.

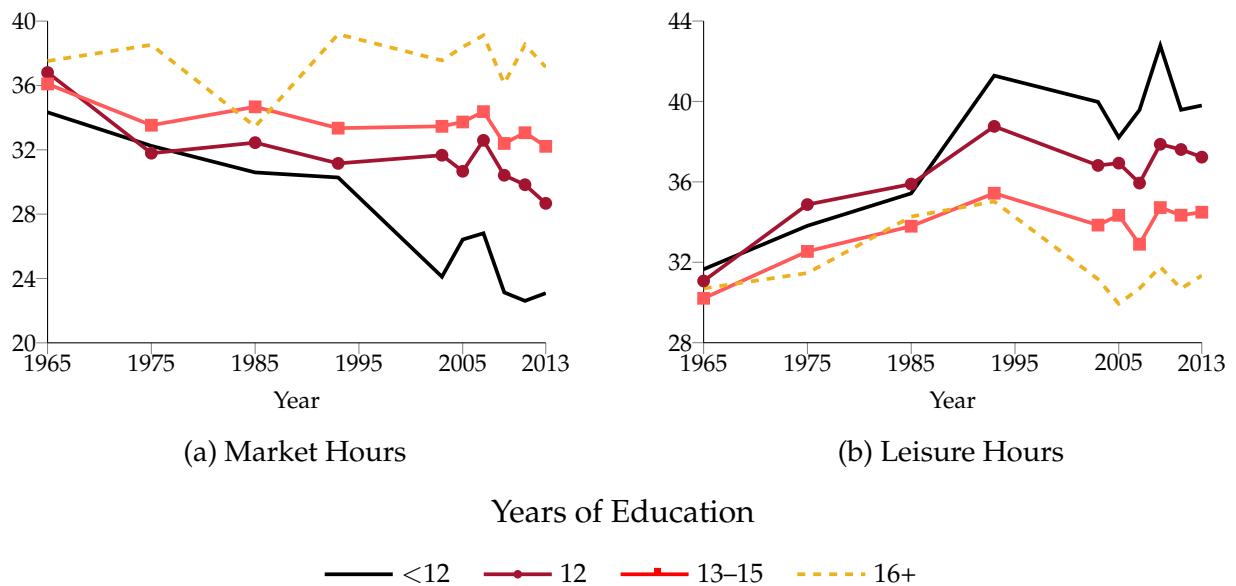


Figure A.6: Market and leisure hours by education group.

Notes: The left panel plots market hours and the right panel plots leisure hours by education group, ATUS dataset. The four groups correspond to less than 12 years of education, 12 years, 13-15 years, and 16 or more years. Source: [Boppart and Ngai \(2021\)](#) and [Aguiar and Hurst \(2007\)](#).

⁸³Following [Aguiar and Hurst \(2007\)](#), the data for the pre-2003 comes from America's Use of Time datasets and the 1975–1976 Time Use in Economics and Social Accounts. See their paper for details.

A.3 Empirical Evidence

The following Figure replicates, using the CPS-ORG sample, an analogous analysis as in [Denning *et al.* \(2022\)](#). The Figure shows clearly that the within-occupation elasticity of wages relative to hours worked, as in their paper, is negative or around zero throughout the recent decades; the across-occupation elasticity, on the other hand, rises sharply and continuously throughout the period, suggesting that returns to hours are primarily a between-occupation phenomenon.

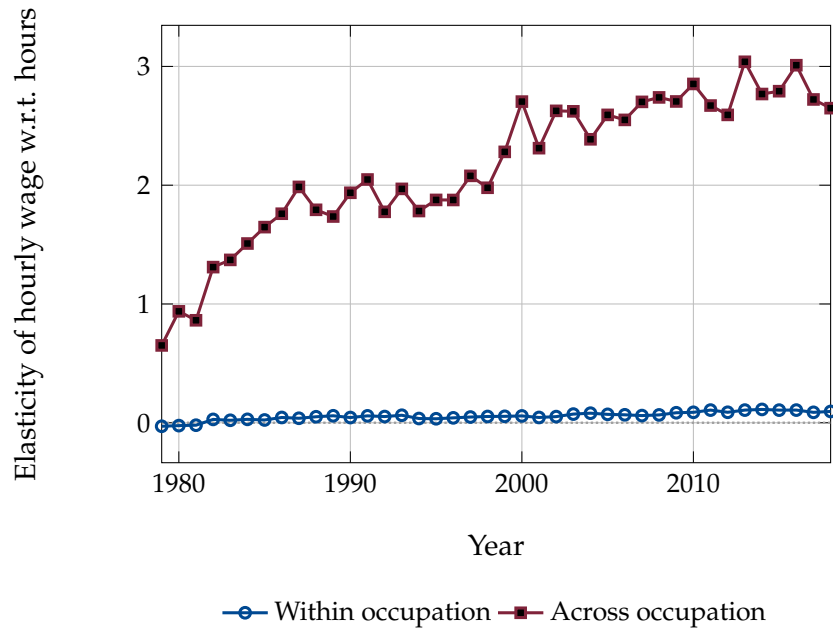


Figure A.7: Denning-style decomposition of the elasticity of hourly wages with respect to hours.

Notes: The figure plots the within-occupation component and the across-occupation component of the elasticity of log hourly wages with respect to log hours. The within component (blue circled line) is estimated from year-specific regressions of log hourly wages on log hours with 3-digit occupation fixed effects. The across component (brown squared line) is estimated from year-specific regressions of occupation-year mean log hourly wages on occupation-year mean log hours.

B Theory Appendix

B.1 Conditions for Assortative Matching

Proof of Proposition 1 As shown in [Chade *et al.* \(2017\)](#), under the regularity conditions in Theorem 3 in [Milgrom and Shannon \(1994\)](#), condition (6) is equivalent to the Spence-Mirrlees single crossing condition. In this setting, this means that the Marginal Rate of Sub-

stitution (MRS) $-\frac{\Phi_y}{\Phi_V}$ is increasing in x . The intuition is that if the MRS , $-\frac{\Phi_y}{\Phi_V}$, increases in x , higher types x are willing to pay more to be matched with a higher type y . Therefore, higher types x can outbid lower x in the competition to be matched with high y , and positive sorting obtains in equilibrium.

Following this monotone comparative statics argument, we now check that indeed the MRS is increasing in x . We have:

$$\Phi_y = u_c f_y, \quad \Phi_V = -u_c$$

Therefore, $-\frac{\Phi_y}{\Phi_V}$ increasing in x implies that $f_y(x, y, h^*(x, y, V))$ is increasing in x . Then, differentiating $f_y(x, y, h^*(x, y, V))$ with respect to x , we directly obtain (7), which completes the proof. ■

Proof of Proposition 2 The first-order condition for the optimal hours choice in (5) is given by:

$$u_c(f(x, y, h) - V, h) f_h(x, y, h) + u_h(f(x, y, h) - V, h) = 0.$$

Define

$$\mathcal{F}(x, y, V, h) \equiv u_c(f(x, y, h) - V, h) f_h(x, y, h) + u_h(f(x, y, h) - V, h).$$

Then $h^*(x, y, V)$ satisfies the FOC

$$\mathcal{F}(x, y, V, h^*(x, y, V)) = 0. \tag{B.1}$$

By the implicit function theorem,

$$h_x^* = -\frac{\mathcal{F}_x}{\mathcal{F}_h}. \tag{B.2}$$

The numerator of (B.2) is

$$\mathcal{F}_x = u_{cc} f_x f_h + u_c f_{xh} + u_{hc} f_x. \tag{B.3}$$

The denominator of (B.2) is

$$\mathcal{F}_h = u_{cc} f_h^2 + (u_{ch} + u_{hc}) f_h + u_c f_{hh} + u_{hh}. \tag{B.4}$$

Recalling that, from Proposition 1, PAM obtains if and only if

$$f_{xy} + f_{hy} h_x^* > 0, \tag{B.5}$$

we can write h_x^* in (B.5) explicitly by substituting the numerator and denominator terms (B.3) and (B.4), this is equivalent to

$$f_{xy} - f_{hy} \frac{u_{cc}f_x f_h + u_c f_{xh} + u_{hc} f_x}{u_{cc}f_h^2 + (u_{ch} + u_{hc})f_h + u_c f_{hh} + u_{hh}} > 0.$$

Since the denominator is negative by the strict second-order condition⁸⁴, multiplying by it reverses the inequality. Therefore PAM obtains if and only if

$$f_{xy} \left[- \left(u_{cc}f_h^2 + 2u_{ch}f_h + u_c f_{hh} + u_{hh} \right) \right] + f_{hy} [u_{cc}f_x f_h + u_c f_{xh} + u_{hc} f_x] > 0. \quad (\text{B.6})$$

for all (x, y, h, c) , which is the condition in the main text. The opposite inequality gives the condition for NAM. The simplified condition (9) is immediately obtained when $u_{ch} = 0$ is assumed. ■

Alternative derivation

We can obtain the same condition by following the joint-decision argument. Let

$$\Phi(x, y, h) = u(f(x, y, h) - V(y), h),$$

where $V(y)$ is the equilibrium payoff schedule for the job side. The first-order conditions with respect to h and y are

$$\Phi_h = u_c f_h + u_h = 0,$$

and

$$\Phi_y = u_c (f_y - V_y) = 0.$$

Since $u_c > 0$, the second condition implies

$$V_y = f_y$$

along the equilibrium allocation.

Differentiating the two first-order conditions with respect to x gives

$$\begin{pmatrix} \Phi_{hh} & \Phi_{hy} \\ \Phi_{yh} & \Phi_{yy} \end{pmatrix} \begin{pmatrix} h_x \\ \mu_x \end{pmatrix} + \begin{pmatrix} \Phi_{hx} \\ \Phi_{yx} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

⁸⁴The second-order condition (SOC) for the hours choice is

$$u_{cc}f_h^2 + (u_{ch} + u_{hc})f_h + u_c f_{hh} + u_{hh} < 0.$$

Using $V_y = f_y$, the relevant derivatives are

$$\begin{aligned}
\Phi_{hh} &= u_{cc}f_h^2 + 2u_{ch}f_h + u_cf_{hh} + u_{hh}, \\
\Phi_{hy} &= u_{cc}(f_y - V_y)f_h + u_cf_{hy} = u_cf_{hy} \\
\Phi_{yh} &= u_{cc}(f_y - V_y)f_h + u_cf_{yh} = u_cf_{yh} \\
\Phi_{yy} &= u_{cc}(f_y - V_y)(f_y - V_y) + u_c(f_{yy} - V_{yy}) = u_c(f_{yy} - V_{yy}) \\
\Phi_{hx} &= u_{cc}f_xf_h + u_cf_{xh} + u_{hc}f_x \\
\Phi_{yx} &= u_{cc}f_x(f_y - V_y) + u_cf_{xy} = u_cf_{xy}
\end{aligned}$$

Hence

$$\begin{pmatrix} u_{cc}f_h^2 + 2u_{ch}f_h + u_cf_{hh} + u_{hh} & u_cf_{hy} \\ u_cf_{yh} & u_c(f_{yy} - V_{yy}) \end{pmatrix} \begin{pmatrix} h_x \\ \mu_x \end{pmatrix} + \begin{pmatrix} u_{cc}f_xf_h + u_cf_{xh} + u_{hc}f_x \\ u_cf_{xy} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Solving for μ_x gives

$$\mu_x = \frac{u_cf_{hy}(u_{cc}f_xf_h + u_cf_{xh} + u_{hc}f_x) - u_cf_{xy}(u_{cc}f_h^2 + 2u_{ch}f_h + u_cf_{hh} + u_{hh})}{\begin{vmatrix} u_{cc}f_h^2 + 2u_{ch}f_h + u_cf_{hh} + u_{hh} & u_cf_{hy} \\ u_cf_{yh} & u_c(f_{yy} - V_{yy}) \end{vmatrix}}.$$

By optimality, the joint second-order condition implies that

$$\begin{vmatrix} u_{cc}f_h^2 + 2u_{ch}f_h + u_cf_{hh} + u_{hh} & u_cf_{hy} \\ u_cf_{yh} & u_c(f_{yy} - V_{yy}) \end{vmatrix} > 0.$$

which is to say the determinant of the Hessian of the joint hours and type decision needs to be positive.

Since $u_c > 0$, PAM, i.e. $\mu_x > 0$, is therefore equivalent to

$$f_{hy}(u_{cc}f_xf_h + u_cf_{xh} + u_{hc}f_x) - f_{xy}(u_{cc}f_h^2 + 2u_{ch}f_h + u_cf_{hh} + u_{hh}) > 0.$$

Equivalently,

$$f_{xy} \left[- \left(u_{cc}f_h^2 + 2u_{ch}f_h + u_cf_{hh} + u_{hh} \right) \right] + f_{hy} [u_{cc}f_xf_h + u_cf_{xh} + u_{hc}f_x] > 0.$$

Thus this derivation delivers the same primitive condition as the main derivation, and it is more similar to the one used in [Eckhout and Kircher \(2018\)](#). The usefulness of this

derivation is that it makes explicit the joint nature of the hours and matching problem. ■

Proof of Proposition 3 To prove the result, we use monotone comparative statics. Recall that the first order condition for hours choice:

$$u_c f_h + u_h = 0 \quad (\text{B.7})$$

Then by monotone comparative statics, to prove that h^* weakly increases in the worker type x , we need that:

$$\frac{\partial(u_c f_h + u_h)}{\partial x} > 0 \quad (\text{B.8})$$

Hence we need that

$$u_{cc} f_x f_h + u_c f_{xh} + u_{ch} f_x > 0 \quad (\text{B.9})$$

Using the FOC $u_c f_h + u_h = 0$ into (B.9), and rearranging terms, we get that

$$-f_h f_x \left(-\frac{u_{cc}}{u_c} + \frac{u_{ch}}{u_h} \right) + f_{xh} > 0 \quad (\text{B.10})$$

which proves the result. ■

Proof of Corollary 1 Condition (11) directly follows from (10) after assuming $u_{ch} = 0$ and rearranging terms. To derive (12), start from (11) and multiply both sides by $f - V$ and rearrange terms.

For the second part of the corollary, it is useful to remind that log-supermodularity of a continuous, twice-differentiable function f in x, y implies:⁸⁵

$$f_{xy} f \geq f_x f_y$$

Therefore, log-supermodularity of the term $f - V$ in the corollary implies

$$f_{xh}(f - V) \geq f_h f_x \quad (\text{B.11})$$

or

⁸⁵Log-supermodularity is a stronger condition than supermodularity if $f_x, f_y \geq 0$.

$$\frac{f_{xh}(f - V)}{f_h f_x} \geq 1 \quad (\text{B.12})$$

Hence, if $f - V$ is log-supermodular, then the RHS in Corollary 1 is bigger than the LHS - and the condition is satisfied - if the LHS is lower than 1. Using the budget constraint of the household $c = w$ and the fact that $f - V = w$ implies that the LHS is simply $\left(-c \frac{u_{cc}}{u_c}\right)$ which is the coefficient of relative risk aversion. It follows that if the coefficient of risk aversion is lower than 1, the condition is satisfied - and hours worked are increasing in the worker attributes - as long as $f - V$ is log-supermodular, i.e. $\left(-c \frac{u_{cc}}{u_c}\right) < 1$ is a sufficient condition for $h_x^* > 0$. ■

B.2 Additional Figures: Allocation and Comparative Statics

Equilibrium Allocation We now illustrate how the equilibrium allocation varies according to technology and preference parameters. For PAM allocation to be an equilibrium (for any type distribution), condition (8) must hold. The same condition with opposite sign gives NAM. If the condition holds for some points in the domain but not others, we define the allocation to be neither PAM nor NAM (NEITHER). Figure B.1 illustrates, for a grid of ρ and γ values, what type of allocation is supported by a given parameter combination. The left panel plots it for linear utility (no income effects, $\sigma = 0$) while the right panel plots the equilibrium allocations for the case in which $\sigma = 1.4$.⁸⁶

The Figure contains several insights into the model's workings. First, regardless of the value of σ , a PAM allocation is obtained when γ is not too strong relative to ρ (i.e. hours complementarities are not too strong relative to type-complementarities). This happens because when γ is large (in absolute value), hours become a powerful lever for low-skilled types to reach top jobs, therefore potentially undermining the x - y type complementarity that normally pushes towards PAM, and governed by ρ . Second, a change in σ from 0 to 1.4 can shift the equilibrium allocation and NEITHER region eats into the PAM region. This happens because with $\sigma = 1.4$, high-skilled are more likely to work less; when hours matter for the allocation ($f_{yh} > 0$), this implies high- x are less attractive for top jobs, and the allocation is pushed away from PAM. Lastly, for sufficiently negative values of ρ, γ ⁸⁷, a positive σ makes no difference, and the allocation is purely governed by the relative strength of the type complementarities..⁸⁸

⁸⁶Unless otherwise noted, the values for σ, ψ, θ in the utility function are those used in the quantitative application.

⁸⁷Recall that the more negative ρ and γ are, the stronger the complementarity.

⁸⁸Under the calibration used in the Figure, this happens when γ and ρ are around -2 or lower, in which case the line dividing the PAM region from the NEITHER region is essentially a diagonal.

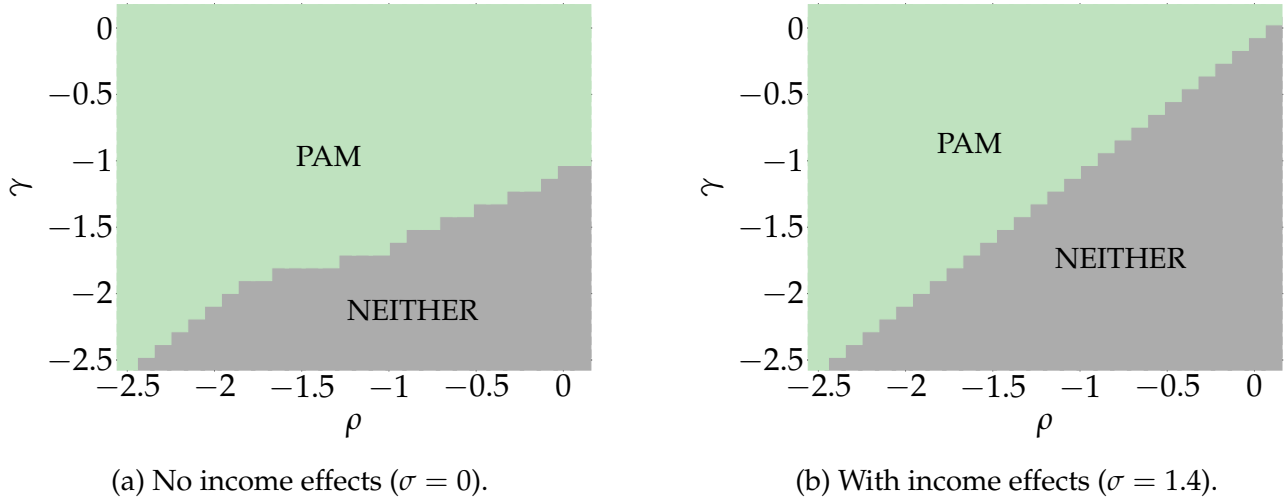


Figure B.1: Allocation regions in the (γ, ρ) space.

Notes: For both panels, green cells correspond to parameter combinations that satisfy PAM, while gray cells correspond to parameter combinations that do not. Panel (a) plots the resulting allocation for different combinations of ρ and γ when utility is linear in consumption ($\sigma = 0$). Panel (b) plots the resulting allocation for different combinations of ρ and γ when utility is concave in consumption and specifically when income effects exceed substitution effects ($\sigma = 1.4$). The remaining parameter values are those corresponding to year 2010 in the estimated model (see Figure 9).

Equilibrium Effects of lower ρ Figure B.2 plots the equilibrium effects of lower ρ , complementing the analysis in the main text.

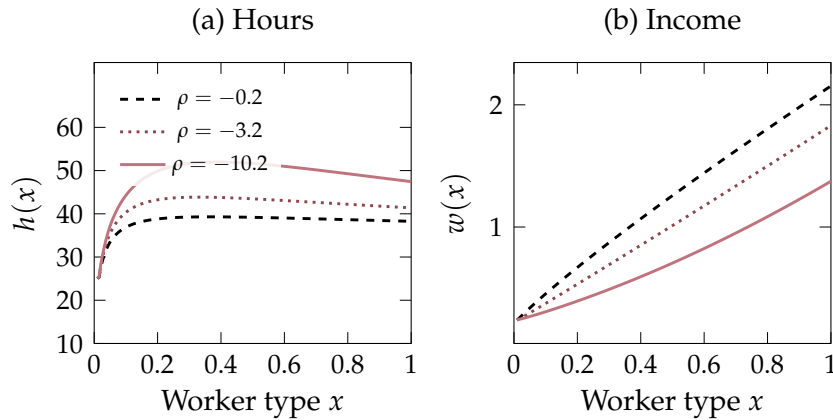


Figure B.2: Equilibrium policy functions for alternative values of ρ .

Notes: Income $w(x)$ is in thousands. The remaining parameter values are those corresponding to year 2010 in the estimated model (see Figure 9).

Decomposition of the PAM allocation for ρ and α The decomposition exercises for α and ρ are plotted in Figure B.3 and Figure B.4, respectively. Similarly to the case of γ , a lower ρ makes the income effect more negative for high-skilled workers; the substitution effect, however, barely moves at the top and only changes a bit at the bottom. By contrast, as shown

in the right panel, the sorting term increases, reinforcing the direct type-complementarity effect governed by ρ . Increasing α , on the other hand, operates mostly at the bottom of the distribution, where all terms are amplified. However, these effects jointly reduce the incentives to work at the top relative to the bottom, and therefore the hours function becomes more decreasing in skills as α increases, as shown in the main text.

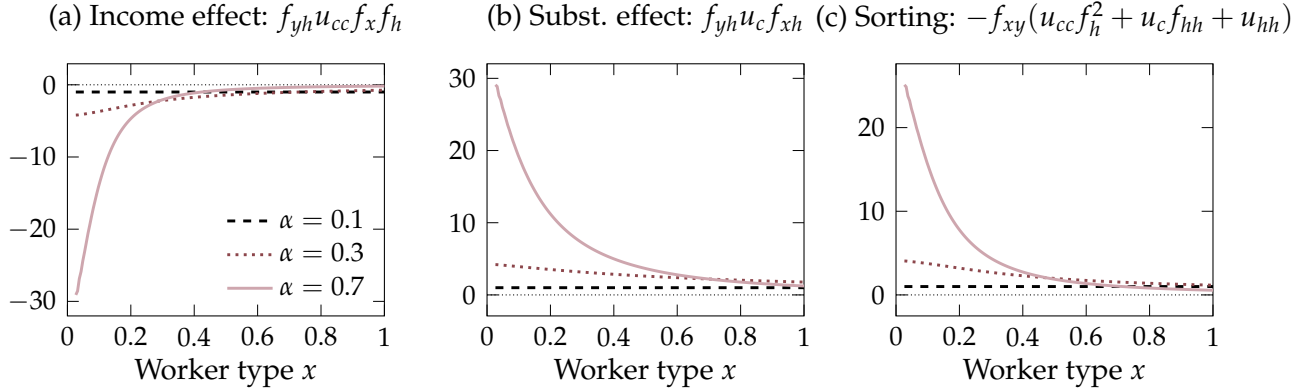


Figure B.3: Decomposition of (9) for alternative values of α .

Notes: Each panel plots, along the equilibrium matching path $y = \mu(x)$, one of the three terms that govern the sign of positive assortative matching from Figure 9 : the income effect $f_{yh} u_{cc} f_x f_h$ (Panel a), the substitution effect $f_{yh} u_c f_{xh}$ (Panel b), and the sorting term $-f_{xy}(u_{cc} f_h^2 + u_c f_{hh} + u_{hh})$ (Panel c). Each curve is normalized pointwise by the absolute value of the corresponding term evaluated at the baseline value of α , so the baseline (dashed) is flat at ± 1 and other curves show the force relative to baseline. The remaining parameter values are those corresponding to year 2010 in the estimated model (see Figure 9).

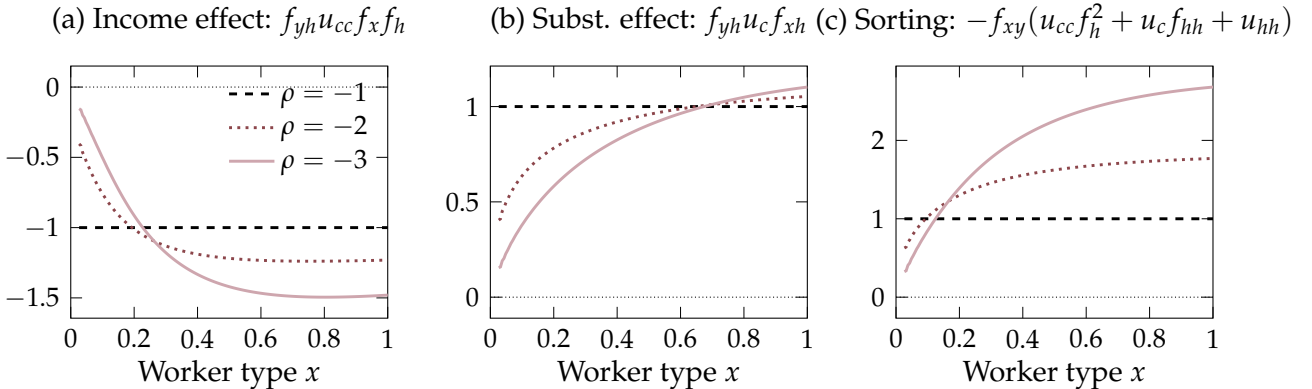


Figure B.4: Decomposition of (9) for alternative values of ρ .

Notes: Each panel plots, along the equilibrium matching path $y = \mu(x)$, one of the three terms that govern the sign of positive assortative matching from Figure 9 : the income effect $f_{yh} u_{cc} f_x f_h$ (Panel a), the substitution effect $f_{yh} u_c f_{xh}$ (Panel b), and the sorting term $-f_{xy}(u_{cc} f_h^2 + u_c f_{hh} + u_{hh})$ (Panel c). Each curve is normalized pointwise by the absolute value of the corresponding term evaluated at the baseline value of ρ , so the baseline (dashed) is flat at ± 1 and other curves show the force relative to baseline. The remaining parameter values are those corresponding to year 2010 in the estimated model (see Figure 9).

B.3 Special Cases: Proofs and Derivations

This section provides proofs and derivations used in the section on the model's link to existing literature.

B.3.1 Neoclassical Macro Frameworks

The goal is to show that in the present framework, the substitution effect is stronger in better matches when better jobs raise the marginal product of hours and f_h is log-supermodular in (y, h) . To prove this, fix a worker x and hold consumption $c(x)$ fixed. Define the conditional hours schedule $h(x, y)$ by

$$MRS(c(x), h) = f_h(x, y, h).$$

Differentiating with respect to job quality y gives

$$MRS_h h_y = f_{hy} + f_{hh} h_y,$$

so that

$$(MRS_h - f_{hh}) h_y = f_{hy}.$$

Hence, if

$$MRS_h - f_{hh} > 0 \quad \text{and} \quad f_{hy} > 0,$$

then

$$h_y = \frac{f_{hy}}{MRS_h - f_{hh}} > 0.$$

Thus, the substitution effect is such that better jobs induce higher hours (holding consumption fixed). Now define the proportional effect of job quality on the return to hours as

$$\eta(y, h) \equiv \partial_y \log f_h(x, y, h) = \frac{f_{hy}}{f_h}.$$

If f_h is log-supermodular in (y, h) , then

$$\eta_h(y, h) = \frac{\partial^2}{\partial y \partial h} \log f_h(x, y, h) \geq 0.$$

Therefore, for any fixed job quality y , the proportional response of the return to an extra hour to a marginal improvement in job quality is larger when hours are higher:

$$h' > h \implies \eta(y, h') \geq \eta(y, h).$$

Since better jobs induce higher hours (holding income fixed), and since log-supermodularity implies that the proportional return to job quality is larger at higher hours, the hours response amplifies the substitution effect of job quality on the return to an extra hour. This is the sense in which the substitution effect is stronger higher up the job ladder.

B.3.2 Efficiency Units

Assignment in the simple efficiency units model First, we prove that in an efficiency-units version of this paper's framework, PAM simply depends on the sign of $\tilde{f}_{zy} > 0$. Suppose production takes the form

$$f(x, y, h) = \tilde{f}(z, y), \quad z \equiv xh.$$

Assume separable utility,

$$\bar{u}(c) - u(h), \quad c = \tilde{f}(z, y) - V,$$

with the usual properties that $u_h > 0$ and $u_{hh} \geq 0$. By Equation (7), PAM obtains if

$$f_{xy} + f_{hy}h_x^* > 0.$$

Under $f(x, y, h) = \tilde{f}(xh, y)$,

$$\begin{aligned} f_x &= h\tilde{f}_z, \\ f_h &= x\tilde{f}_z, \\ f_{xy} &= h\tilde{f}_{zy}, \\ f_{hy} &= x\tilde{f}_{zy}. \end{aligned}$$

Hence the PAM condition becomes

$$\tilde{f}_{zy}(h + xh_x^*) > 0. \tag{B.13}$$

Now, defining equilibrium efficiency units

$$z^*(x, y) = xh^*(x, y).$$

Differentiating,

$$z_x^*(x, y) = h + xh_x^*.$$

Now, rewrite the worker's problem in terms of z :

$$\max_z \bar{u}(\tilde{f}(z, y) - V) - u(z/x).$$

The first-order condition is

$$\bar{u}_c(\tilde{f}(z, y) - V)\tilde{f}_z(z, y) = \frac{1}{x}u_h(z/x).$$

Define

$$H(z, x; y, V) \equiv \bar{u}_c(\tilde{f}(z, y) - V)\tilde{f}_z(z, y) - \frac{1}{x}u_h(z/x).$$

At the optimum, $H(z^*, x; y, V) = 0$. By the implicit function theorem,

$$z_x^* = -\frac{H_x}{H_z}.$$

Now

$$H_x = -\frac{\partial}{\partial x} \left(\frac{1}{x}v'(z/x) \right) = \frac{u_h(z/x) + (z/x)u_{hh}(z/x)}{x^2} > 0.$$

Under the second-order condition, $H_z < 0$. Therefore

$$z_x^* > 0.$$

Combining the previous expressions,

$$h + xh_x^* = z_x^* > 0.$$

Hence a sufficient condition for PAM, from (B.13), is

$$\tilde{f}_{zy} > 0.$$

The intuition is that in an exact efficiency-units model, workers differ only through the scalar $z = xh$. A more skilled worker can generate any given z with fewer hours, so the utility cost of supplying effective labor is lower. Therefore equilibrium efficiency units rise with skill, even if high-skill workers choose fewer hours. If better jobs value efficiency units more ($\tilde{f}_{zy} > 0$), PAM follows.

Next, we show that by itself, an extension to [Krusell *et al.* \(2000\)](#) (KORV) to have hours in production doesn't change the fact that preferences govern which skill type works more in equilibrium. Consider a framework as in KORV in which labor of type $i \in \{u, s\}$ (unskilled,

skilled) enters production through efficiency units

$$L_i = x_i N_i g(h_i),$$

where x_i is worker efficiency, N_i is the mass of workers of type i , and $g(h)$ is common across types. Let the representative firm price one unit of effective labor of type i at p_i , so that earnings of type i are

$$e_i(h_i) = p_i x_i g(h_i) \equiv \omega_i g(h_i), \quad \omega_i \equiv p_i x_i.$$

This is still an efficiency-units model, since the firm only cares about the bundle $x_i g(h_i)$.

Now suppose each type solves

$$\max_{h_i} u(c_i) - v(h_i) \quad \text{subject to} \quad c_i = \omega_i g(h_i),$$

and let

$$u(c) = \frac{c^{1-\sigma} - 1}{1-\sigma}.$$

The first-order condition is

$$u'(\omega_i g(h_i)) \omega_i g'(h_i) = v'(h_i).$$

Since $u'(c) = c^{-\sigma}$,

$$(\omega_i g(h_i))^{-\sigma} \omega_i g'(h_i) = v'(h_i),$$

or equivalently

$$\omega_i^{1-\sigma} g(h_i)^{-\sigma} g'(h_i) = v'(h_i).$$

Define

$$\Phi(h, \omega) \equiv \omega^{1-\sigma} g(h)^{-\sigma} g'(h) - v'(h).$$

At the optimum, $\Phi(h^*(\omega), \omega) = 0$. By the implicit function theorem,

$$\frac{dh^*}{d\omega} = -\frac{\Phi_\omega}{\Phi_h}.$$

Now

$$\Phi_\omega = (1-\sigma) \omega^{-\sigma} g(h)^{-\sigma} g'(h).$$

Hence the sign of Φ_ω is the sign of $1-\sigma$. Under the second-order condition, $\Phi_h < 0$, so $\frac{dh^*}{d\omega}$ has the same sign as $1-\sigma$.

Therefore:

- if $\sigma < 1$, higher-wage / higher-skill workers work more;
- if $\sigma = 1$, hours do not vary with wages;
- if $\sigma > 1$, higher-wage / higher-skill workers work less.

Thus, in the canonical decentralized KORV-with-hours benchmark with a common hours technology $g(h)$, who works more depends only on the preference parameter σ .⁸⁹

Embodied technical change inside the CES specification. We now show in what sense the current model can capture embodied technical change as in [Krusell *et al.* \(2000\)](#) through changes in A, α, β for given γ and ρ . The importance of this derivation is not only to clarify the meaning of these parameters in this context relative to papers where technological change is modeled as embodied in capital, but also to show the usefulness of allowing for flexible complementarity patterns, captured by γ and ρ .

Consider the production function

$$f(x, y, h) = A \left[\beta (\alpha x^\rho + (1 - \alpha) y^\rho)^{\gamma/\rho} + (1 - \beta) h^\gamma \right]^{1/\gamma}.$$

Suppose technical change acts on the job side through a multiplicative shifter q_t , so that production (13) becomes

$$f_t(x, y, h) = A \left[\beta (\alpha x^\rho + (1 - \alpha) (q_t y)^\rho)^{\gamma/\rho} + (1 - \beta) h^\gamma \right]^{1/\gamma}.$$

Define

$$\kappa_t = \alpha + (1 - \alpha) q_t^\rho, \quad \alpha_t = \frac{\alpha}{\kappa_t}, \quad 1 - \alpha_t = \frac{(1 - \alpha) q_t^\rho}{\kappa_t}.$$

Then

$$\alpha x^\rho + (1 - \alpha) (q_t y)^\rho = \kappa_t (\alpha_t x^\rho + (1 - \alpha_t) y^\rho),$$

so

$$(\alpha x^\rho + (1 - \alpha) (q_t y)^\rho)^{\gamma/\rho} = \kappa_t^{\gamma/\rho} (\alpha_t x^\rho + (1 - \alpha_t) y^\rho)^{\gamma/\rho}.$$

Hence

$$f_t(x, y, h) = A \left[\beta \kappa_t^{\gamma/\rho} (\alpha_t x^\rho + (1 - \alpha_t) y^\rho)^{\gamma/\rho} + (1 - \beta) h^\gamma \right]^{1/\gamma}.$$

Now define

$$D_t = \beta \kappa_t^{\gamma/\rho} + (1 - \beta), \quad A_t = A D_t^{1/\gamma}, \quad \beta_t = \frac{\beta \kappa_t^{\gamma/\rho}}{D_t}.$$

⁸⁹Of course, this conclusion can change if the model is extended to have e.g. insurance motives, or if one imposes exogenous type-specific hour technologies $g_s(h) \neq g_u(h)$.

Then

$$1 - \beta_t = \frac{1 - \beta}{D_t},$$

and therefore

$$f_t(x, y, h) = A_t \left[\beta_t (\alpha_t x^\rho + (1 - \alpha_t) y^\rho)^{\gamma/\rho} + (1 - \beta_t) h^\gamma \right]^{1/\gamma}.$$

This shows that embodied technical change on the job side (q_t) can be mapped directly in this framework as time variation in A_t, α_t, β_t , holding ρ and γ fixed.

Why ρ and γ govern the return to hours relative to skill across jobs. Let

$$C(x, y) = (\alpha x^\rho + (1 - \alpha) y^\rho)^{1/\rho},$$

so that

$$f(x, y, h) = A \left[\beta C(x, y)^\gamma + (1 - \beta) h^\gamma \right]^{1/\gamma}.$$

Differentiating with respect to hours,

$$f_h = A(1 - \beta) h^{\gamma-1} \left[\beta C(x, y)^\gamma + (1 - \beta) h^\gamma \right]^{1/\gamma-1}.$$

Next, we have that

$$f_x = A\beta\alpha x^{\rho-1} C(x, y)^{\gamma-\rho} \left[\beta C(x, y)^\gamma + (1 - \beta) h^\gamma \right]^{1/\gamma-1}.$$

Taking the ratio,

$$\frac{f_h}{f_x} = \frac{1 - \beta}{\beta\alpha} h^{\gamma-1} x^{1-\rho} C(x, y)^{\rho-\gamma}.$$

Hence

$$\log\left(\frac{f_h}{f_x}\right) = \log\left(\frac{1 - \beta}{\beta\alpha}\right) + (\gamma - 1) \log h + (1 - \rho) \log x + (\rho - \gamma) \log C(x, y).$$

Since

$$\log C(x, y) = \frac{1}{\rho} \log(\alpha x^\rho + (1 - \alpha) y^\rho),$$

we obtain

$$\frac{\partial \log C(x, y)}{\partial y} = \frac{(1 - \alpha) y^{\rho-1}}{\alpha x^\rho + (1 - \alpha) y^\rho}.$$

Therefore

$$\frac{\partial}{\partial y} \log\left(\frac{f_h}{f_x}\right) = (\rho - \gamma) \frac{(1 - \alpha) y^{\rho-1}}{\alpha x^\rho + (1 - \alpha) y^\rho}.$$

This shows that:

- if $\rho = \gamma$, the trade-off between hours and skill is job-invariant;
- if $\rho \neq \gamma$, the return to hours relative to skill varies systematically across jobs.

Hence ρ and γ are the parameters that determine whether the hours-skill trade-off is job-specific.

B.3.3 Bundle models

Suppose utility is separable in consumption and hours, so that $u_{ch} = 0$, and let

$$f(x, y, h) = \tilde{f}(g(x, h), y).$$

Then

$$\begin{aligned} f_x &= \tilde{f}_g g_x, \\ f_h &= \tilde{f}_g g_h, \\ f_{xy} &= \tilde{f}_{gy} g_x, \\ f_{hy} &= \tilde{f}_{gy} g_h, \\ f_{xh} &= \tilde{f}_{gg} g_x g_h + \tilde{f}_g g_{xh}, \\ f_{hh} &= \tilde{f}_{gg} g_h^2 + \tilde{f}_g g_{hh}. \end{aligned}$$

Recall that, under separable utility,

$$h_x = \frac{u_{cc} f_x f_h + u_c f_{xh}}{\varphi},$$

where, because of optimality

$$\varphi = -u_{cc} f_h^2 - u_c f_{hh} - u_{hh} > 0.$$

Condition (9) can be written as

$$f_{xy} \varphi + f_{hy} (u_{cc} f_x f_h + u_c f_{xh}) > 0.$$

Substituting the expressions above gives

$$f_{xy} \varphi + f_{hy} (u_{cc} f_x f_h + u_c f_{xh}) = \tilde{f}_{gy} [g_x \varphi + g_h (u_{cc} f_x f_h + u_c f_{xh})].$$

Using the formulas for f_x , f_h , f_{xh} , and f_{hh} ,

$$\begin{aligned} g_x \varphi + g_h (u_{cc} f_x f_h + u_c f_{xh}) &= g_x \left[-u_{cc} \tilde{f}_g^2 g_h^2 - u_c (\tilde{f}_{gg} g_h^2 + \tilde{f}_g g_{hh}) - u_{hh} \right] \\ &\quad + g_h \left[u_{cc} \tilde{f}_g^2 g_x g_h + u_c (\tilde{f}_{gg} g_x g_h + \tilde{f}_g g_{xh}) \right]. \\ &= -g_x u_{hh} + u_c \tilde{f}_g (g_h g_{xh} - g_x g_{hh}). \end{aligned}$$

Therefore,

$$f_{xy} \varphi + f_{hy} (u_{cc} f_x f_h + u_c f_{xh}) = \tilde{f}_{gy} [-g_x u_{hh} + u_c \tilde{f}_g (g_h g_{xh} - g_x g_{hh})].$$

If $u_c > 0$, $\tilde{f}_g > 0$, $\tilde{f}_{gy} > 0$, $g_x > 0$, and $u_{hh} < 0$, a sufficient condition for Condition (8) to hold is

$$g_h g_{xh} - g_x g_{hh} > 0,$$

that is,

$$g_{hh} < \frac{g_h}{g_x} g_{xh}.$$

For the hours condition, Proposition (3) becomes (recalling that for simplicity we assumed $u_{ch} = 0$)

$$u_{cc} f_x f_h + u_c f_{xh} > 0.$$

Substituting the same derivatives yields

$$u_{cc} \tilde{f}_g^2 g_x g_h + u_c \tilde{f}_{gg} g_x g_h + u_c \tilde{f}_g g_{xh} > 0.$$

Dividing by $u_c \tilde{f}_g g_x g_h > 0$, we obtain

$$\frac{g_{xh}}{g_x g_h} > \tilde{f}_g \left(-\frac{u_{cc}}{u_c} \right) - \frac{\tilde{f}_{gg}}{\tilde{f}_g}.$$

This is the condition for hours to be increasing in skills in the bundle case. If $\tilde{f}_{gg} \leq 0$, then the right-hand side is positive, which implies that $g_{xh} > 0$ is necessary for $h_x > 0$.

B.3.4 Multiplicatively separable hours.

Suppose production is

$$f(x, y, h) = \bar{f}(x, y) z(y, h),$$

and preferences are separable CRRA,

$$u(c, h) = \frac{c^{1-\sigma}}{1-\sigma} - u(h), \quad c = f(x, y, h) - V(y),$$

with $V(y) \geq 0$. The relevant derivatives are

$$\begin{aligned} f_x &= \bar{f}_x z, \\ f_h &= \bar{f}_z z_h, \\ f_{xh} &= \bar{f}_{xz} z_h, \\ f_{xy} &= \bar{f}_{xy} z + \bar{f}_x z_{y}, \\ f_{hy} &= \bar{f}_y z_h + \bar{f}_z z_{hy}, \\ f_{hh} &= \bar{f}_z z_{hh}. \end{aligned}$$

By Corollary (1), $h_x > 0$ if and only if

$$\sigma < \frac{f_{xh}(f - V)}{f_x f_h}.$$

Under multiplicative separability,

$$\frac{f_{xh}(f - V)}{f_x f_h} = \frac{\bar{f}_x z_h (f - V)}{(\bar{f}_x z)(\bar{f}_z z_h)} = \frac{f - V}{f} \leq 1,$$

where the last inequality uses $V(y) \geq 0$. Hence, $h_x > 0$ is impossible for any $\sigma \geq 1$.

For sorting, Proposition (1) implies that PAM obtains if and only if

$$f_{xy} + f_{hy} h_x > 0.$$

Substituting the derivatives above gives

$$\bar{f}_{xy} z + \bar{f}_x z_y + (\bar{f}_y z_h + \bar{f}_z z_{hy}) h_x > 0.$$

This is the PAM condition stated in the text. In the more restrictive subcase $z(y, h) = z(h)$, it simplifies to

$$\bar{f}_{xy} z + \bar{f}_y z_h h_x > 0.$$

B.3.5 Supervision-time microfoundation with endogenous technology

This subsection shows how a simplified one-layer knowledge hierarchy in the spirit of [Garicano \(2000\)](#) and [Garicano and Rossi-Hansberg \(2006\)](#) can be embedded in the general production environment $f(x, y, h)$ used in the text, and how a simple extension with endogenous supervision technology provides a microfoundation for the technological change in hours complementarities similar to those emphasized in the main text.

Baseline supervision structure. Consider a worker of skill x who solves a fraction $F(x)$ of problems on her own and escalates the remaining share $1 - F(x)$ to the managerial side of the match, indexed by y . Each escalated problem requires $\tau > 0$ units of supervisory time. If a team of n workers is active for h hours⁹⁰, total supervisory time required over the shift is $\tau n h [1 - F(x)]$, while total supervisory time available is h . Equivalently, in per-hour terms, the supervision constraint is

$$1 \geq \tau n [1 - F(x)].$$

Hence feasible span is

$$n(x) = \frac{1}{\tau [1 - F(x)]}.$$

If each supervised worker-hour produces value y , total output over h hours is

$$f^{GRH}(x, y, h) = y n(x) h = \frac{y h}{\tau [1 - F(x)]}.$$

Defining

$$A(x, y) \equiv \frac{y}{1 - F(x)}$$

this can be written as

$$f^{GRH}(x, y, h) = \frac{1}{\tau} A(x, y) h.$$

The derivatives of $A(x, y)$ are

$$A_y(x, y) = \frac{1}{1 - F(x)} \quad A_x(x, y) = \frac{y \varphi(x)}{[1 - F(x)]^2} \quad A_{xy}(x, y) = \frac{\varphi(x)}{[1 - F(x)]^2}$$

where $\varphi(x) = F'(x)$. It follows that

$$f_{yh}^{GRH}(x, y, h) = \frac{1}{\tau [1 - F(x)]} > 0,$$

⁹⁰Here, h should be interpreted as the duration for which the entire worker-supervisor production unit is active. Thus, when the team works longer, both the flow of problems requiring help and the available supervisory time scale proportionally with h .

$$f_{xh}^{GRH}(x, y, h) = \frac{y\varphi(x)}{\tau[1 - F(x)]^2} > 0,$$

$$f_{xy}^{GRH}(x, y, h) = \frac{h\varphi(x)}{\tau[1 - F(x)]^2} > 0.$$

Therefore, the baseline supervision-time model (stripped down of the layers' endogenous decision) can be seen as a special case of the general environment $f(x, y, h)$ and directly generates the complementarity patterns used in the main text to characterize the equilibrium allocations. In particular, in this model, these complementarities are positive.

Endogenous supervision technology. To connect the supervision structure to technological change, we introduce a richer extension of the original [Garicano and Rossi-Hansberg \(2006\)](#) framework. Suppose now that the firm can choose an internal supervision technology H . Let the supervisory time required for each escalated problem be

$$\tau(H, \lambda) = \frac{\tau}{1 + \lambda H}$$

where $\lambda > 0$ indexes the technology frontier. A larger λ means that a given unit of supervision technology H is more effective at reducing supervisory time. Assume the firm pays a convex cost

$$v(H) = \frac{\kappa}{2}H^2, \quad \kappa > 0.$$

With supervision technology H , feasible span per hour becomes

$$n(x, H, \lambda) = \frac{1 + \lambda H}{\tau[1 - F(x)]}.$$

Hence, if the match is active for h hours, output net of technology cost is

$$\tilde{q}(x, y, h, H; \lambda) = y n(x, H, \lambda) h - \frac{\kappa}{2}H^2 = \frac{y h}{\tau[1 - F(x)]}(1 + \lambda H) - \frac{\kappa}{2}H^2.$$

Using $A(x, y) = y/[1 - F(x)]$, this becomes

$$\tilde{q}(x, y, h, H; \lambda) = \frac{A(x, y)h}{\tau}(1 + \lambda H) - \frac{\kappa}{2}H^2.$$

For a given match (x, y) and hours choice h , the firm chooses H to maximize

$$f(x, y, h; \lambda) = \max_{H \geq 0} \left\{ \frac{A(x, y)h}{\tau}(1 + \lambda H) - \frac{\kappa}{2}H^2 \right\}.$$

The first-order condition is

$$\frac{A(x, y)h}{\tau}\lambda - \kappa H = 0,$$

so the optimal supervision technology is

$$H^*(x, y, h; \lambda) = \frac{\lambda A(x, y)h}{\tau\kappa}.$$

Substituting back gives the reduced-form production function

$$f(x, y, h; \lambda) = \frac{A(x, y)h}{\tau} + \frac{\lambda^2}{2\tau^2\kappa} (A(x, y)h)^2.$$

Hours complementarities. Differentiating with respect to h yields

$$f_h(x, y, h; \lambda) = \frac{A(x, y)}{\tau} + \frac{\lambda^2}{\tau^2\kappa} A(x, y)^2 h.$$

Hence

$$f_{yh}(x, y, h; \lambda) = \frac{A_y(x, y)}{\tau} + \frac{2\lambda^2}{\tau^2\kappa} A(x, y) A_y(x, y) h = \left[\frac{1}{\tau} + \frac{2\lambda^2 A(x, y) h}{\tau^2\kappa} \right] A_y(x, y),$$

and

$$f_{xh}(x, y, h; \lambda) = \frac{A_x(x, y)}{\tau} + \frac{2\lambda^2}{\tau^2\kappa} A(x, y) A_x(x, y) h = \left[\frac{1}{\tau} + \frac{2\lambda^2 A(x, y) h}{\tau^2\kappa} \right] A_x(x, y).$$

Using the expressions for A_y and A_x above,

$$f_{yh}(x, y, h; \lambda) = \left[\frac{1}{\tau} + \frac{2\lambda^2 A(x, y) h}{\tau^2\kappa} \right] \frac{1}{1 - F(x)} > 0,$$

$$f_{xh}(x, y, h; \lambda) = \left[\frac{1}{\tau} + \frac{2\lambda^2 A(x, y) h}{\tau^2\kappa} \right] \frac{y\varphi(x)}{[1 - F(x)]^2} > 0.$$

Likewise,

$$f_{xy}(x, y, h; \lambda) = \frac{h}{\tau} A_{xy}(x, y) + \frac{\lambda^2 h^2}{\tau^2\kappa} \left(A_x(x, y) A_y(x, y) + A(x, y) A_{xy}(x, y) \right) > 0.$$

Each of the cross-partial above decomposes into two pieces. The first piece, A_y/τ , A_x/τ , and hA_{xy}/τ , respectively, is the baseline GRH complementarity, independent of λ . The second piece, $2\lambda^2 AA_y h/(\tau^2\kappa)$, $2\lambda^2 AA_x h/(\tau^2\kappa)$, and $\lambda^2 h^2(A_x A_y + AA_{xy})/(\tau^2\kappa)$, is the contribution of endogenous supervision technology and is scaled by λ^2 . The first piece is unchanged by improvements in the technology frontier; the second piece grows with λ and is

what drives the rise in f_{yh} , f_{xh} , and f_{xy} documented below. In this sense, the model has two layers: a GRH baseline that pins down the sign of the complementarities, and a technology-choice layer that determines how those complementarities respond to technological change.

Increasing complementarities and technological change. The role of the frontier parameter λ is now explicit. Differentiating the cross-partials with respect to λ gives

$$\frac{\partial f_{yh}}{\partial \lambda} = \frac{4\lambda A(x, y)h}{\tau^2 \kappa} A_y(x, y) = \frac{4\lambda A(x, y)h}{\tau^2 \kappa} \frac{1}{1 - F(x)} > 0,$$

and

$$\frac{\partial f_{xh}}{\partial \lambda} = \frac{4\lambda A(x, y)h}{\tau^2 \kappa} A_x(x, y) = \frac{4\lambda A(x, y)h}{\tau^2 \kappa} \frac{y\varphi(x)}{[1 - F(x)]^2} > 0.$$

Similarly,

$$\frac{\partial f_{xy}}{\partial \lambda} = \frac{2\lambda h^2}{\tau^2 \kappa} \left(A_x(x, y)A_y(x, y) + A(x, y)A_{xy}(x, y) \right) > 0.$$

Therefore, an improvement in the supervision technology frontier λ increases the strength of the hours complementarities f_{yh} and f_{xh} , as well as the type-complementarity f_{xy} . The intuition is simple: when supervision technology becomes more effective, managerial attention stretches further. This raises the return to an additional hour particularly in high-value matches, where workers require little help and the managerial side of the match is especially productive. Importantly, the purpose of this extension was not to derive the exact nested-CES specification used in the quantitative section. Rather, it provides a simple supervision-based microfoundation for two ideas emphasized in the text: first, hours are more valuable in better matches; second, improvements in organizational and communication technologies can increase the strength of these complementarities over time.

B.4 Mechanism Details

Linking Global and Local Returns to hours The model mechanism sheds light on the link between the nature of hours in production, summarized by the complementarity parameter γ , and the convexity of earnings in hours. This link operates through sorting. To see this, recall that

$$f(x, y, h) = A \left[\beta C(x, y)^\gamma + (1 - \beta)h^\gamma \right]^{1/\gamma}, \quad C(x, y) = (\alpha x^\rho + (1 - \alpha)y^\rho)^{1/\rho}.$$

Define

$$S(x, y, h) \equiv \beta C(x, y)^\gamma + (1 - \beta)h^\gamma.$$

Then the marginal product of hours is

$$f_h(x, y, h) = A(1 - \beta)h^{\gamma-1}S^{1/\gamma-1}.$$

Differentiate with respect to the skill-occupation bundle C:

$$f_{hC} = A\beta(1 - \beta)(1 - \gamma)h^{\gamma-1}C^{\gamma-1}S^{1/\gamma-2}.$$

So if $\gamma < 1$, and in particular if $\gamma < 0$,

$$f_{hC} > 0.$$

This is the first key point: better occupations, or better skill-job matches, have a higher marginal product of hours.

Now, consider the role of sorting. Let equilibrium assignment be $y = \mu(x)$, and assume hours are increasing in skill, so that higher-hour workers are also higher-skill workers in better occupations. Then along the equilibrium allocation you can write

$$x = \eta(h), \quad y = \mu(\eta(h)),$$

and therefore

$$C(h) \equiv C(\eta(h), \mu(\eta(h))).$$

With positive assortative matching and $h_x^* > 0$,

$$C'(h) > 0.$$

In fact, since

$$C(h) = C(\eta(h), \mu(\eta(h))),$$

by the chain rule,

$$C'(h) = C_x(\eta(h), \mu(\eta(h))) \eta'(h) + C_y(\eta(h), \mu(\eta(h))) \mu'(\eta(h)) \eta'(h).$$

Using $\eta'(h) = 1/h_x(\eta(h))$, this becomes

$$C'(h) = \frac{C_x + C_y \mu'(\eta(h))}{h_x(\eta(h))}.$$

Since f_h depends on (x, y) only through $C(x, y)$,

$$f_{xh} = f_{hC}C_x, \quad f_{yh} = f_{hC}C_y,$$

so equivalently

$$C'(h) = \frac{f_{xh} + f_{yh}\mu'(\eta(h))}{f_{hC}h_x(\eta(h))}.$$

Under positive assortative matching, $\mu'(x) > 0$, and if hours are increasing in skill, $h_x(x) > 0$; since $C_x, C_y > 0$, this implies $C'(h) > 0$.

Define the equilibrium marginal return to hours

$$m(h) \equiv f_h(C(h), h).$$

Then

$$m'(h) = f_{hh} + f_{hC}C'(h). \tag{B.14}$$

- f_{hh} is the within-occupation curvature of the hours return.
- $f_{hC}C'(h)$ is the sorting amplification term.

Because $f_{hC} > 0$ and $C'(h) > 0$, sorting makes the return to hours rise as one moves to workers with higher hours. In other words, even if within a given occupation the marginal return to hours is flat or diminishing, the global earnings-hours schedule can be much steeper at the top because higher-hour workers are endogenously in better occupations.

C Estimation Appendix

C.1 Estimation Details

Algorithm We minimize the SMM objective with a two-stage global–local procedure, separately for each year. In the global stage, we evaluate the objective at $N_1 = 2 \times 10^5$ low-discrepancy Sobol points spread quasi-uniformly over the parameter space; this gives broad coverage of the domain and avoids the clustering of purely random draws, locating the most promising regions. In the local stage, the $N^* = 10$ Sobol points with the lowest objective are used as starting values for a derivative-free Nelder–Mead simplex search, and we retain the parameter vector attaining the lowest objective across these restarts.

Weighting matrix and moment covariance The sampling uncertainty of these moments is estimated by bootstrap. In each bootstrap replication, I resample individuals from the cleaned CPS-ORG sample, recompute the same moment vector using CPS weights and the

same sample restrictions and trimming rules as in the baseline moment construction, and then compute the covariance matrix across bootstrap replications. Denote this full bootstrap covariance matrix by

$$\widehat{C}_t = \widehat{\text{Var}}(\bar{m}_t).$$

The weighting matrix used in estimation is diagonal and uses the bootstrap variances of the targeted moments:

I then set

$$\Omega_{t,jj} = \frac{1}{\widehat{\text{Var}}(\bar{m}_{jt})}$$

Thus, the estimation objective uses a diagonal inverse-variance weighting matrix, while the standard-error calculation below uses the full bootstrap covariance matrix $\widehat{C}_t$. All objects are expressed in the same units and moment scaling as the SMM objective.

SMM diagnostic checks For each year, I compute the numerical Jacobian of the vector of targeted model moments with respect to the estimated parameters,

$$J_t = \frac{\partial m_t(\theta)}{\partial \theta'}.$$

Since the SMM criterion weights moments by the matrix Ω_t , the relevant object for local identification is the weighted Jacobian $\sqrt{\Omega_t}J_t$. The rank tests indicate that $\sqrt{\Omega_t}J_t$ has full column rank in every year, so the six estimated parameters satisfy the local rank condition given the seven targeted moments, as shown in Tables C.1, C.2, C.3, and C.4.⁹¹ Because the productivity scale parameter A is measured in levels, I also report the same diagnostic after rescaling the A -column of the Jacobian to correspond to derivatives with respect to $\log A$. This rescaling does not change the model, but it makes the numerical conditioning of the local rank test easier to interpret.

Table C.1: Local rank condition, year $t = 1980$

Jacobian	Moments	Parameters	Rank	Min. singular value
$\sqrt{\Omega}J$	7	6	6	0.00224
$\sqrt{\Omega}J_{\log A}$	7	6	6	0.31402

Notes: The table reports the numerical rank of the weighted Jacobian of targeted moments with respect to estimated parameters. Full column rank requires rank equal to the number of estimated parameters. The second row rescales the A column to correspond to derivatives with respect to $\log A$.

⁹¹I report only selected years for brevity, but the results are analogous for every year used in the estimation.

Table C.2: Local rank condition, year $t = 1990$

Jacobian	Moments	Parameters	Rank	Min. singular value
$\sqrt{\Omega}J$	7	6	6	0.00162
$\sqrt{\Omega}J_{\log A}$	7	6	6	0.86284

Notes: The table reports the numerical rank of the weighted Jacobian of targeted moments with respect to estimated parameters. Full column rank requires rank equal to the number of estimated parameters. The second row rescales the A column to correspond to derivatives with respect to $\log A$.

Table C.3: Local rank condition, year $t = 2000$

Jacobian	Moments	Parameters	Rank	Min. singular value
$\sqrt{\Omega}J$	7	6	6	0.00185
$\sqrt{\Omega}J_{\log A}$	7	6	6	3.2112

Notes: The table reports the numerical rank of the weighted Jacobian of targeted moments with respect to estimated parameters. Full column rank requires rank equal to the number of estimated parameters. The second row rescales the A column to correspond to derivatives with respect to $\log A$.

Table C.4: Local rank condition, year $t = 2010$

Jacobian	Moments	Parameters	Rank	Min. singular value
$\sqrt{\Omega}J$	7	6	6	0.00166
$\sqrt{\Omega}J_{\log A}$	7	6	6	4.11713

Notes: The table reports the numerical rank of the weighted Jacobian of targeted moments with respect to estimated parameters. Full column rank requires rank equal to the number of estimated parameters. The second row rescales the A column to correspond to derivatives with respect to $\log A$.

I also report [Andrews, Gentzkow and Shapiro \(2017\)](#) sensitivity measures. Given the moment condition

$$g_t(\theta) = m_t(\theta) - \bar{m}_t,$$

the local sensitivity of the estimated parameters to perturbations in the data moments is computed in a reparameterized space used for the standard errors (see below), to deal with the fact that because the estimated A_t is very large, $J_t' \Omega_t J_t$ is mechanically poorly conditioned, as shown above. This is then mapped back to the original parameter units. Let

$$z_t = (\log A_t, \alpha_t, \beta_t, \gamma_t, \rho_t, b_t^y), \quad G_t = \frac{\partial \theta_t}{\partial z_t'} = \text{diag}(A_t, 1, 1, 1, 1, 1),$$

and let $J_t^z = J_t G_t$. The sensitivity matrix reported in the tables is⁹²

$$\Lambda_t = G_t (J_t^{z'} \Omega_t J_t^z)^{-1} J_t^{z'} \Omega_t. \quad (\text{C.1})$$

Each entry of Λ_t measures how much a parameter estimate would change, locally, in response to a change in one data moment, holding fixed the local SMM first-order conditions. To make sensitivities comparable across moments measured in different units, I focus on standardized sensitivities, which scale each column by the bootstrap standard deviation of the corresponding data moment. The results show that the most influential moments are wage dispersion and wage-hours moments rather than level moments. In particular, $\text{sd}(\log w)$, w_{90}/w_{10} , and w_{90}/w_{50} are central for the distributional and curvature parameters, while the hours-by-wage moments are especially important for γ , ρ , and b_y . This indicates that the joint wage-hours profile helps discipline the technology and assignment parameters. Results for selected years are reported in Tables C.5, C.6, C.7, C.8.

Table C.5: Top-3 sensitivity moments by parameter, year $t = 1980$.

Parameter	Top 1	Top 2	Top 3
A	sd'log'w	w90w50	h'wdec2
α	w90w10	w90w50	h'wdec5
β	w90w10	w90w50	h'wdec5
γ	h'wdec9	sd'log'w	w90w10
ρ	w90w10	h'wdec5	sd'log'w
b_y	w90w10	h'wdec5	sd'log'w

Notes: The table reports, for each parameter, the three moments with the largest absolute standardized sensitivity in year 1980, computed using C.1.

⁹²When the raw-parameter matrix is nonsingular, this is algebraically equivalent to $(J_t' \Omega_t J_t)^{-1} J_t' \Omega_t$; the reparameterization only improves numerical conditioning.

Table C.6: Top-3 sensitivity moments by parameter, year $t = 1990$.

Parameter	Top 1	Top 2	Top 3
A	w90w50	sd'log'w	h'wdec5
α	w90w50	sd'log'w	h'wdec5
β	w90w50	sd'log'w	h'wdec2
γ	h'wdec5	h'wdec9	w90w10
ρ	h'wdec5	w90w10	h'wdec2
b_y	h'wdec5	w90w10	h'wdec2

Notes: The table reports, for each parameter, the three moments with the largest absolute standardized sensitivity in year 1990, computed using C.1.

Table C.7: Top-3 sensitivity moments by parameter, year $t = 2000$.

Parameter	Top 1	Top 2	Top 3
A	w90w10	w90w50	h'wdec5
α	w90w10	h'wdec5	h'wdec2
β	w90w10	h'wdec5	w90w50
γ	h'wdec9	sd'log'w	h'wdec5
ρ	w90w10	h'wdec5	sd'log'w
b_y	h'wdec5	w90w10	sd'log'w

Notes: The table reports, for each parameter, the three moments with the largest absolute standardized sensitivity in year 2000, computed using C.1.

Table C.8: Top-3 sensitivity moments by parameter, year $t = 2010$.

Parameter	Top 1	Top 2	Top 3
A	w90w50	h'wdec5	h'wdec9
α	w90w50	h'wdec5	w90w10
β	w90w50	h'wdec5	h'wdec9
γ	h'wdec9	h'wdec2	w90w50
ρ	h'wdec5	h'wdec9	h'wdec2
b_y	h'wdec5	h'wdec2	h'wdec9

Notes: The table reports, for each parameter, the three moments with the largest absolute standardized sensitivity in year 2010, computed using C.1.

Standard Errors SMM standard errors are computed using

$$\widehat{\text{Var}}(\hat{\theta}_t) = (J_t' \Omega_t J_t)^{-1} J_t' \Omega_t \widehat{C}_t \Omega_t J_t (J_t' \Omega_t J_t)^{-1},$$

where $\widehat{C}_t$ is the full bootstrap covariance matrix of the data moments. The weighting matrix Ω_t is the diagonal inverse-variance matrix used in estimation, constructed from the diagonal of $\widehat{C}_t$. Thus, Ω_t uses only the bootstrap variances, while the sandwich formula uses the full covariance matrix of the data moments.⁹³

C.2 Additional Model Validation

Figure C.1 plots average hours worked by wage decile for two selected years. Overall, the Figure shows that the model tracks data closely, with the exception of the 10th decile in 2018, for which the model underpredicts hours worked.

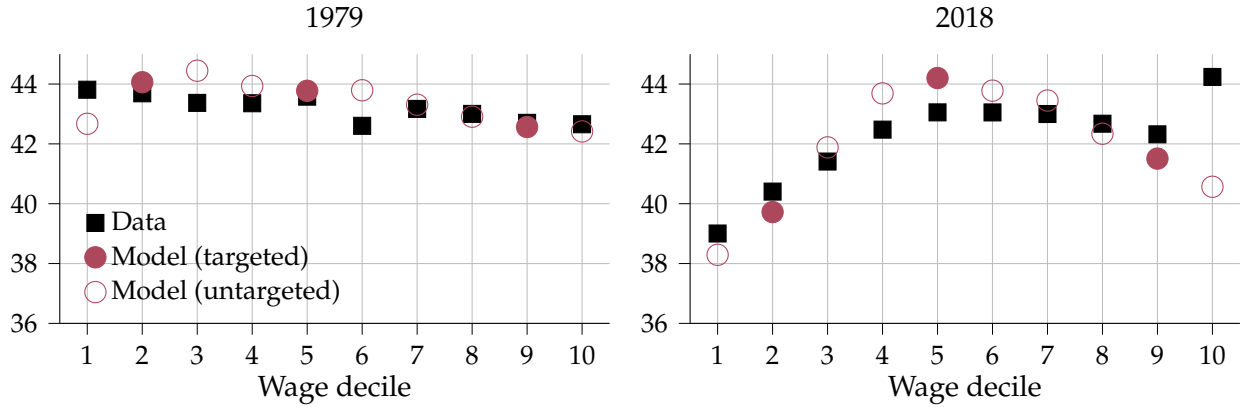


Figure C.1: Hours worked by wage decile: model vs. data.

Notes: The figure compares average hours worked by wage decile in the model and in the data for 1979 and 2018. Black squares denote the data. Red filled circles denote the model moments that are directly targeted in the estimation (wage deciles 2, 5, and 9). Red open circles denote untargeted model implications for the remaining wage deciles.

⁹³Because A_t enters in levels and is large, $J_t' \Omega_t J_t$ is poorly conditioned in the raw parameters (the same scaling issue visible in the rank diagnostics above). We therefore use the reparameterized vector $z_t = (\log A_t, \alpha_t, \beta_t, \gamma_t, \rho_t, b_t^y)$, i.e. with J_t replaced by $J_t^z = J_t G_t$, where $G_t = \partial \theta_t / \partial z_t = \text{diag}(A_t, 1, 1, 1, 1, 1)$ is evaluated at the estimate and rescales only the A -column. In z -space the matrix $J_t^{z'} \Omega_t J_t^z$ is well-conditioned and I recover the raw-parameter covariance by the delta method,

$$\widehat{\text{Var}}(\hat{\theta}_t) = G_t \widehat{\text{Var}}(\hat{z}_t) G_t', \quad \widehat{\text{Var}}(\hat{z}_t) = (J_t^{z'} \Omega_t J_t^z)^{-1} J_t^{z'} \Omega_t \widehat{C}_t \Omega_t J_t^z (J_t^{z'} \Omega_t J_t^z)^{-1}.$$

This is algebraically identical to the raw-parameter sandwich estimator and only improves the numerical conditioning; in particular $\text{se}(\hat{A}_t) = A_t \text{se}(\log \hat{A}_t)$.

C.3 Additional Counterfactuals

Interaction of HBTC and sorting: decomposition Figure C.2 contains the decomposition of income inequality in Figure 13 in each of its three channels. As clear from the Figure, the effect of HBTC given $\Delta\mu(x)$ is that it increases wage dispersion steadily (as opposed to a positive but relatively flat contribution when HBTC is considered on its own).

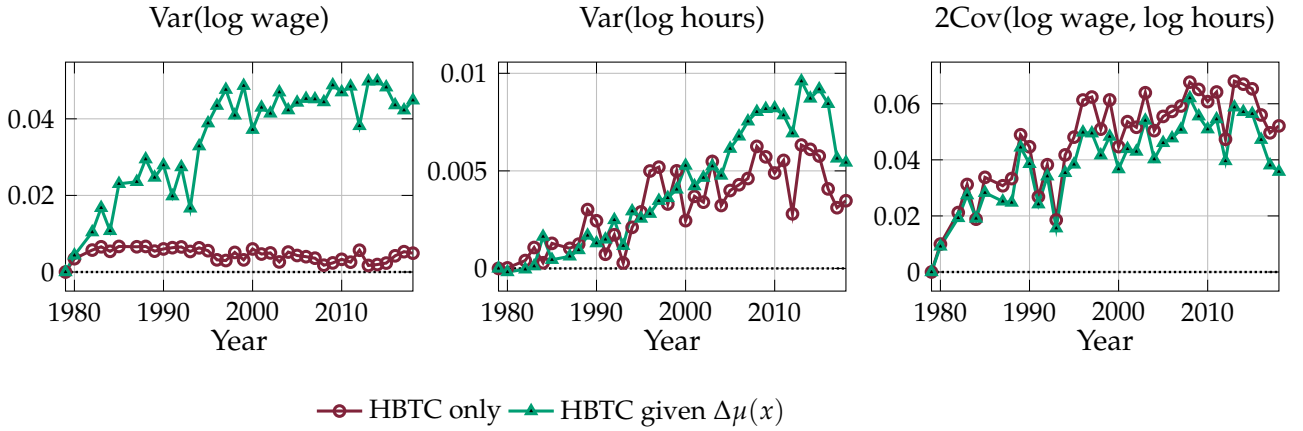


Figure C.2: Marginal effects: decomposition.

Notes: The figure displays the changes in the three components of the variance of log income in the model economy when estimated changes in γ are fed into the 1980 estimated economy, leaving all other parameters unchanged, relative to the level of the variance of log income in 1980 ("HBTC only"); the change in the same components when both estimated changes in γ and in ρ, b^y are fed into the 1980 estimated economy relative to when only ρ, b^y are changing ("HBTC given $\Delta\mu$ ").

D Implications Appendix

D.1 Stochastic Extension

This subsection presents the stochastic extension used in the counterfactual exercises. The extension follows the distance-to-frontier approach (Galichon *et al.*, 2019) to matching with imperfectly transferable utility (ITU). The key additional ingredient is that a contract is now a pair (y, h) , where y is the (discrete) job type and $h \in \mathcal{H}$ is hours worked, also discrete. The extension generates non-degenerate allocations across worker types, job types, and hours, and thus allows for misallocation in the sense of departures from the output-maximizing assignment. We follow (Galichon *et al.*, 2019) in deriving the equilibrium of the model.

Hours wedge. We study a reduced-form wedge that raises the nonpecuniary cost of supplying hours:

$$\psi_\tau(h) = (1 + \tau)\psi(h), \quad \tau \geq 0. \quad (\text{D.1})$$

When $\tau = 0$, the economy coincides with the baseline. When $\tau > 0$, long hours become more costly in utility terms.

Timing and shocks. The stochastic model is a simultaneous contract-choice model, i.e. hours are part of the contract. For each worker type x , the choice set is

$$\{(y, h) : y \in Y, h \in \mathcal{H}\} \cup \{\emptyset\},$$

where $\emptyset$ denotes remaining unmatched. Likewise, for each job type y , the choice set is

$$\{(x, h) : x \in X, h \in \mathcal{H}\} \cup \{\emptyset\}.$$

Each worker of type x receives i.i.d. type-I extreme-value shocks

$$\{\varepsilon_{xyh}\}_{y \in Y, h \in \mathcal{H}} \quad \text{and} \quad \varepsilon_{x0}$$

over the contract alternatives and the outside option. Each job of type y receives i.i.d. type-I extreme-value shocks

$$\{\eta_{xyh}\}_{x \in X, h \in \mathcal{H}} \quad \text{and} \quad \eta_{0y}.$$

All shocks are independent across sides and alternatives and have common scale parameter $\sigma_\delta > 0$. Thus, a single scale parameter governs both the dispersion of matching across partner types and the dispersion of hours choices within matches.

Bargaining sets. Fix a worker type x , a job type y , and an hours contract $h \in \mathcal{H}$. If the worker consumes $c \geq 0$, then worker utility is $u(c) - \psi_\tau(h)$ and the job receives the residual $f(x, y, h) - c$. The corresponding contract-level bargaining set is

$$F^\tau(x, y, h) = \left\{ (\tilde{u}, \tilde{v}) \in \mathbb{R}^2 : \exists c \geq 0 \text{ s.t. } \tilde{u} \leq u(c) - \psi_\tau(h), \quad \tilde{v} \leq f(x, y, h) - c \right\}. \quad (\text{D.2})$$

Distance-to-frontier. For any $(U, V) \in \mathbb{R}^2$, define the contract-level distance-to-frontier as

$$D_{xyh}^\tau(U, V) = \inf \{z \in \mathbb{R} : (U - z, V - z) \in F^\tau(x, y, h)\}. \quad (\text{D.3})$$

Using (D.2), this distance can be written equivalently as

$$D_{xyh}^\tau(U, V) = \min_{c \geq 0} \max \{U - (u(c) - \psi_\tau(h)), V - (f(x, y, h) - c)\}. \quad (\text{D.4})$$

Equation (D.4) is the contract-level analogue of the distance representation in Galichon *et al.* (2019): it is the smallest diagonal shift that makes (U, V) feasible under contract (x, y, h) .

Systematic utilities. Let U_{xyh} denote the worker-side systematic utility associated with contract (x, y, h) and let V_{xyh} denote the corresponding job-side systematic payoff. Under the EV1 specification, the standard log-odds formulas imply

$$U_{xyh} = u_0 + \sigma_\delta \log \left(\frac{\mu(x, y, h)}{\mu(x, \emptyset)} \right), \quad (\text{D.5})$$

and

$$V_{xyh} = v_0 + \sigma_\delta \log \left(\frac{\mu(x, y, h)}{\mu(\emptyset, y)} \right). \quad (\text{D.6})$$

Define the type-level potentials

$$U(x) = u_0 - \sigma_\delta \log \mu(x, \emptyset), \quad V(y) = v_0 - \sigma_\delta \log \mu(\emptyset, y). \quad (\text{D.7})$$

Then (D.5) and (D.6) can be rewritten as

$$U_{xyh} = U(x) + \sigma_\delta \log \mu(x, y, h), \quad V_{xyh} = V(y) + \sigma_\delta \log \mu(x, y, h). \quad (\text{D.8})$$

At equilibrium, any contract (x, y, h) that is chosen with positive mass must lie on its contract-level bargaining frontier:

$$D_{xyh}^\tau(U_{xyh}, V_{xyh}) = 0. \quad (\text{D.9})$$

Using the translation property of the distance-to-frontier function,

$$D_{xyh}^\tau(a + t, b + t) = D_{xyh}^\tau(a, b) + t,$$

and substituting (D.8) into (D.9), we obtain

$$\begin{aligned} 0 &= D_{xyh}^\tau(U(x) + \sigma_\delta \log \mu(x, y, h), V(y) + \sigma_\delta \log \mu(x, y, h)) \\ &= D_{xyh}^\tau(U(x), V(y)) + \sigma_\delta \log \mu(x, y, h). \end{aligned}$$

Hence the equilibrium mass of contracts (x, y, h) is

$$\mu(x, y, h) = \exp \left(- \frac{D_{xyh}^\tau(U(x), V(y))}{\sigma_\delta} \right). \quad (\text{D.10})$$

Pair masses. The mass of matches of type (x, y) is the sum of contract masses across hours:

$$\mu(x, y) = \sum_{h \in \mathcal{H}} \mu(x, y, h). \quad (\text{D.11})$$

Substituting (D.10) into (D.11) yields

$$\mu(x, y) = \sum_{h \in \mathcal{H}} \exp \left(-\frac{D_{xyh}^\tau(U(x), V(y))}{\sigma_\delta} \right).$$

Define the pair-level distance

$$D_{xy}^\tau(U, V) = -\sigma_\delta \log \sum_{h \in \mathcal{H}} \exp \left(-\frac{D_{xyh}^\tau(U, V)}{\sigma_\delta} \right). \quad (\text{D.12})$$

Then the pair mass can be written compactly as

$$\mu(x, y) = \exp \left(-\frac{D_{xy}^\tau(U(x), V(y))}{\sigma_\delta} \right). \quad (\text{D.13})$$

Equation (D.12) is the EV1 regularization of the deterministic union-of-contracts object. As σ_δ vanishes, it converges to its deterministic version

$$D_{xy}^\tau(U, V) = \min_{h \in \mathcal{H}} D_{xyh}^\tau(U, V).$$

Matching equations. The unmatched masses are recovered from the type-level potentials:

$$\mu(x, \emptyset) = \exp \left(\frac{u_0 - U(x)}{\sigma_\delta} \right), \quad \mu(\emptyset, y) = \exp \left(\frac{v_0 - V(y)}{\sigma_\delta} \right). \quad (\text{D.14})$$

Combining (D.13) with (D.14), equilibrium potentials solve

$$\exp \left(\frac{u_0 - U(x)}{\sigma_\delta} \right) + \sum_{y \in Y} \exp \left(-\frac{D_{xy}^\tau(U(x), V(y))}{\sigma_\delta} \right) = n_x, \quad \forall x \in X, \quad (\text{D.15})$$

and

$$\exp \left(\frac{v_0 - V(y)}{\sigma_\delta} \right) + \sum_{x \in X} \exp \left(-\frac{D_{xy}^\tau(U(x), V(y))}{\sigma_\delta} \right) = m_y, \quad \forall y \in Y. \quad (\text{D.16})$$

Equivalently, one may write the market-clearing conditions at the contract level as

$$\mu(x, \emptyset) + \sum_{y \in Y} \sum_{h \in \mathcal{H}} \mu(x, y, h) = n_x, \quad \forall x \in X, \quad (\text{D.17})$$

and

$$\mu(\emptyset, y) + \sum_{x \in X} \sum_{h \in \mathcal{H}} \mu(x, y, h) = m_y, \quad \forall y \in Y. \quad (\text{D.18})$$

Grouping contract masses by pair gives the equivalent pair-level version

$$\mu(x, \emptyset) + \sum_{y \in Y} \mu(x, y) = n_x, \quad \forall x \in X, \quad (\text{D.19})$$

and

$$\mu(\emptyset, y) + \sum_{x \in X} \mu(x, y) = m_y, \quad \forall y \in Y. \quad (\text{D.20})$$

Hours. Conditional on the pair (x, y) being formed, the probability of choosing hours h is

$$\pi^\tau(h \mid x, y) = \frac{\exp\left(-\frac{D_{xyh}^\tau(U(x), V(y))}{\sigma_\delta}\right)}{\sum_{k \in \mathcal{H}} \exp\left(-\frac{D_{xyk}^\tau(U(x), V(y))}{\sigma_\delta}\right)}. \quad (\text{D.21})$$

Hence the mass of contracts (x, y, h) is

$$\mu^\tau(x, y, h) = \mu(x, y) \pi^\tau(h \mid x, y). \quad (\text{D.22})$$

Expected hours and expected output conditional on pair (x, y) are therefore

$$\bar{h}^\tau(x, y) = \sum_{h \in \mathcal{H}} \pi^\tau(h \mid x, y) h, \quad (\text{D.23})$$

and

$$\bar{f}^\tau(x, y) = \sum_{h \in \mathcal{H}} \pi^\tau(h \mid x, y) f(x, y, h). \quad (\text{D.24})$$

Solution method. Under logit heterogeneity, the stochastic extension can be written as a matching-function equilibrium in the sense of [Galichon *et al.* \(2019\)](#). In particular, the theory above defines a matching function

$$M_{xy}^\tau(\mu(x, \emptyset), \mu(\emptyset, y)) = \exp\left(-\frac{D_{xy}^\tau(U(x), V(y))}{\sigma_\delta}\right),$$

where $U(x) = u_0 - \sigma_\delta \log \mu(x, \emptyset)$ and $V(y) = v_0 - \sigma_\delta \log \mu(\emptyset, y)$. Hence equilibrium reduces to solving the nonlinear system (D.15)–(D.16) in the unmatched masses. Following [Galichon *et al.* \(2019\)](#), we initialize one side's unmatched masses at a sufficiently high value and iterate between row and column updates: holding $\mu(\emptyset, y)$ fixed, we solve each worker

equation for $\mu(x, \emptyset)$; holding the resulting $\mu(x, \emptyset)$ fixed, we solve each job equation for $\mu(\emptyset, y)$. This is an application of the iterative projective fitting procedure (IPFP) adapted to the ITU-logit matching-function system. Under the same full-support and regularity assumptions as in [Galichon *et al.* \(2019\)](#), the induced matching function inherits the properties needed for monotone convergence, and the resulting equilibrium is unique. This is the same logic underlying the computational procedure in [Bandiera *et al.* \(2024\)](#).

Aggregate output. Aggregate output under wedge τ is

$$Y(\tau) = \sum_{x \in X} \sum_{y \in Y} \mu(x, y) \bar{f}^\tau(x, y) = \sum_{x \in X} \sum_{y \in Y} \sum_{h \in \mathcal{H}} \mu^\tau(x, y, h) f(x, y, h). \quad (\text{D.25})$$

The output cost of the hours wedge is measured relative to the baseline $\tau = 0$ economy as

$$\mathcal{L}(\tau) = 1 - \frac{Y(\tau)}{Y(0)}. \quad (\text{D.26})$$

With and without HBTC. The paper's baseline counterfactuals take the terminal-year parameter vector as the benchmark technology. Denote this parameter vector by θ^{HBTC} . The no-HBTC comparison replaces only the hours-complementarity parameter γ with its 1979 estimate while keeping all other parameters at their terminal-year values. Denote the resulting vector by θ^{noHBTC} . Formally,

$$\theta^{\text{HBTC}} = (A_T, \alpha_T, \beta_T, \rho_T, \gamma_T, b_T^y, \dots), \quad \theta^{\text{noHBTC}} = (A_T, \alpha_T, \beta_T, \rho_T, \gamma_{1979}, b_T^y, \dots). \quad (\text{D.27})$$

The hours-wedge counterfactuals are then evaluated under both technology regimes:

$$Y^{\text{HBTC}}(\tau), \quad Y^{\text{noHBTC}}(\tau),$$

with corresponding losses

$$\mathcal{L}^{\text{HBTC}}(\tau) = 1 - \frac{Y^{\text{HBTC}}(\tau)}{Y^{\text{HBTC}}(0)}, \quad \mathcal{L}^{\text{noHBTC}}(\tau) = 1 - \frac{Y^{\text{noHBTC}}(\tau)}{Y^{\text{noHBTC}}(0)}.$$

This comparison isolates how the same hours distortion differs in its aggregate consequences depending on whether the economy features the strong hours complementarities associated with HBTC.

Results The first result is reported in [Figure D.1](#). Output losses are increasing in τ , and the same wedge is more costly under the HBTC benchmark than under the no-HBTC counterfactual. The intuition follows directly from the mechanism: When γ is lower, better jobs have a higher marginal return to hours, so the sorting amplification term $f_{hC}'(h)$ is stronger. In

that environment, increasing the cost of hours depresses the value of the high-hour jobs in top matches, and therefore has larger aggregate consequences. By contrast, when γ is held at its 1979 value, the complementarity between hours and the skill-job bundle is weaker and the same wedge destroys less output.

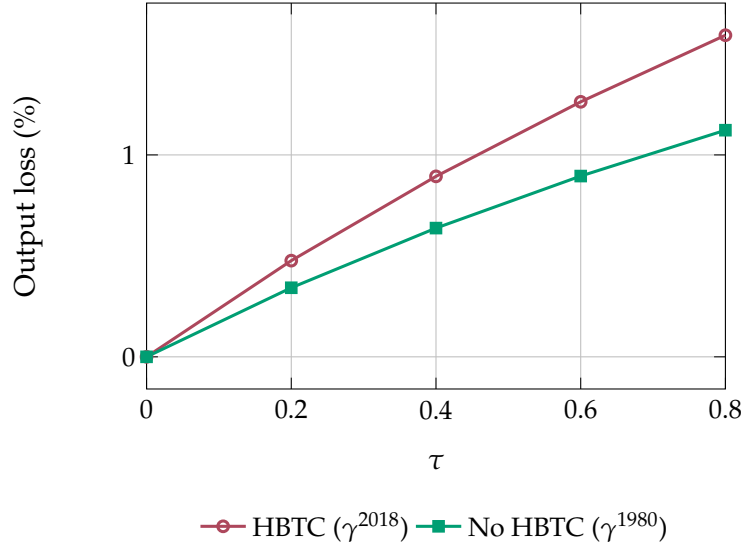


Figure D.1: Output loss as a function of τ .

Notes: The figure plots output loss as a function of the hours wedge τ under the HBTC benchmark and under the no-HBTC counterfactual. The HBTC benchmark is defined as the simulated model using the estimated value of γ in 2018. The no-HBTC counterfactual is defined as the simulated model using the estimated value of γ in 1979. The remaining parameter values are held at their 2018 estimated values. The shocks dispersion parameter σ_{δ} is fixed at 0.5.

Figure D.2 goes deeper in the mechanism by comparing outcomes for bottom- and top-skill workers under the HBTC benchmark and under the no-HBTC counterfactual. Under HBTC, the wedge reduces the probability that top workers occupy top jobs and raises the corresponding probability for bottom workers (middle panel). Hence, the distortion worsens the allocation of talent exactly in the region where long hours are most productive. Without HBTC, by contrast, the reallocation pattern is much weaker and can even go in the opposite direction. In other words, whether the same hours wedge worsens or attenuates mismatch depends on the production technology governing the value of hours in top matches. Moreover, under HBTC, the same wedge reduces income proportionally more at the top than at the bottom; the difference under the no-HBTC case is much smaller (left panel). This occurs despite the reduction in long hours happen quite uniformly across skills (right panel). In other words, similar hours reductions can have very heterogeneous effects along the skill distribution. Depending on whether hours are particularly complementary in top jobs, workers in those jobs are those who suffer the largest consequences.

A useful way to summarize the mechanism is the following. Absent HBTC, long hours can partially act as a substitute margin, allowing lower-skill workers to compensate for lower skill by supplying more hours. A wedge on hours then weakens that compensation margin. Under HBTC, however, long hours are part of the comparative advantage of high-skill workers in high-quality jobs. Raising the cost of hours therefore reduces the value of precisely those top contracts, pushes some high-skill workers away from them, and allows lower-skill workers to move in. The resulting output losses are thus not merely the consequence of fewer hours being supplied; they also reflect a deterioration in the assignment of workers to jobs.

Finally, the patterns in Figure D.2 suggest a natural connection to group-specific wedges. If a common hours-cost wedge already compresses top incomes and reallocates top jobs away from the workers who would otherwise benefit most from them, then a group-specific version of the same wedge (as in, for example, gender-specific time constraints that could originate from unequal division of labor at home) would naturally generate group differences in access to high-return, long-hours contracts. These differences would then be more pronounced the stronger (i.e. more negative) the complementarity term γ .

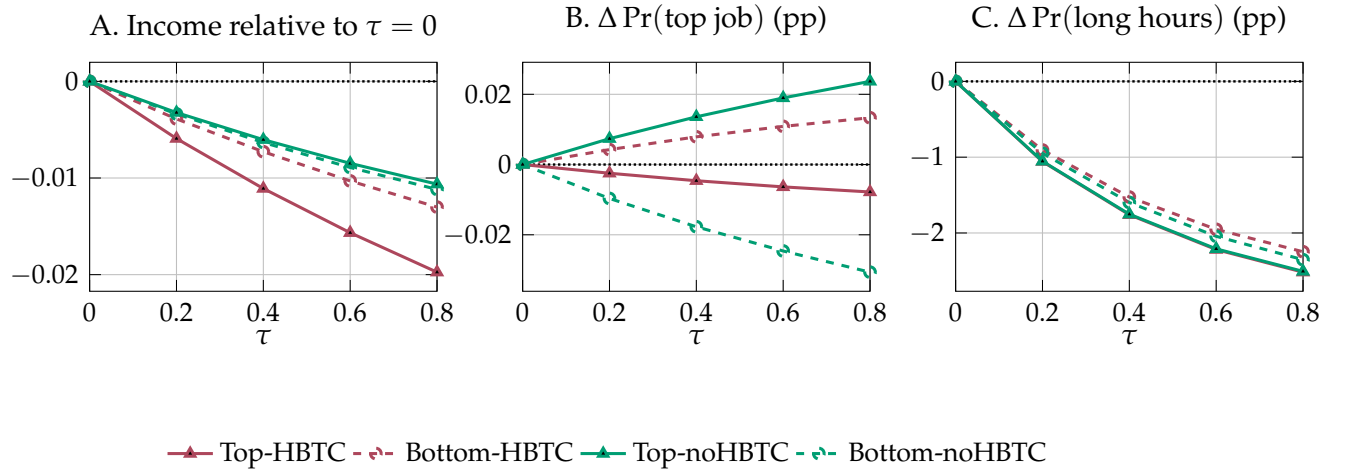


Figure D.2: Top versus bottom workers: effects of τ .

Notes: The figure compares bottom- and top-skill workers under the HBTC benchmark and under the no-HBTC counterfactual. It plots income relative to $\tau = 0$, the change in the unconditional probability of obtaining a top job, and the change in the unconditional probability of working long hours. The HBTC benchmark is defined as the simulated model using the estimated value of γ in 2018. The no-HBTC counterfactual is defined as the simulated model using the estimated value of γ in 1979. The remaining parameter values are held at their 2018 estimated values. The shocks dispersion parameter σ_δ is fixed at 0.5.

D.2 Elasticities

We employ elasticity concepts borrowed from [Scheuer and Werning \(2017\)](#). In particular, we use fixed-assignment elasticities (how much earnings respond if the worker changes hours but stays at the current match/job) and full-assignment elasticities (how much earnings respond if the worker changes hours and is also allowed to move along the equilibrium assignment/hedonic schedule). This appendix derives the relationship between fixed-assignment and full-assignment earnings elasticities in the model, following analogous derivation as those in the Appendix of [Scheuer and Werning \(2017\)](#).⁹⁴

Baseline equilibrium and local schedule representation Fix a baseline equilibrium under tax schedule T . For each worker type x in the region of interest, let:

$y_0(x)$ baseline job, $h_0(x)$ baseline hours, $w_0(x)$ baseline pre-tax earnings.

Recall that earnings(income) are defined as

$$w(x, y, h) = f(x, y, h) - V(y),$$

where $V(y)$ is the equilibrium value associated with job y , or job/firm profits.

For a fixed worker type x , define:

- *Fixed-assignment* schedule. Holding the baseline job $y_0(x)$ fixed,

$$\hat{W}_x(h) \equiv f(x, y_0(x), h) - V(y_0(x)).$$

This object will be useful because it essentially captures the case in which workers can vary effort, but that has no effect on the assignment.

- *Full* schedule. Allowing the worker to move along the baseline job menu,

$$W_x(h) \equiv \max_y \{f(x, y, h) - V(y)\}.$$

This schedule captures the effect of workers moving to better matches. Therefore, elasticities computed using this full-schedule will capture the effect that raising effort has on the allocation of workers to jobs.

⁹⁴In the spirit of [Scheuer and Werning \(2017\)](#), the elasticities derived below are partial-equilibrium elasticities, obtained by perturbing the worker's local tax wedge while holding the relevant baseline earnings schedules fixed. This is sufficient for the mechanism discussed in the main text, but it's possible to analyze how sorting itself is impacted by progressive taxation. This is beyond the scope of this exercise, and results on the topic can be found in [Vereshchagina \(2021\)](#).

Let $y_x(h)$ denote the argmax in the definition of $W_x(h)$. At the baseline point (y_0, h_0, w_0) ,

$$W_x(h_0) = \widehat{W}_x(h_0) = w_0.$$

Moreover, the two schedules are tangent at h_0 by the envelope theorem:

$$W'_x(h_0) = \widehat{W}'_x(h_0).$$

Local tax perturbation and elasticity definitions Let $T(w)$ be the baseline nonlinear tax schedule. Introduce the standard local perturbation (Saez (2001))

$$c = \nu w - T(w) + I, \quad \nu \equiv 1 - \tau,$$

where ν is the local net-of-tax-rate shifter and I is a virtual-income shifter. As standard in the literature, these objects will be used to derive the uncompensated earnings elasticity, the income effects, and the compensate earnings elasticity.

For any schedule defined above $S \in \{W_x, \widehat{W}_x\}$, the worker solves

$$\max_h u(c, h) \quad \text{s.t.} \quad c = \nu S(h) - T(S(h)) + I.$$

Define

$$MRS(c, h) \equiv -\frac{u_h(c, h)}{u_c(c, h)}.$$

The first-order condition is

$$MRS(c, h) = (\nu - T'(w)) S'(h), \quad w = S(h). \quad (\text{D.28})$$

Let $h(\nu, I; S)$ solve (D.28) and define

$$w(\nu, I; S) = S(h(\nu, I; S)).$$

At the baseline $(\nu, I) = (1, 0)$, define the uncompensated elasticity, income-effect term, and compensated elasticity (see also Scheuer and Werning (2017)):

$$\varepsilon_u^S(w_0) = \left. \frac{\partial w}{\partial \nu} \right|_{(1,0)} \frac{1 - T'(w_0)}{w_0}, \quad (\text{D.29})$$

$$\eta^S(w_0) = -(1 - T'(w_0)) \left. \frac{\partial w}{\partial I} \right|_{(1,0)}, \quad (\text{D.30})$$

$$\varepsilon_c^S(w_0) = \varepsilon_u^S(w_0) + \eta^S(w_0). \quad (\text{D.31})$$

When $S = W_x$, these are the full-assignment elasticities. When $S = \widehat{W}_x$, they are the fixed-assignment elasticities.

Generic elasticity formulas We now derive these elasticity formulas explicitly. Let

$$F(h, \nu, I; S) \equiv MRS(c, h) - (\nu - T'(w))S'(h),$$

with $w = S(h)$ and $c = \nu S(h) - T(S(h)) + I$.

Define

$$MRS_c \equiv \frac{\partial MRS}{\partial c}, \quad MRS_h \equiv \frac{\partial MRS}{\partial h}.$$

Differentiating F yields

$$F_h = MRS_c(\nu - T'(w))S'(h) + MRS_h - (\nu - T'(w))S''(h) + T''(w)(S'(h))^2, \quad (D.32)$$

$$F_\nu = MRS_c w - S'(h), \quad (D.33)$$

$$F_I = MRS_c. \quad (D.34)$$

At the baseline, define $\Delta_S \equiv F_h|_{(\nu, I)=(1, 0)}$. By the implicit function theorem,

$$h_\nu = -\frac{F_\nu}{\Delta_S} = -\frac{S'(h_0) - MRS_c(c_0, h_0) w_0}{\Delta_S}, \quad (D.35)$$

$$h_I = -\frac{F_I}{\Delta_S} = -\frac{MRS_c(c_0, h_0)}{\Delta_S}. \quad (D.36)$$

Since $w = S(h)$,

$$w_\nu = S'(h_0)h_\nu, \quad w_I = S'(h_0)h_I.$$

Substituting these expressions for h_ν and h_I into the definitions of ε_u^S , η^S , and ε_c^S , we get explicit expressions for elasticities. In fact, for any differentiable schedule S ,

$$\varepsilon_u^S(w_0) = \frac{(1 - T'(w_0))S'(h_0)[S'(h_0) - MRS_c(c_0, h_0) w_0]}{w_0 \Delta_S}, \quad (D.37)$$

$$\eta^S(w_0) = \frac{(1 - T'(w_0))S'(h_0)MRS_c(c_0, h_0)}{\Delta_S}, \quad (D.38)$$

$$\varepsilon_c^S(w_0) = \frac{(1 - T'(w_0))(S'(h_0))^2}{w_0 \Delta_S}. \quad (D.39)$$

Equivalently,

$$\frac{\eta^S(w_0)}{\varepsilon_c^S(w_0)} = MRS_c(c_0, h_0) \frac{w_0}{S'(h_0)} \quad (D.40)$$

and, substituting out Δ_S using (D.32),

$$\frac{1}{w_0 \varepsilon_c^S(w_0)} = \frac{1}{S'(h_0)} \left[MRS_c(c_0, h_0) + \frac{MRS_h(c_0, h_0)}{MRS(c_0, h_0)} - \frac{S''(h_0)}{S'(h_0)} \right] + \frac{T''(w_0)}{1 - T'(w_0)}. \quad (\text{D.41})$$

Fixed-assignment versus full-assignment elasticities Apply the previous derivation to the two schedules W_x and $\widehat{W}_x$. Because the two schedules are tangent at h_0 ,

$$W'_x(h_0) = \widehat{W}'_x(h_0),$$

equation (D.40) immediately implies

$$\frac{\eta(w_0)}{\varepsilon_c(w_0)} = \frac{\widehat{\eta}(w_0)}{\widehat{\varepsilon}_c(w_0)}. \quad (\text{D.42})$$

To see the subtraction explicitly, first apply (D.41) to the full-assignment schedule $S = W_x$:

$$\frac{1}{w_0 \varepsilon_c(w_0)} = \frac{1}{W'_x(h_0)} \left[MRS_c(c_0, h_0) + \frac{MRS_h(c_0, h_0)}{MRS(c_0, h_0)} - \frac{W''_x(h_0)}{W'_x(h_0)} \right] + \frac{T''(w_0)}{1 - T'(w_0)}.$$

Applying the same expression to the fixed-assignment schedule $S = \widehat{W}_x$ gives

$$\frac{1}{w_0 \widehat{\varepsilon}_c(w_0)} = \frac{1}{\widehat{W}'_x(h_0)} \left[MRS_c(c_0, h_0) + \frac{MRS_h(c_0, h_0)}{MRS(c_0, h_0)} - \frac{\widehat{W}''_x(h_0)}{\widehat{W}'_x(h_0)} \right] + \frac{T''(w_0)}{1 - T'(w_0)}.$$

Since $W'_x(h_0) = \widehat{W}'_x(h_0)$, all terms are common across the two expressions except the curvature term. Subtracting the second expression from the first gives

$$\frac{1}{w_0 \varepsilon_c(w_0)} = \frac{1}{w_0 \widehat{\varepsilon}_c(w_0)} - \frac{W''_x(h_0) - \widehat{W}''_x(h_0)}{(W'_x(h_0))^2}. \quad (\text{D.43})$$

For any worker type x and baseline earnings level w_0 , rearranging (D.43) yields

$$\varepsilon_c^{full}(w_0) = \frac{\widehat{\varepsilon}_c(w_0)}{\widehat{\Phi}_x(w_0)}, \quad (\text{D.44})$$

where

$$\widehat{\Phi}_x(w_0) = 1 - \widehat{\varepsilon}_c(w_0) \mathcal{G}_x(w_0), \quad \mathcal{G}_x(w_0) \equiv \frac{(W''_x(h_0) - \widehat{W}''_x(h_0)) w_0}{(W'_x(h_0))^2}. \quad (\text{D.45})$$

The above derivations are the direct analogue of the fixed-assignment adjustment in

Scheuer and Werning (2017). The form is the same because the geometry is the same: the full schedule is the upper envelope of fixed-assignment schedules, and is therefore more convex. What differs in the present model is the set of primitives that determine the curvature wedge $\mathcal{G}_x(w_0)$, which we characterize next.

Primitive expression for the curvature wedge Since

$$\widehat{W}_x(h) = f(x, y_0, h) - V(y_0),$$

we have

$$\widehat{W}'_x(h_0) = f_h(x, y_0, h_0), \quad \widehat{W}''_x(h_0) = f_{hh}(x, y_0, h_0). \quad (\text{D.46})$$

For the full schedule, the optimizer $y_x(h)$ solves

$$f_y(x, y_x(h), h) = V'(y_x(h)). \quad (\text{D.47})$$

By the envelope theorem,

$$W'_x(h) = f_h(x, y_x(h), h),$$

and differentiating,

$$W''_x(h) = f_{hh}(x, y_x(h), h) + f_{yh}(x, y_x(h), h) y'_x(h). \quad (\text{D.48})$$

Differentiate (D.47) with respect to h :

$$(f_{yy}(x, y_0, h_0) - V''(y_0)) y'_x(h_0) + f_{yh}(x, y_0, h_0) = 0,$$

so

$$y'_x(h_0) = \frac{f_{yh}(x, y_0, h_0)}{V''(y_0) - f_{yy}(x, y_0, h_0)}. \quad (\text{D.49})$$

Using this expression for $y'_x(h)$ in (D.48)

$$W''_x(h) = f_{hh}(x, y_x(h), h) + \frac{f_{yh}(x, y_x(h), h)^2}{V''(y_x(h)) - f_{yy}(x, y_x(h), h)}.$$

Using (D.48) in this last expression, the curvature wedge is⁹⁵

$$W''_x(h_0) - \widehat{W}''_x(h_0) = \frac{f_{yh}(x, y_0, h_0)^2}{V''(y_0) - f_{yy}(x, y_0, h_0)}. \quad (\text{D.50})$$

⁹⁵Since $W'_x(h_0) = f_h(x, y_0, h_0)$, equation (D.51) follows immediately from (D.45).

Hence

$$\mathcal{G}_x(w_0) = \frac{w_0 f_{yh}(x, y_0, h_0)^2}{f_h(x, y_0, h_0)^2 [V''(y_0) - f_{yy}(x, y_0, h_0)]}. \quad (\text{D.51})$$

The equilibrium objects $V''(y_0)$, y_0 , h_0 , and w_0 all depend on sorting and on hours choices, and therefore on the richer set of primitives in the model.

Let $\chi(y) \equiv \mu^{-1}(y)$ denote the inverse matching function, and let $h^*(x, y, V)$ denote the worker's hours choice at a match (x, y) under value $V(y)$. Along the equilibrium matching locus,

$$V'(y) = f_y(\chi(y), y, h^*(\chi(y), y, V(y))).$$

Differentiating,

$$V''(y) - f_{yy} = f_{xy}\chi'(y) + f_{yh}(h_y^* + h_x^*\chi'(y) + h_V^*V'(y)). \quad (\text{D.52})$$

Equation (D.52) makes clear that

- f_{yh} enters the adjustment wedge directly;
- f_{xy} enters through the equilibrium matching slope $\chi'(y)$;
- f_{xh} enters through the hours response h_x^* ;
- and preferences enter through the hours derivatives h_x^*, h_y^*, h_V^* .

It is in this sense that the present framework generalizes the effective-labor environments studied in the previous literature: the form of the envelope correction is generic, but the mapping from primitives to the correction is richer because hours and sorting are jointly determined.

Quantitative relevance We now quantify how much larger the required full-assignment elasticity is than the fixed-assignment elasticity in the estimated economy, and how this correction changes when the hours-biased technology channel is shut down.

To do so, I consider two exercises. The first is a time-series exercise in the spirit of the main quantitative analysis. For each year, I compute the top-tail average adjustment

$$\overline{\mathcal{R}}_{t,p} \equiv \mathbb{E}_t \left[\frac{\varepsilon_c(x)}{\widehat{\varepsilon}_c(x)} \mid x \in \text{top } p\% \text{ of the earnings distribution} \right],$$

with $p = 1$ in the baseline specification, and compare it to the corresponding object in a no-HBTC counterfactual. The no-HBTC economy is constructed exactly as in the main text: I hold fixed the parameter governing the complementarity between hours and the skill-job bundle at its initial level and re-solve the equilibrium in each year. This exercise isolates how

much HBTC changes the wedge between the required elasticity and the within-job one in the upper tail of the earnings distribution, which is the region most relevant for discussions of progressivity.

Figure D.3 shows that the full elasticity always exceeds the fixed-assignment elasticity in both economies (both plotted lines are always above one), so that the assignment margin matters quantitatively throughout the sample. At the same time, the top-tail adjustment is systematically larger in the estimated economy than in the no-HBTC counterfactual, and particularly so in recent decades. This shows that HBTC can raise the extent to which sorting matters for elasticity calculations.

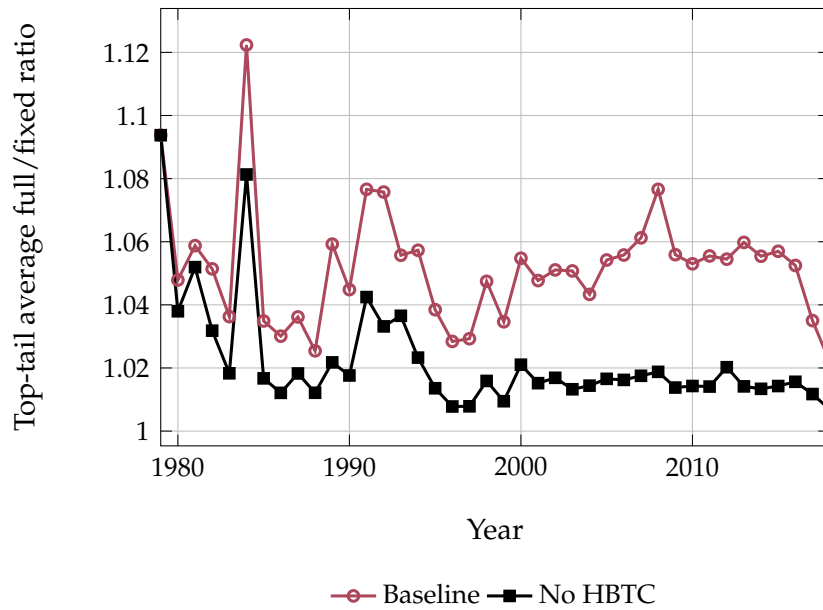


Figure D.3: Top-tail elasticity adjustment over time.

Notes: The figure plots the top-tail average ratio of the full compensated elasticity to the fixed-assignment compensated elasticity over time in the baseline model and in the no-HBTC counterfactual.

The second exercise is cross-sectional and focuses on the most recent year. Results are plotted in Figure D.4. I partition the earnings distribution into deciles and compute both fixed-assignment and full-assignment compensated elasticities in the baseline and in the no-HBTC economy. The results reveal a stark difference in the cross-sectional elasticity profile. In the baseline economy, compensated elasticities are increasing in earnings deciles, both with and without assignment adjustment. In the no-HBTC counterfactual, by contrast, the elasticity profile is inverse-U shaped and declines at the top. Intuitively, with HBTC top earnings remain closely tied to the marginal return to hours, whereas in the no-HBTC counterfactual top earnings are increasingly driven by worker-job levels rather than by the marginal payoff to additional hours.

Turning to the role of assignment, the full elasticity exceeds the fixed-assignment elasticity throughout the distribution in both economies, but the proportional amplification is larger under HBTC for the middle and upper deciles (while it vanishes at the top for the no-HBTC case). Thus, HBTC affects the relevant earnings elasticity through two distinct channels. First, it reshapes the within-job elasticity profile itself. Second, it increases the extent to which sorting magnifies those within-job responses. Put differently, the effect of HBTC on the policy-relevant elasticity is not only that workers at the top become more responsive within jobs, but also that their response is more strongly amplified by the ability to move along the assignment schedule.

Taken together, these exercises show that the elasticity object relevant once assignment is allowed to respond can differ materially from the fixed-assignment elasticity, and that this difference is itself sensitive to the technology governing the return to hours.

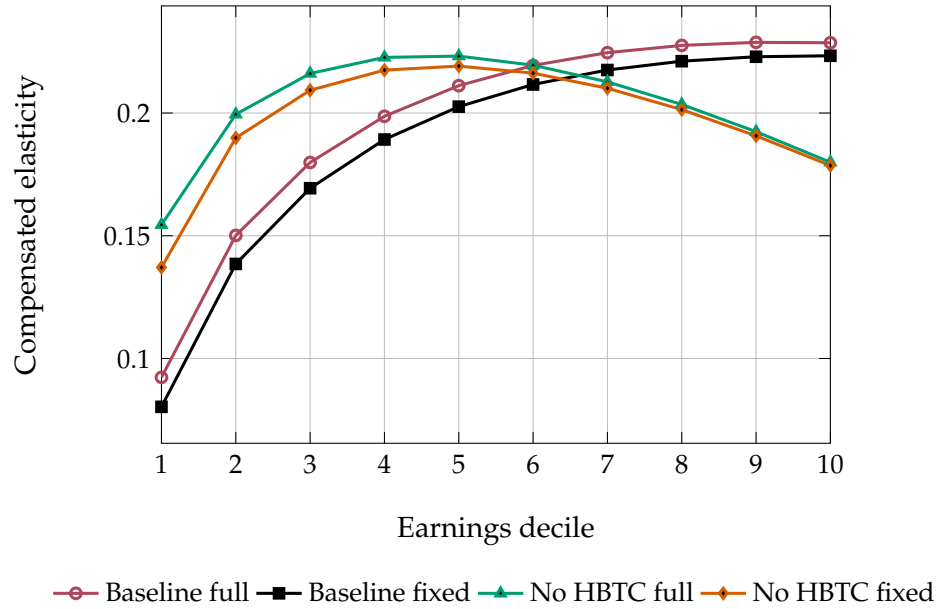


Figure D.4: Compensated elasticities by earnings decile.

Notes: The figure plots decile-level averages of the compensated elasticity under the baseline model and under the no-HBTC counterfactual, distinguishing between the fixed-assignment elasticity and the full elasticity.